# *syngo*.via Administrator Manual

## Administrator Manual – Basics

VB40A

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SIEMENS Healthineers

# syngo.via Administrator Manual

## Administrator Manual – Basics

VB40A

# Legend

| | |
|---|---|
| i | Indicates a hint<br>Is used to provide information on how to avoid operating errors or information emphasizing important details |
| » | Indicates the solution of a problem<br>Is used to provide troubleshooting information or answers to frequently asked questions |
| ● | Indicates a list item |
| ✓ | Indicates a prerequisite<br>Is used for a condition that has to be fulfilled before starting a particular operation |
| ◆ | Indicates a one-step operation |
| 1<br>2<br>3 | Indicates steps within operating sequences |
| *Italic* | Is used for references and for table or figure titles |
| → | Is used to identify a link to related information as well as previous or next steps |
| **Bold** | Is used to identify window titles, menu items, function names, buttons, and keys, for example, the Save button |
| **Orange** | Is used to emphasize **particularly** important sections of the text |
| **Courier** | Is used for on-screen output of the system including code-related elements or commands |
| `Courier` | Is used to identify inputs you need to provide |
| **Menu > Menu Item** | Is used for the navigation to a certain submenu entry |
| **<variable>** | Is used to identify variables or parameters, for example, within a string |
| ⚠ **CAUTION** | CAUTION<br>Used with the safety alert symbol, indicates a hazardous situation which, if not avoided, could result in minor or moderate injury or material damage.<br>CAUTION consists of the following elements:<br>• Information about the nature of a hazardous situation<br>• Consequences of not avoiding a hazardous situation<br>• Methods of avoiding a hazardous situation |

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

WARNING consists of the following elements:

- Information about the nature of a hazardous situation
- Consequences of not avoiding a hazardous situation
- Methods of avoiding a hazardous situation

**WARNING**

# Legend

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# 1 Introduction

This document gives an overview on administrative tasks and tools of *syngo*.via.

For better readability, we refer to the user in the masculine form.

(➔ Page 13 *Intended use*)

(➔ Page 14 *syngo.via Interfaces*)

(➔ Page 15 *syngo.via system overview*)

(➔ Page 18 *Integration of syngo.via clients*)

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(➔ Page 23 *Documentation overview*)

(➔ Page 19 *Required user qualifications*)

(➔ Page 21 *Education and training*)

(➔ Page 45 *General tasks of the administrator*)

(➔ Page 51 *Tools for Administration*)

## 1.1 Intended use

*syngo*.via is a software solution intended to be used for viewing, manipulation, communication, and storage of medical images.

It can be used as a stand-alone device or together with a variety of cleared and unmodified *syngo* based software options.

*syngo*.via supports interpretation and evaluation of examinations within healthcare institutions, for example, in Radiology, Nuclear Medicine, and Cardiology environments.

The system is not intended for the displaying of digital mammography images for diagnosis in the U.S.

## 1.2 *syngo*.via Interfaces

The following interfaces are included in *syngo*.via:

- Clinical User Interface (*syngo*.via client)

  Interactive user interface to access *syngo*.via functionality.

- Administration and Service User Interface (*syngo*.via Administration Portal)

  Interactive user interface to access *syngo*.via service functionality.

- Direct Image Transfer

  A fast data transfer link between the *syngo* modalities and *syngo*.via.

- HL7/DICOM

  Standard communication interfaces for communication in medical systems. *syngo*.via is thus able to communicate with information systems such as RIS, PACS, and modalities from Siemens Healthineers and other vendors.

- Image call-up

  Interface provided for external systems to initiate an image call-up at *syngo*.via (load and unload).

The following interfaces are used by *syngo*.via:

- Active Directory

  For user authentication and authorization, and security policies.

- SMTP

  For sending important system messages as e-mails or SMS.

- Domain Name System (DNS)

  For resolving names to IP addresses.

- SRS Infrastructure

  Provides access to the Smart Remote Services back-end.

Within your clinical network, *syngo*.via has to rely on trusted entities to enforce its security policy. Generally, a trusted entity is a certificate authority (CA) or a defined trusted source.

*syngo*.via trusts the following entities:

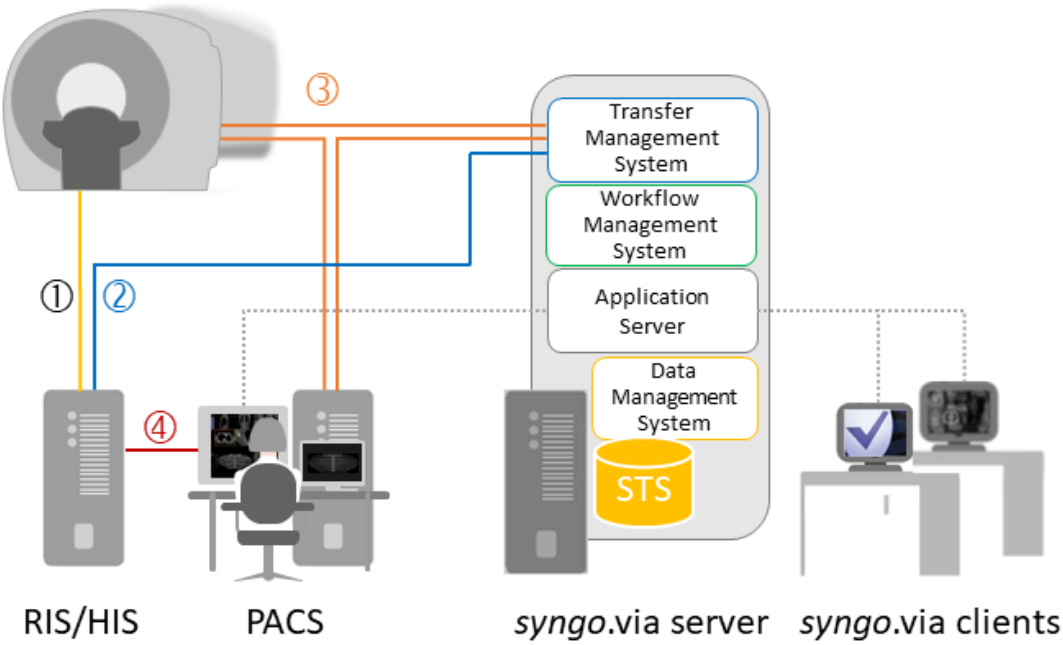
- PKI infrastructure
- Active Directory and Windows user management
- Domain Name System (DNS)

To ensure maximum security of your system, it is essential that these entities (if used) are configured correctly.

## 1.3 *syngo*.via system overview

*syngo*.via comprises physical and functional subsystems which cooperate with the various components of the clinical environment.

# 1 Introduction




(1) DICOM Modality Worklist, Modality Performed Procedure Step

(2) DICOM Modality Worklist, Patient Information Reconciliation, Structured Results

(3) DICOM Image Transfer

(4) DICOM / HL7

*syngo*.via essentially consists of the following components:

- The **Transfer Management System (TMS)** which provides the DICOM interface and the Direct Image Transfer interface of *syngo*.via.
- The **Workflow Management System (WMS)**, which constitutes the set of services that cover the workflow of the *syngo*.via system management functionality. It interacts with an external RIS (DICOM Modality Worklist) or HIS (HL7 Patient Update & Merge, Report Export).

  The main functionality provided by the WMS is administration of scheduled and running workflows, including the triggering of progress messages for external systems. It also triggers workflow creation and monitors its progress.
- The **Application Server (APS)**, which allows access to *syngo*.via applications from a Windows workplace with a *syngo*.via client. The *syngo*.via server and the *syngo*.via client provide 2D, 3D, and hybrid viewing, processing, and reading.
- The **Data Management System (DMS)**, which represents the set of tasks and services that make up the data management functionality of the *syngo*.via system.

  The DMS maintains index data for information stored in STS, and performs automatic STS clean-up functions based on high and low water marks.
- The **Short Term Storage (STS)**, which keeps high-volume data produced by the modalities. This function enables fast data access, for example, for prior study comparison or longitudinal studies processing with high-volume data.
- The **Service** module, that provides the maintenance functionality like error and message handling, system status monitoring, configuration, SRS connectivity, update and upgrade handling, and the Administration Portal.

## 1.4 Integration of *syngo*.via clients

The *syngo*.via client runs as a stand-alone application. It can be installed on a Windows workstation.

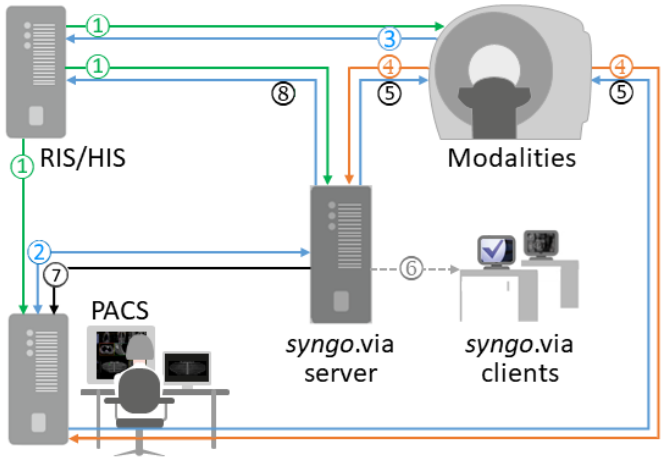
Without image call-up integration, the built-in **Patient Browser** is available when the client is started.

The *syngo*.via client can also be integrated at PACS or RIS workstations by image call-up. The client is then controlled by the PACS patient list or the RIS worklist.

## 1.5 Dataflow within the clinical environment

*syngo*.via is designed to fit into various medical environments.

Therefore, the dataflow varies according to the local configuration. The following illustration shows a scenario with PACS and RIS integration:




(1) Worklist

(2) Prefetching

(3) Procedure Information (MPPS)

(4) Images

(5) Storage Commitment

(6) Reading

(7) Report Data (as DICOM SR)

(8) Report Data

Patients are scheduled at the RIS terminal. The scheduling information is transferred from the RIS to the modality and to *syngo*.via.

Examination and Quality Assurance are performed at the modality. The modality sends the images to PACS for archiving and to *syngo*.via. Once the PACS has successfully archived the data, it sends an acknowledgment to the modality.

Meanwhile, data from previous examinations is prefetched from PACS and sent to *syngo*.via.

Preparation for reading, the reading itself, and reporting are performed in the following environments:

- The second console at the scanner (*syngo* CT workplace) with integrated *syngo*.via client
- A dedicated *syngo* MM WP with integrated *syngo*.via client
- Any PC running a *syngo*.via client
- A PACS workstation with *syngo*.via image call-up
- A RIS workstation with *syngo*.via image call-up

When reading and reporting are completed, *syngo*.via sends the completed structured report to PACS.

## 1.6 Required user qualifications

Usually, two different types of administrators are responsible for the system:

- IT Administrator
- Clinical Administrator

### 1.6.1 IT Administrator

The "IT Administrator" has expert knowledge of networks, operating systems, user administration, and basic knowledge of the RIS/PACS workflow. He is responsible for data security and protection, backup management, client installation, and first level support. He manages the IT infrastructure of the clinical network as well as of the RIS/PACS system. The field of activity includes updating the server and client software, system monitoring, and first-level troubleshooting.

To be able to administrate and maintain the system, the "IT Administrator" must have skills in:

- Microsoft Active Directory (user administration)
- Microsoft Windows OS administration (commands, scripts)
- Microsoft Windows Authorization Manager (user role administration)
- Microsoft Windows Backup & Recovery
- Microsoft Windows Firewall (communication ports)
- IT infrastructure services:
  - DNS
  - SMTP
  - NTP
  - LAN/WLAN
  - VPN
- Backup management systems
- Storage systems (RAID, NAS, etc.)
- English language

### 1.6.2 Clinical Administrator

The "Clinical Administrator" is a medical specialist, for example, a radiographer or radiologist with clinical knowledge, typically someone who works in the radiology department. He is responsible for:

- RIS/PACS interface
- Postprocessing issues on *syngo*.via clients
- Data workflow (DICOM / HL7)
- Clinical workflow (procedures, layouts, workflow mappings)

For more information on user profiles, see (→ *"User profiles" in the Application Online Help*)

## 1.7 Education and training

To empower your staff with expertise, Siemens Healthineers offers continuous education tailored to your institutions needs based on a blended learning approach.

At the time your *syngo*.via is installed, an initial training is provided to guarantee a seamless start with your *syngo*.via software.

The handover training is delivered by the Clinical Education Specialist and includes the following:

- Clinical integration of the main modality
- Pre-training clarification (remote or on-site)
- Dedicated number of *syngo*.via training hours/days, depending on the specific applications and users for your institution

To give you the possibility to increase your knowledge at your pace, you have access to the Siemens Healthineers learning platform **PEPconnect**, where you can benefit from online training and educational videos, focused on the utilization of your *syngo*.via clinical applications. https://pep.siemens-info.com

With **PEPconnections** – our premium subscription for workforce education management – you can assign and manage the education of your staff, ensuring to your employees a continuous education path.

In addition, an **extended education plan** can be provided which may consist of remote trainer or classroom training.

For further information about education plans, please contact your local Siemens Healthineers sales representative.

# 1 Introduction

For general information on classroom training, please see https://training.healthcare.siemens.com.

# 2 Documentation overview

The *syngo*.via software offers several levels of user assistance, beginning with tooltips and extended tooltips on the screen linking into the comprehensive *syngo*.via Online Help. The *syngo*.via software is also accompanied by the *syngo*.via Basic Operator Manual and the *syngo*.via Administrator Manual.

Both manuals constitute the Instructions for Use of *syngo*.via. They are available in local languages as online-version.

Some functions described in the documentation may not be available on your system. Contact your Clinical Administrator or Siemens Healthineers for more information.

## 2.1 Administrator Manual

The *syngo*.via Administrator Manual contains the safety advisories (regarding administration), and provides information about administration and configuration of your *syngo*.via system. It is available in local languages.

Detailed information is available in the *syngo*.via Administration Online Help.

Most configuration tasks of *syngo*.via are performed with the *syngo*.via Administration Portal. For detailed information about the User Interface and the corresponding configuration tasks, see (→ *"Configuration in syngo.via" in the Online Help*).

## 2.2 Basic Operator Manual

The *syngo*.via Basic Operator Manual contains the safety advisories, gives you an overview of the most important tools that are available in your *syngo*.via system, and contains introductory as well as basic information. The *syngo*.via Basic Operator Manual is available in local languages.

## 2.3 Online Help

The comprehensive information for *syngo*.via is the context-based Online Help. You can quickly access the Online Help following the links in an extended tooltip, by pressing the **F1** key (on the client), or by clicking the Question Mark icon on the access bar (both on the client and in the *syngo*.via Administration Portal).

The information range depends on your licenses.

The Online Help is also available as a website. You can access it with a browser with the URL “http://<*syngo*.via-server>:8090”, where <*syngo*.via-server> is the IP address or the host name of your *syngo*.via server.

The Online Help is available for the standard user interface languages.

## 2.4 Supported languages

*syngo*.via supports the following user interface languages:

- English
- German
- French
- Spanish
- Japanese
- Chinese (simplified)

The *syngo*.via Administration Portal is available in English only.

*syngo*.via does not support local differences in languages, for example, Spanish (Mexico) is displayed the same as Spanish (Spain).

If the Online Help is not available in your language, the Online Help as well as the tooltips are displayed in English.

## 2.5 Further documentation

- **Data Sheet**

  Detailed technical data is provided in the *syngo*.via Data Sheet, VB40.

- **Applications and disease-specific workflows**

  Applications and disease-specific workflows are accompanied by their own documentation and safety advisories. The Online Help system comprises the help modules for workflows and applications installed on your individual *syngo*.via system. See (→ *Application Online Help*).

  Contact your Clinical Administrator or Siemens Healthineers for more information.

- **e-Clips**

  These application videos are available in the Online Help as a supplement for the user documentation.

- **MM Reading Quick Guide**

  This Quick Guide Online Help takes you on a tour through the *syngo*.via workflow MM Reading. This Online Help contains descriptions of the main functions of this workflow as well as the basics for getting started quickly.

  The MM Reading Quick Guide is available in English only.

- **Pre-Installation Manual for server virtualization**

  This manual describes the installation procedure of a VMware Container to use *syngo*.via as a virtual machine.

- **Reporting Adapter**

  This manual describes the *syngo*.via Reporting Adapter to integrate *syngo*.via reports into external reporting systems.

# 3 Safety Advisory

Warnings indicate a potential hazard to the health or life of patients or personnel.

Cautions indicate conditions or consequences that you should pay particular attention to when working with *syngo*.via, but no direct danger is involved.

## 3.1 Hardware

**CAUTION**

Unexpected shutdown of the server for users.

**Data loss or data inconsistency possible.**

- Before any scheduled shutdown, inform all users (for example by e-mail or phone) about the scheduled downtime and give them enough time to finish their work and close workplace applications before the server is shut down.

**CAUTION**

Backups onto non-redundant hardware are not sufficient for data security.

**If a hardware failure or other severe failures happen, a massive loss of data can occur if backups have not been performed or if non-redundant hardware was used for backups.**

- Set up a routine for secondary backup of database and configuration items on external (removable) media at regular intervals and based on a backup concept.
- Set up a backup concept for patient data routed from modalities over *syngo*.via to long-term archive (PACS).
- Regularly check that backups are performed properly.
- Ensure that critical data is additionally stored on redundant hardware (RAID).

**CAUTION**

Computers are not infallible and unexpected errors may occur. In addition, scheduled downtimes are necessary to maintain the software.

**The system may not be available for use, for instance, in the operating room or for an emergency case.**

- Work out an emergency plan for response to non-availability of the system or the network, for instance, to use a system on a different network or to use print-outs or films.

**CAUTION**

The installation of unsuitable hardware may cause serious problems.

**Data loss.**

- Do not install hardware which is not adequate in terms of reliability, capacity and performance.
- Problems arising due to unsuitable third-party hardware are not the responsibility of Siemens Healthineers.

**CAUTION**

Hardware failure such as disk crash.

**Data loss.**

- The IT Administrator is responsible for developing a concept for patient data recovery in case of defective hardware, and for the improvement of fail-safe operation of short-term and archive configuration (i.e. use redundant RAID concept).

**CAUTION**

Image on hard-copy does not match the displayed image.

**The diagnosis and treatment may be made on the basis of incorrect information.**

- Do not install hard-copy devices that have not been released for use with the system.

**CAUTION**

Use of inappropriate devices (displays, printers) to review and report radiological images.

**Incorrect review of images.**

- Review and reporting of images require an optimum display of images.
- Only use suitable monitors and approved DICOM printers for review and reporting of images.
- Follow the maintenance and care instructions given in the manufacturer's documentation.

**CAUTION**

Malfunction of system due to hardware not fulfilling manufacturer's specification.

**Hardware failure, execution of tasks may be delayed.**

- Use hardware components only as specified in the manufacturer's documentation (installation and operating instructions, data sheets).

**CAUTION**

Power outage can lead to an unintended server shutdown.

**Data loss or corrupted data as data may be left in an undefined state.**

- Use the system with a UPS to protect your system from data loss in case of power outages.

## 3.2 Software

**CAUTION**

Use of an anti-virus software that is not provided by Siemens Healthineers.

**Malicious software can cause harms to the system and loss of all patient data.**

- The administrator is responsible for the configuration of the anti-virus software.
- Configure and regularly update the anti-virus software.

**CAUTION**

The installation of other non-system software interferes with the system software.

**Harms to the server system and loss of all patient data.**

- The installation of software not provided from Siemens Healthineers is not allowed on the server system.
- Problems arising due to interactions with non-Siemens Healthineers software are not the responsibility of Siemens Healthineers.

Once the server system has been handed over to the customer, no software must be installed on the *syngo*.via server that does not comply with the rules and restrictions described in the “syngo.via Software Blacklist”.

The latest available revision of the Software Blacklist is provided in the LifeNet syngo Information Center, “System Security” section.

**CAUTION**

After a software update of *syngo*.via, RIS, PACS or MMWP the applications/their interfaces can become incompatible to each other.

**Clinical workflow can be interrupted due to misconfiguration.**

- If the system is updated or upgraded, all important and frequently used applications/their interfaces must be checked thoroughly.
- Be aware that also a change of the RIS or PACS may make interfaces incompatible to each other and therefore should be checked.

**CAUTION**

Images processed with radial ranges may be displayed in oblique orientation.

**Wrong diagnosis due to orientation mix-up.**

- Set up a departmental policy for creation of radial ranges regarding the orientation and review of the resulting outcome.

**CAUTION**

Result of true size in printout can depend on printer settings, round-off errors and other factors.

**True Size printout does not correlate exactly to the real anatomy size.**

- Be aware of precision limitations when printing in **True Size**. Always compare the image scalebar within the printed images for validation with a physical measurement unit such as a ruler to ensure that the printout has the real anatomy size.
- If your layout contains small segments because of which the scalebar cannot be applied, either validate true size directly by measuring the printed anatomy or choose a different layout.

**CAUTION**

Failed system updates can be time-consuming.

**System availability can be impacted**

- Always calculate a sufficient time buffer for updates or upgrades.

Do not directly manipulate the database!

Manual manipulation of the *syngo*.via database by a third-party database tool can completely destroy the contents, or part of it. This may lead to missing patient data sets or lost images.

## 3.3 Configuration

**CAUTION**

Data are automatically deleted due to configurable settings.

**Loss of clinically relevant data if rules are not correctly specified.**

- Be very careful when creating "not to be archived" rules for data. This data can be automatically deleted and cannot be recovered.
- Make sure that all data (images and reports) necessary for medical purposes are completely and successfully sent to an archive.
- Do not use automatic deletion if the archive node does not support Storage Commitment.

**CAUTION**

Configurable automatic rules can become complex.

**Unexpected system behavior or loss of data due to definition of complex automatic rules.**

- Test all new rules to ensure that the results conform to your expectations.

**CAUTION**

There is no mechanism to recognize an emergency patient in *syngo*.via.

**The user may not be aware of a pending emergency case.**

- Establish and apply a method to identify emergency patients, for instance, by adding a corresponding notice to the patient identification.

**CAUTION**

Limited system access due to security measures (for example, licensing issues)

**System access might be hindered or restricted in emergency cases.**

- Set up an emergency access environment if necessary.

  This may include:

  - Creating an emergency user account with limited access rights and ensuring that this account is available to appropriate personnel only
  - Disabling the screen saver at the designated emergency treatment clients
  - Establishing a license strategy to ensure that sufficient licenses are always available at the designated emergency treatment clients

**CAUTION**

Unauthorized access to the system.

**System can become non-operational; loss of patient data.**

- This medical device is designed to be operated in a protected network environment. We strongly recommend to not directly connect the device to public networks.
- The IT Administrator is responsible for the network security at the site and for the security of optional infrastructure, such as desktop-virtualization environments. Consult the corresponding manuals for secure setup, and update as required.
- Ensure that only authenticated devices, i.e. belonging to the healthcare enterprise, are connected to the network.
- Set up firewalls and user-account password protections for both server and client.
- Do not allow users to change configuration files.
- Update virus protection software as required.

**CAUTION**

Short Term Storage (STS) or system disk is full.

**System not available, no new image storage possible, system lock-up.**

- Verify the settings for high and low watermark, and check frequency in the *syngo*.via Administration Portal in the **Technical Configuration** workspace, **DICOM Data Handling** > **Archiving and Deletion**.
- Regularly check the **Status Monitoring**, especially the system partition and the fill level of the STS.
- Regularly check the system log for messages regarding storage status, auto-deletion, and auto-archiving.

**CAUTION**

*syngo*.via provides a mode that allows users to load local studies for an examination as priors, based on a site-specific unique Enterprise Master Patient Index (EMPI).

**Wrong diagnosis because the EMPI for a certain patient is not unique.**

- EMPI mode is disabled by default and should only be configured if the unique Enterprise Master Patient Index (EMPI) for each patient in the DICOM attribute OtherPatientID (0010,1000) can be ensured by the site. EMPI mode can only be configured by the Customer Care Center.
- Always consider that prior patient identification attributes displayed in the image text and in the patient tab on the access bar may differ, for example, the patient name or the patient ID may not be the same. Use the Other Patient ID value in the Patient Browser to verify that the current study and the prior study belong to the same patient.
- EMPI is only supported for studies assigned as priors from the short-term storage (STS). Searches based on an Enterprise Master Patient Index are not supported for pre-fetching studies and querying/retrieving studies.
- If the EMPI option is enabled but causes issues with prior loading, contact the Customer Care Center immediately to have the EMPI settings adjusted.

## 3.4 User Management

After installation of the *syngo*.via server, the IT Administrator must change the default passwords of the administrative user accounts (for example, for AdminUser/Administrator and RemoteAdmin).

Due to security reasons, it is not recommended to use shared or group accounts. Additionally, these accounts do not allow for proper auditing of who is accessing the application, and security incidents cannot be attributed to specific individuals (required by some regulations).

## 3.5 Data Transfer/Communication

**CAUTION**

Unencrypted client-server transfer of patient health information.

**Patient health information will be vulnerable in case of unauthorized network access.**

- Set up encrypted client/server communication.
- Set up encrypted DICOM communication.
- Protect your network by a firewall.

**CAUTION**

Security certificates may expire.

**Encrypted client/server communication will be blocked, when the server certificate expires.**

- Renew security certificates in time.

**CAUTION**

Data transfers between systems are not guaranteed.

**Loss of data if data are deleted locally before they have been successfully transferred to another system.**

- Only systems and communication protocols supporting Storage Commitment guarantee successful transfer to another system.
- In other transfers, it is necessary to verify the correct arrival of the data at the remote system itself.
- Do not delete local data until the transfer to the remote system is verified.

**CAUTION**

Archiving has been configured without using Storage Commitment. Thus, the archive flag is set based only on a transfer response.

**Data is marked with the archive flag even if it has not been archived successfully. If this data is subsequently deleted from the local system, it is irretrievably lost.**

- Only use DICOM nodes configured with Storage Commitment as an archive. If this is not possible, verify the storage of the data at the remote system. Do not delete local data until its storage at the remote system has been verified.

**CAUTION**

Long-term archiving is not provided with this system.

**Loss of data (including reports) can occur if data have not been transferred to a long-term archive.**

- Verify that local data has been transferred successfully to a long-term archive, before deleting it.
- Check the status bar for archiving failures (indicated by an error icon).
- Check the **Job View** to locate the error and take appropriate measures (for example, re-start failed archive jobs).

**CAUTION**

Patient merge conflicts can occur without user's knowledge.

**Wrong diagnosis.**

- Check **Status Monitoring** for potential patient merge conflicts and resolve them. Use the e-mail notification system to be notified of potential patient merge conflicts.

For details on using email notification, see (→ *syngo.via Administration Online Help*)

**CAUTION**

Patients may be incorrectly merged if only Patient ID is used to identify patients.

**Incorrect diagnosis basis.**

- The types of identification used by the system when automatically merging patients can be configured. It is recommended that at least two of the following forms of identification be used: Patient Name, Patient ID (required), and Date of Birth. If the default configuration is changed, test the new configuration to avoid unintended consequences.

**CAUTION**

Labels on CDs and DVDs created by the *syngo*.via system do not include the patient identification.

**CDs or DVDs may be mixed up.**

- If high volumes of CDs or DVDs are created at your site, it is recommended to use a media burning system that uses information from the DICOM header to create the media labels. The media burning system must be configured as a DICOM node and images must be exported as DICOM objects to this node. This is the only way to include patient identifying information on the label of the CD or DVD.

**CAUTION**

Deletion or modification of a DICOM node while not completed jobs are in queue.

**Jobs may fail. Archiving states of data objects may be no longer sufficient. The data may be no longer subject for further archiving or auto routing jobs and thus unintendedly deleted after a time period.**

- Check the usage of the DICOM node in the rule definitions, and the existence and status of jobs using this node prior to deletion or modification of the DICOM node.
- Check regularly for DICOM objects in state "Archive failed".

**CAUTION**

The connection between a remote node and *syngo*.via is temporarily unavailable, due to a system shutdown/crash or network problems.

**Messages from a remote node are not applied to data (no patient update) or data availability is not notified to remote node.**

- Only connect to remote nodes that can buffer and return messages.
- If patient (personal) data has been corrected but not propagated properly to remote nodes receiving images from *syngo*.via, although the respective study is available, resend HL7 messages from *syngo*.via to the remote node, so both systems are in sync again.
- Configure an appropriate amount of time between retries for HL7 messages on both communicating systems to ensure high probability of HL7 message application.
- Contact your Customer Service Engineer for adapting the configuration of the remote node interfaces.

**CAUTION**

An HL7 message could not be applied to data in the workplace.

**Inconsistencies between HIS/RIS and *syngo*.via may result in hampered image callup from HIS/RIS or other external system, or data may not be found at all.**

- Regularly check the event log and scan for messages concerning unsuccessful processing of HL7 messages.

The HL7 interface provides access to sensitive patient data.

As an administrator you have to ensure that only information systems (such as RIS, HIS) which are allowed to access these sensitive data can connect to the provided service.

Access control can be done, for example, by configuring the local firewall so that it restricts the access of the HL7 interface to the dedicated IP address of the information system.

## 3.6 *syngo*.via Reporting

**CAUTION**

User treats report as official report although it has not been signed off.

**The diagnosis and treatment may be made on the basis of incorrect information.**

- Configure a disclaimer that states that the report is not valid without signature.

**CAUTION**

Insecure storage of patient reports.

**Patient privacy is compromised**

- Only store reports on secure systems with controlled access.

**CAUTION**

Incorrect information in user-customized templates

**Incorrect diagnosis basis**

- Be very careful when including information in templates. If a template provided by Siemens Healthineers is changed, Siemens Healthineers is no longer responsible for its content.

**CAUTION**

The printed report could be mixed up by mistake with other printed reports, e.g. if not thoroughly filed or handled.

**Wrong diagnosis due to incorrect patient identification.**

- If the report is intended to be printed, configure for every page attributes identifying the patient.

**CAUTION**

Unexpected report findings and text due to copy-paste errors, imported or automatically filled-in data.

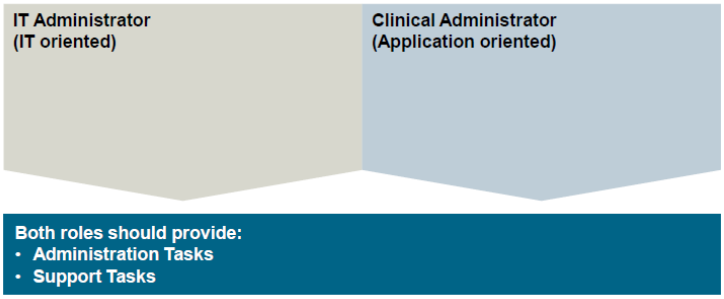
**The diagnosis and treatment may be based on incorrect information.**

- Read the report text carefully before signing it off or storing it.

# 4 General tasks of the administrator

The customer is regarded as a partner in the service support process, given that the customer's administrator assumes responsibility for the operation and for the first line support of *syngo*.via.

This contributes not only to a fuller and more efficient usage of the customer's *syngo*.via system, but also to maximizing system uptime.




This section summarizes the tasks of the administrator regarding *syngo*.via server and client. See (➔ Page 49 *Service Support*)

The task description is based on two roles an administrator may have:

- IT Administrator
  - administration tasks (➔ Page 46 *Administration tasks of the IT Administrator*)
  - support tasks (➔ Page 47 *Support tasks of the IT Administrator*)

- Clinical Administrator
  - administration tasks (➔ Page 48 *Administration tasks of the Clinical Administrator*)
  - support tasks (➔ Page 48 *Support tasks of the Clinical Administrator*)

## 4.1 Administration tasks of the IT Administrator

The following list contains the main tasks of the "IT Administrator":

- Installation and update of *syngo*.via client prerequisites (for example, Microsoft .NET Framework) and application — **regularly and on demand**. (➔ Page 177 *Installing the prerequisites for syngo.via clients*).
- Update of Microsoft Windows on clients **regularly and on demand**.
- Update of Microsoft Windows operating system on *syngo*.via server — **regularly**.
- Update of *syngo*.via server with Siemens Healthineers hotfixes and Service Packs (using the Software Update) — **on demand**.
- Update of *syngo*.via client BIOS, firmware and drivers — **on demand**.
- Configuration of system backup — **once**. (➔ Page 150 *Configuring backup settings*)
- Check for successful execution of backups — **daily**.
- Archiving of Audit Trail logs using optical media or network shares (HIPAA Audit Controls, USA only) — **weekly**. (➔ Page 163 *Audit trail*)
- Configuration of DICOM nodes (for example, printers, PACS, modalities) — **on demand**. (➔ *See Configuration of DICOM nodes in syngo.via Configuration Online Help*)

- License Management (import, check availability of *syngo*.via application licenses, assign to dedicated users or clients) — **on demand**. (➔ *See License configuration in syngo.via Configuration Online Help*).
- User Account and Role Management (manage domain and local user accounts using Active Directory and/or Windows Authorization Manager, assign roles to users and user groups using Windows Authorization Manager) — **on demand**.
- Network Management (allow remote access for the SRS, configure to send important messages to the IT Administrator by e-mail or SMS) — **once**. (➔ Page 223 *Smart Remote Services*)
- Data security and data protection (install, configure and update firewalls, virus scanners, and Microsoft operating system hotfixes on clients and servers) — **regularly and on demand**. (➔ Page 193 *Data and system security*).
- Management of Device Guard server protection (if switched on) — **on demand**.
- Exchange of existing certificates — **on demand**. (➔ Page 221 *Replacement of self-signed syngo.via certificates*)

## 4.2 Support tasks of the IT Administrator

- Provide help to clinical users regarding IT topics (use trouble-shooting tools, escalate issues to the Customer Care Center, if required) — **on demand**.
- Assist the Customer Care Center during trouble-shooting of software issues (provide access and configuration data) — **on demand**. (➔ Page 223 *Smart Remote Services*)
- Assist the hardware vendor during trouble-shooting of hardware issues (provide access to server hardware and diagnostic tool results) — **on demand**.
- Check *syngo*.via server systems for working properly (use Status Monitoring and e-mail notifications) — **daily**.
- Solve *syngo*.via server issues (*syngo*.via application server, operating system, network, and firmware) — **on demand**.

- Solve *syngo*.via client issues (user management, network, hardware, and operating system issues) — **on demand**.
- Use Remote Assistance for desktop sharing with Customer Care Center — **on demand**.

## 4.3 Administration tasks of the Clinical Administrator

The following list contains the administration tasks of the "Clinical Administrator":

- Configuration of application settings (for example, configuration of Display Layouts, Report Templates) — **on demand**. (→ *See Layout Gallery in syngo.via Basic Application Online Help*).
- Configuration of data-related settings (auto data deletion, auto routing, exclude from archiving rules) — **on demand**. (→ *See Configuring automatic data deletion from STS in syngo.via Configuration Online Help*).
- Configuration of workflow-related settings (workflow assignment rules, auto pre-fetching rules) — **on demand**. (→ *See Workflow assignment in syngo.via Configuration Online Help*).
- Customize client software options (for example, Patient Browser)

## 4.4 Support tasks of the Clinical Administrator

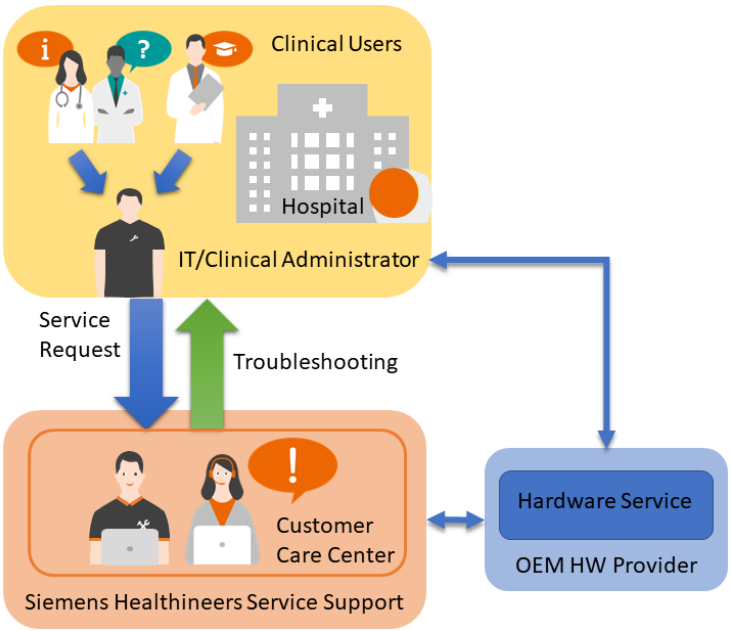
The following list contains the support tasks of the "Clinical Administrator":

- Provide help to clinical users regarding application topics (use trouble-shooting tools, Online Help, escalate issues to the Customer Care Center, if required) — **on demand**.
- Train clinical users in handling the *syngo*.via client (knowledge transfer on *syngo*.via applications to clinical users, e-Clips) — **on demand**. (→ *See syngo.via Basic Application Online Help*).

- Assist Siemens Healthineers application specialists during trouble-shooting of software issues (for example, provide anonymous patient examination data for reproducing a software issue and help to reproduce reported issues) — **on demand**.
- Solve *syngo*.via application-related issues (for example, delete examination data, layouts, or worklists) — **on demand**.
- Check archive states in the Patient Browser (for example, for not archived data) — **regularly**

## 4.5 Service Support

The following diagram shows the support chain between clinical users, IT or Clinical Administrators, the Customer Care Center and Hardware Provider(s):



# 4 General tasks of the administrator

# 5 Tools for Administration

Several tools are available for the administration of *syngo*.via.

You can:

- Work in *syngo*.via Administration Portal
- Work on operating system level of the *syngo*.via server

With the help of Smart Remote Services (SRS), Siemens Healthineers service professionals are able to access your administrative tools and provide remote support. (➔ Page 223 *Smart Remote Services*)

## 5.1 Work in *syngo*.via Administration Portal

Most administrative tasks are done through the *syngo*.via Administration Portal, which can be accessed from a Windows workstation over Microsoft Internet Explorer.

(➔ Page 55 *Logging on to the syngo.via Administration Portal*).

Other browsers may work, but they are not explicitly tested by Siemens Healthineers. For details on versions, see the *syngo*.via Data Sheet.

## 5.2 Work on operating system level of the *syngo*.via server

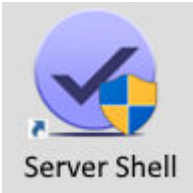
General server administration as well as certain *syngo*.via-specific administrative tasks are performed with standard tools provided by the operating system.

Administrative tasks on the operating system level are usually performed using a Remote Desktop Connection.

See (➔ Page 87 *Logging on to syngo.via server operating system*).

# 5 Tools for Administration

Most Windows administration tasks are performed using the Microsoft Management Console. For example, the **Server Manager** (ServerManager.msc) provides access to the **Diagnostics** and **Server Roles** management console.




Command-line tools provided with *syngo*.via need to be started in the ***syngo*.via Server Shell**. The shell can be launched using the corresponding icon on the Windows Desktop of the *syngo*.via server.

Certain *syngo*.via system variables are only available in the shell. Therefore, always start *syngo*.via command-line tools in this environment.

# 6 *syngo*.via Administration Portal

The ***syngo*.via Administration Portal** is used to perform administrative tasks. It can be accessed directly from your client, or remotely through Microsoft Internet Explorer.

It provides access to:

- System Configuration

  (→ *System configuration*)
- Interface Configuration

  (→ *Configuration of DICOM nodes*)
- Workflow Configuration

  (→ *Workflow configuration*)

  (→ *Configuration of the DICOM modality worklist query*)

  (→ *Prior rules and prior handling*)
- Archive Configuration

  (→ *Configuration of data archiving*)
- Data Management Configuration

  (→ *Configuration of autorouting rules*)

  (→ *Configuring automatic data deletion from STS*)
- Backup Configuration

  (→ *Configuring backup settings*)
- Software Update

  (→ *Updating the syngo.via application server*)
- Status Monitoring

  (→ *Status Monitoring*)

- Message Viewer

  (→ *Message Viewer*)

As the content is structured in workflow-oriented workspaces, some contents can be accessed from more than one workspace.

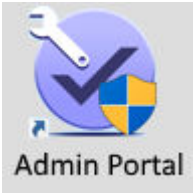
The range of functionality offered by the ***syngo*.via Administration Portal** depends on the service level of the user account.

- (→ *Opening the syngo.via Administration Portal*)
- (→ *Logging on to the syngo.via Administration Portal*)
- (→ *Screen layout of the syngo.via Administration Portal*)

## 6.1 Opening the *syngo*.via Administration Portal

The range of functionality offered by the ***syngo*.via Administration Portal** depends on the service level of the user account.

You can access the ***syngo*.via Administration Portal** in different ways:




◆ On the desktop of the server, double-click the **Admin Portal** icon.

– or –

On the access bar of a client, click the **Settings** icon.

– or –

From a computer with network access to the server, start the Internet Explorer and enter **`https://<server>/adminportal`**.

**`<server>`** is the FQDN (fully qualified domain name) or the IP address of the server.

The ***syngo*.via Administration Portal** login page is displayed.
(→ Page 55 *Logging on to the syngo.via Administration Portal*)

Some functions of the ***syngo*.via Administration Portal** may not be available over network access.

If you encounter the message **There is a problem with this website's security certificate. [...] Continue to this website (not recommended)**, you can ignore it and continue.

To avoid the message, add the certificate to the trusted certificates store or use the FQDN for access.

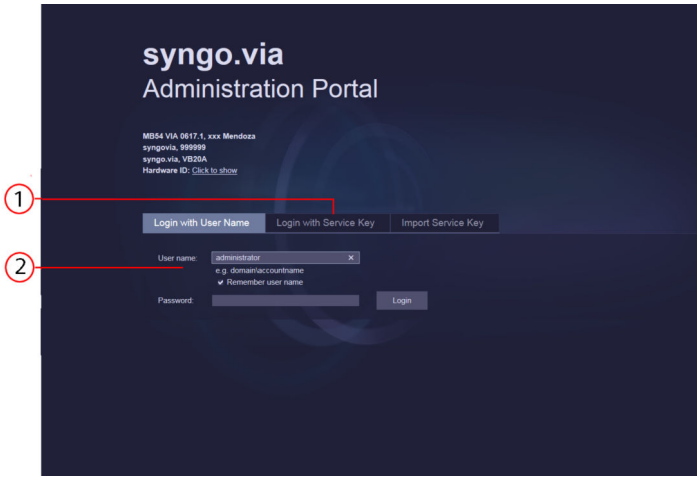
## 6.2 Logging on to the *syngo*.via Administration Portal

You can log on to the ***syngo*.via Administration Portal** with a Windows user account. The range of functionality offered by the ***syngo*.via Administration Portal** depends on the service level of the user account.

Logging on with a service key is intended for Smart Remote Services. (→ Page 85 *Access control to syngo.via Administration Portal based on service levels*)

1 Call up the ***syngo*.via Administration Portal**. (→ Page 54 *Opening the syngo.via Administration Portal*)

The login page is displayed.



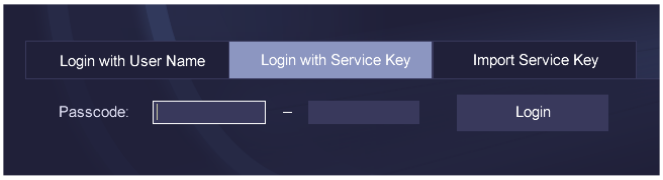

(1) Log on with service key

(2) Log on with user account

2 Click **Login with User name**, enter the user credentials of your Windows user account in the **User name** and **Password** field and click the **Login** button.

– or –

As a service user, click **Login with Service Key**, enter the last 10 digits of the service key and click the **Login** button.




The password is case-sensitive.

If available, you can click the password eye to check the typing of your password. The password is only displayed as long as the mouse button is pressed.

Before you can log on with a service key, you must import the service key once.

(➔ Page 57 *Importing the service key*)

After a certain period of inactivity (default 30 minutes), you are automatically logged off from the ***syngo*.via Administration Portal**.

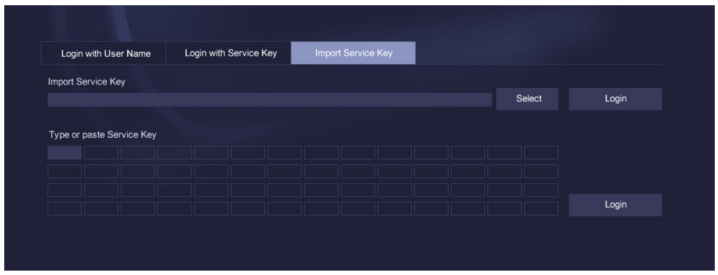
### 6.2.1 Importing the service key

Prior to logging on to the ***syngo*.via Administration Portal** with a service key, you must import the service key file or enter the key manually **once**.

1 Call up the ***syngo*.via Administration Portal**. (➔ Page 54 *Opening the syngo.via Administration Portal*)

The login page is displayed.

2 Click the **Import Service Key** tab.




3 Click **Select**, browse to the service key file and import it.

– or –

Enter the service key in the **Type or paste Service Key** area.

4 Click **Login**.

Service keys expire after a defined time. The **Service Key Expiry Date** is displayed on the status bar of the ***syngo*.via Administration Portal**.

## 6.3 Screen layout of the *syngo*.via Administration Portal

In the ***syngo*.via Administration Portal**, workspaces group the content according to workflows.

The selected workspace determines the content of the navigation area on the left. Clicking an item of the navigation tree opens a window in the content area.

As the content is structured in workflow-oriented workspaces, some contents can be accessed from more than one workspace.

**Example: Layout of the *syngo*.via Administration Portal**

(1) Toolbar

Shows the name of the connected server as well as the name of the current user and his access mode. Icons provide further functions.

(2) Workspaces for primary navigation

Provide a workflow-oriented grouping of the content.

(3) Navigation tree

Shows and admits access to the administrative content. The navigation tree changes according to the workspace selected.

(4) Content area

Shows information and configurable items.

(5) Status bar

Shows the following elements:

– Number of active ***syngo*.via Administration Portal** users

– Number of active client users

– Service key expiry date

– System time

– Status of Event notification

## 6.4 Status Monitoring

On the **Status Monitoring** window, you can monitor states and failures of hardware and software components of your system. It provides you an overview of the system health and shows you components which need your attention.

(→ Page 60 *Accessing Status Monitoring*)

The following tabs are available for different views:

- **Component view**

  Displays the current status of important system components, for example hardware, database, DICOM interfaces, third-party components.

- **Asset view**

  Provides general environmental data collected regularly from your system, for example, hardware, graphics card, hotfixes, services, and so on.

### 6.4.1 Accessing Status Monitoring

You can access **Status Monitoring** in the ***syngo*.via Administration Portal**.

1 Log on to the ***syngo*.via Administration Portal**.

2 On the title bar, click the **Open Status Monitoring** icon to open the window:

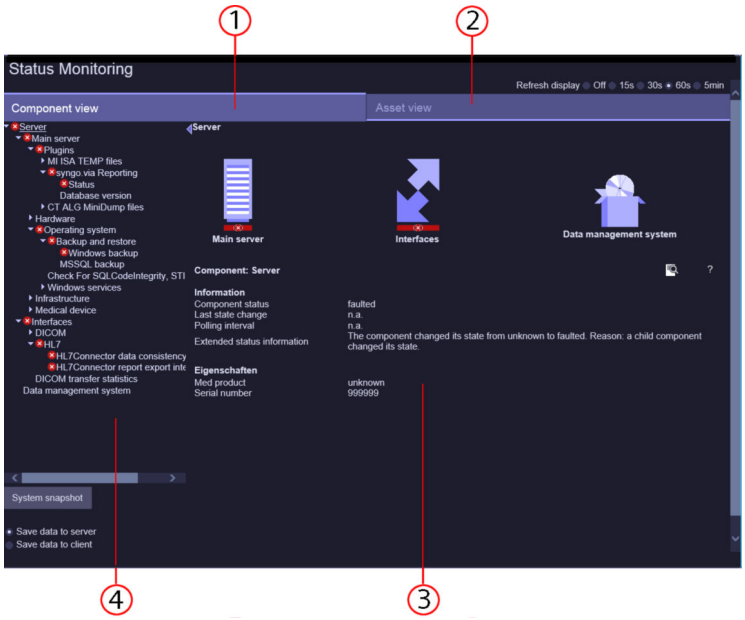
The **Status Monitoring** window opens.

(➔ Page 61 *Screen layout of Status Monitoring*)

A yellow mark on the **Open Status Monitoring** icon indicates that at least one warning is pending; a red mark indicates that at least one error is pending.

## 6.4.2 Screen layout of Status Monitoring

The **Status Monitoring** window consists of two different views.




(1) **Component view** tab

(2) **Asset view** tab

(3) Content area

(4) Component navigation tree

## 6.5 Message Viewer

Use the **Message Viewer** to find the corresponding message to an identified error condition. In addition, you receive suggestions for further analysis and corrective actions.

In the **Message Viewer** window, you can display and filter system and application-relevant messages written in the event log and in the central Message repository.

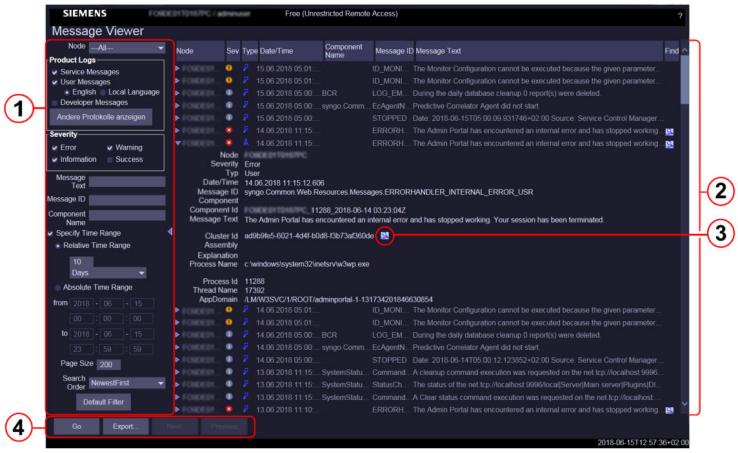
The **Message Viewer** can be accessed by clicking the corresponding icon on the toolbar of the ***syngo*.via Administration Portal**, or from the system **Status Monitoring** UI. If you access the **Message Viewer** from **Status Monitoring**, only messages related to the selected component are displayed.

Note that some functions of the **Message Viewer** are only available with service level 7 access rights.

(→ Page 62 *Screen layout of the Message Viewer*)

### 6.5.1 Screen layout of the Message Viewer

The **Message Viewer** window contains the following elements:




(1) Filter options (log type, severity, contents, date)
(➔ Page 63 *Filter options for Message Viewer*)

(2) Messages list / contents area

Lists the messages according to the given filter set. Each message is expandable/collapsible for details.

(3) Icon to view related messages

(4) **Go** button to apply the filter, and further control buttons

## 6.5.2 Filter options for Message Viewer

The following filter options are available in the **Message Viewer**:

| | |
|---|---|
| **Node** | To search for messages of a certain server or clients. |

| | |
|---|---|
| **Product Logs** | Different messages created by your system's applications can be selected for viewing:<br>• Developer messages, such as program exceptions.<br>• User messages, displayed on the client screen.<br>For user messages, you can choose between *English* or *Local Language*.<br>• Service messages, such as component status messages. |
| **Show Other Logs** | Allows you to select further messages for example, from ADAM (*syngo*Configuration), Application, Key Management Service |
| **Severity** | To select the classification of errors/messages:<br>• **Error**:<br>• **Warning**:<br>• **Information**:<br>• **Success**: |
| **Message Text** | To filter event logs for certain message texts. Use asterisk (*) as wildcard character for zero or more characters. |
| **Message ID** | To filter event logs for a specific message ID. Use asterisk (*) as wildcard character for zero or more characters. The search term is case-sensitive. |

| | |
|---|---|
| **Component Name** | To filter event logs for a specific component name. Use asterisk (*) as wildcard character for zero or more characters. The search term is case-sensitive. |
| **Specify Time Range** | Limits the output to the number of messages that are valid for the desired date and time range.<br>You can select a relative or absolute time range. |
| **Page Size** | To limit the number of results displayed on one page. |
| **Search Order** | **Newest first**: The most recent filtered message is displayed at the top.<br>**Oldest first**: The oldest filtered message is displayed at the top. |

# 7 User management

*syngo*.via uses local users or groups from the server operating system for authentication or authorization. Additionally, it is possible to integrate *syngo*.via in your clinical IT infrastructure to combine the existing user authentication with the authorization of *syngo*.via.

(➔ Page 68 *Authentication*)

(➔ Page 69 *Authorization*)

(➔ Page 159 *Adding a server to a domain*)

The assignment of users and user groups to *syngo*.via roles is done with Authorization Store. This configuration is stored in an SQL database on the *syngo*.via server.

There are a few typical situations which require adaptations of the user management.

- *Add new users (for example, after installation)*
  - Create users with the Windows or Active Directory user management.
  - Create user groups with the Windows or Active Directory user management.
  - Assign users to user groups with the Windows or Active Directory user management. The users inherit the roles and permissions associated with the user group.
  - Assign users or user groups to *syngo* roles with Authorization Store.
- *Modify user or user group roles*
  - Remove *syngo* roles assigned to users or user groups with Authorization Store.
  - Assign users or user groups to new *syngo* roles with Authorization Store.

- *Remove users*
  - Delete users with the Windows or Active Directory user management.
  - Delete users from user groups with the Windows or Active Directory user management.
  - Remove users from the *syngo* role assignment with Authorization Store.

For adapting the user management, see:

(→ Page 70 *Creating local user accounts for syngo.via*)

(→ Page 72 *Predefined administrative user accounts*)

(→ Page 73 *Access rights and roles*)

(→ Page 80 *Role manager*)

## 7.1 Authentication

For authentication, *syngo*.via relies on standardized user management solutions.

Authentication means to identify a user. Unique identification of the user is the basis and prerequisite for access control and logging of relevant user activities. In principle, *syngo*.via users are identified by their user name, password, and their corresponding domain.

For authentication, *syngo*.via uses local Windows user accounts, managed by the Security Accounts Manager (SAM). Additionally, it is possible to authenticate Active Directory (AD) domain accounts.

Single sign-on is only available for AD domain users with a configured *syngo* role. (→ Page 80 *Role manager*)

Furthermore, *syngo*.via allows access to the system in emergency cases.

The *syngo*.via Administration Portal supports an additional authentication system based on service keys which is only used by the Customer Care Center. See (→ Page 85 *Access control to syngo.via Administration Portal based on service levels*).

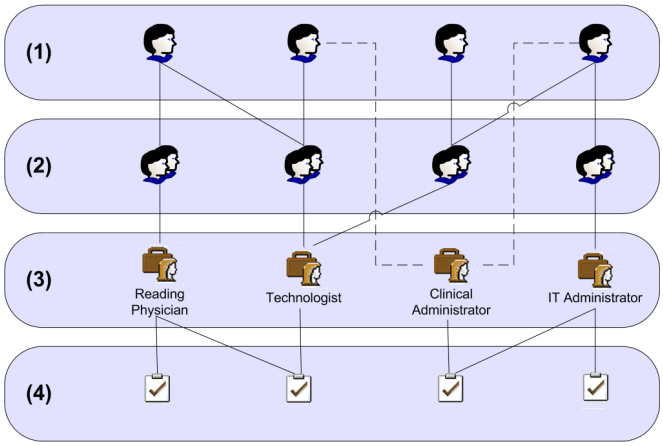
## 7.2 Authorization

Authorization is the act of specifying user permissions for dedicated tasks.

The mapping between user roles and *syngo* tasks is configured with the *syngo*.via server operating system Authorization Store. The Authorization Store allows you to assign Windows users and user groups as well as AD domain users and groups to designated *syngo* roles. A *syngo*.via user can only invoke a *syngo* task if his *syngo* role matches the role assigned to the *syngo* task.

The Authorization Store stores the configuration in an SQL database on the *syngo*.via server.

The following picture shows the relationship between users, roles, and tasks:




(1) Clinical user – Windows or Active Directory user

(2) Clinical role – Windows or Active Directory user group

(3) *syngo* role – Authorization Store role

(4) *syngo* task – Authorization Store task

Users with more than one user role are able to define their preferred user role with the *syngo*.via client.

See (→ *syngo.via Configuration Online Help, Defining the preferred user role*).

## 7.3 Creating local user accounts for *syngo*.via

To create a local user account, perform the following steps:

1. Log on to the *syngo*.via server operating system and open the **Computer Management** console.
2. Expand the tree down through **Computer Management (Local)** / **System Tools** / **Local Users and Groups** / **Users**.

   Defined users are shown in the content area.
3. Right-click **Users** and choose **New User...** from the context menu.
4. In the **New User** dialog box, fill in the new user information.

There are no *syngo*.via-specific rules for user names or passwords. But the Windows password policy enforces complexity requirements by default. New passwords must meet the following minimum requirements:

- Passwords cannot contain the user's account name or parts of the user's full name which exceed two consecutive characters.
- For an improved system security, you should set the password length for user accounts to a minimum of 14 characters.
- Passwords must contain characters from three of the following four categories:
  - English uppercase characters (A through Z).
  - English lowercase characters (a through z).
  - Digits (0 through 9).
  - Non-alphabetic characters (for example, !, $, #, %).

You can disable or modify the complexity requirements at **Computer Configuration** > **Windows Settings** > **Security Settings** > **Account Policies** > **Password Policy** in the Group Policy Object Editor (**gpedit.msc**).

5. Select the password settings according to the policies in your location.
6. Click the **Create** button.
7. Fill in user information for an additional user or click **Close**.
8. Optionally, add users to user groups in **Computer Management (Local)** / **System Tools** / **Local Users and Groups** / **Groups**.

## 7.4 Predefined administrative user accounts

### Predefined administrative accounts

After the installation of *syngo*.via, the following predefined administrative user accounts are available:

- "RemoteAdmin"

  This is the default administrative account used for Windows administration purposes with a Remote Desktop Connection.

  If you want to use another user account for Remote Desktop Connections, this account must be a member of the "Remote Desktop Users" user group.

- "AdminUser" (or, depending on the operating system, "Administrator")

  This is the default administrative account for local logon (i.e. console session). It cannot be used to logon with Remote Desktop Connection.

The "Administrator" / "AdminUser" account is essential for the proper operation of the *syngo*.via server.

Do **not log off the administrative account** regardless of its name. If you log off, the 3D rendering performance decreases.

After installation of the *syngo*.via server, the IT Administrator must change the default passwords both of the "Administrator" / "AdminUser" and "RemoteAdmin" user accounts.

Other predefined user accounts are available for the operating system, the MSSQL database, for OPENLink, etc.

### Default service accounts

Additionally, the following service accounts are created which are used exclusively by the Customer Care Center:

- "aremote"

  This account is intended for Remote Desktop Connections to the server.

  Access is only possible if it has been explicitly granted in the **Remote Access Control**.

- "alocal"

  This is the default service account for local login.

Both accounts are activated and passwords are set automatically after each logon to the *syngo*.via Administration Portal with service key (Level 7) . It may take up to 2 minutes until these accounts are activated.

Both accounts cannot be used to log on to the *syngo*.via Administration Portal.

Do not change any settings of these accounts. You may hinder service activities.

## 7.5 Access rights and roles

Roles are a central element of the authorization system. A *syngo* role is assigned to a set of tasks. A user will therefore only be allowed to perform a task if his user account was assigned to a *syngo* role.

Only user accounts which are assigned to at least one *syngo* role are able to log on to the *syngo*.via client.

The mapping of tasks and roles is part of the *syngo*.via distribution. You only have to assign users to the preset *syngo* roles.

*syngo*.via is delivered with the following *syngo* roles:

- "Technologist"
- "Reading Physician"
- "Clinical Administrator"
- "IT-Administrator"

You can change the names of the preset *syngo* roles that are displayed in the user interface.

Users with more than one user role are able to define their preferred user role with the *syngo*.via client.

(→ *syngo.via Configuration Online Help, Defining the preferred user role*)

By default, the *syngo* roles "ITAdministrator" and "ClinicalAdministrator" are mapped to the Windows user group "Administrators", whereas the other roles are mapped to the Windows user group "Everyone".

Deleting or changing the name of a pre-defined *syngo* role or task may break the correlation with work items or prevent users from performing specific tasks.

The system will not give warnings when deleting a role or task, even if there are objects linked to that *syngo* role or task.

The following table gives an overview of the basic tasks, their default role assignment, and their relevance for authorization.

| Task Name | Default Roles | Authorized function |
|---|---|---|
| ACCESSCONTROL_REQUIRE_AUTHENTICATION | ITAdministrator | Allows enabling or disabling explicit authentication in the Configuration Panel. |

| Task Name | Default Roles | Authorized function |
|---|---|---|
| AdvancedLayoutOperations | ClinicalAdministrator, ITAdministrator | Allows to perform the following operations with public layouts in the **Layout Gallery**:<br>• switch to a layout<br>• set a layout as default<br>• sort layouts |
| COMPLETE_READ_WORKITEM | no default roles | Allows saving a workflow and sending the results to the archive. |
| CONF_WORKPLACE_SETTINGS_PERMISSION | no default roles | Allows configuring the idle time after which the workplace is locked and a lock screen is activated to protect the system against unauthorized access. |
| DATA_CORRECTION_AND_ REARRANGEMENT | ClinicalAdministrator, ITAdministrator | Allows correction and rearrangement of patient and study data. |
| DATA_PERMANENT_DELETION | ClinicalAdministrator, ITAdministrator | Allows data to be permanently deleted from the local database. |
| Expert_i_Direct_login | ReadingPhysician | Allows for connecting to another workplace with direct login (Expert-i collaboration) |
| FavoritesToolbox.Site.Edit | ClinicalAdministrator, ITAdministrator | Allows customizing the content of the **Favorite Tools** area for all users. |
| FavoritesToolbox.User.Edit | ReadingPhysician, Technologist | Allows customizing the content of **Favorite Tools** area for the current user. |
| INSTALL_SOFTWARE | ClinicalAdministrator, ITAdministrator | Allows installation of software updates and upgrades provided by Siemens Healthineers. |
| NAV_ADMINISTRATION_MODE | ClinicalAdministrator, ITAdministrator | Allows access to the following functions of the Patient Browser:<br>• display of internal DICOM objects<br>• display of TaskflowID in the result list |

| Task Name | Default Roles | Authorized function |
|---|---|---|
| NAV_ARCHIVE_STATE_CHANGE | ClinicalAdministrator, ITAdministrator | Prevents or allows marking of data for automatic archiving. |
| NAV_EXTERNAL_DICOM_ACCESS | Technologist, ReadingPhysician, ClinicalAdministrator, ITAdministrator | Allows access to DICOM Query/Retrieve from the Patient Browser. |
| NAV_PUBLIC_FILTER_MANIPULATION | ClinicalAdministrator, ITAdministrator | Allows creation, modification, and deletion of public work lists. |
| NAV_COMMUNICATION_EXPORT | Technologist, ReadingPhysician, ClinicalAdministrator, ITAdministrator | Allows export of data from local database to a remote node or to removable media. |
| NAV_COMMUNICATION_IMPORT | Technologist, ClinicalAdministrator, ITAdministrator, ReadingPhysician | Allows import of data from the file system to the local database in the Patient Browser. |
| NAV_COMMUNICATION_SEND_TO_ARCHIVE | Technologist, ClinicalAdministrator, ITAdministrator, ReadingPhysician | Allows access to directly sending data from the local database to the configured default archive in the Patient Browser. |
| NAV_WORKFLOW_ADMINISTRATION | ClinicalAdministrator, ITAdministrator | Allows access to the following workflow administration functions in the Patient Browser<br>• cancel workflow<br>• batch assign workflow |
| NAV_THIRD_PARTY_APPLICATION_ CALLUP | no default roles | Allows to start third-party applications from within the Patient Browser (by icon and context menu). |
| PERFORM_CLINICAL_ADMIN_TASK | ClinicalAdministrator | Allows configuration of clinical settings in the *syngo*.via Administration Portal (e.g. configuration of anatomy processing). |

| Task Name | Default Roles | Authorized function |
|---|---|---|
| PERFORM_IT_ADMIN_TASK | ITAdministrator | Allows configuration of IT settings in the *syngo*.via Administration Portal (e.g. license configuration). |
| PresetShortcut.Site.Edit | ClinicalAdministrator, ITAdministrator | Allows to modify the keyboard shortcuts for windowing presets for all users. |
| PresetShortcut.Site.Restore | ClinicalAdministrator, ITAdministrator | Allows to restore factory settings of the keyboard shortcuts for windowing presets for all users. |
| PresetShortcut.User.Edit | Technologist | Allows to modify the keyboard shortcuts for windowing presets for the current user. |
| PresetShortcut.User.Restore | Technologist | Allows to restore factory settings of the keyboard shortcuts for windowing presets for the current user. |
| Print.ReArrangeLayouts | ClinicalAdministrator, ITAdministrator | Allows to rearrange layouts within the **Layout Gallery** of the **Print** step. |
| PRIOR_QR_SOURCE_AND_ FILTER_MANIPULATION | ClinicalAdministrator, ITAdministrator | Allows modifying the data sources and filter conditions in the **Add Study** dialog. |
| RangesTools.SharePreset | ClinicalAdministrator, ITAdministrator | Allows to provide a range preset for all users. |
| SaveDefaultLayoutForAllUsers | ClinicalAdministrator, ITAdministrator | Allows to configure if all clinical users can create and save private layout collections. |
| SEE_HIDDEN_JOBS | ClinicalAdministrator | Allows to see jobs that are marked as hidden in the **Job View** (e.g. DICOM communication jobs). |
| TROUBLESHOOT_SYSTEM | ClinicalAdministrator, ITAdministrator | Allows access to functions for error analysis and troubleshooting (e.g. Message Viewer). |
| VrtPreset.Site.Create | ClinicalAdministrator, ITAdministrator | Allows to create VRT presets for all users. |

| Task Name | Default Roles | Authorized function |
|---|---|---|
| VrtPreset.Site.Edit | ClinicalAdministrator, ITAdministrator | Allows to modify VRT presets for all users. |
| VrtPreset.Site.Restore | ClinicalAdministrator, ITAdministrator | Allows to restore factory settings of VRT presets for all users. |
| VrtPreset.User.Create | Technologist | Allows to create VRT presets for the current user. |
| VrtPreset.User.Edit | Technologist | Allows to modify VRT presets for the current user. |
| VrtPreset.User.Restore | Technologist | Allows to restore factory settings of VRT presets for the current user. |
| WindowingPreset.Site.Create | ClinicalAdministrator, ITAdministrator | Allows to create windowing presets for all users. |
| WindowingPreset.Site.Edit | ClinicalAdministrator, ITAdministrator | Allows to modify windowing presets for all users. |
| WindowingPreset.Site.Restore | ClinicalAdministrator, ITAdministrator | Allows to restore factory settings of windowing presets for all users. |
| WindowingPreset.User.Create | Technologist | Allows to create windowing presets for the current user. |
| WindowingPreset.User.Edit | Technologist | Allows to modify windowing presets for the current user. |
| WindowingPreset.User.Restore | Technologist | Allows to restore factory settings of windowing presets for the current user. |
| WorkflowManagerSaveAddTask | ClinicalAdministrator | Allows to add and save workflow steps. |

## 7.6 Assigning users to roles in the *syngo*.via Administration Portal

A user is only allowed to perform a task if his user account was assigned to a role.

In the ***syngo*.via Administration Portal**, you can assign users/groups to *syngo* roles, or remove the assignment.

✓ You have administrator rights.

✓ You know the name of the Windows user or group to be authorized.

1 Log on to the ***syngo*.via Administration Portal**.

2 First select the **Technical Configuration** workspace and then choose **User and Role Administration** from the navigation tree.

The **User and Role Administration** window opens.

A table with the current role assignments is displayed.

3 Click the **Add** button to add a new assignment.

4 From the **Role** list, select the role that you want to assign.

5 In the **Windows User or Group** field, type the name of the Windows user or group.

6 Click the **Add** icon.

The new role assignment is displayed in the table.

To delete a role assignment, click the **Delete** icon at the end of the corresponding table row.

The changes take effect when the user logs on again.

## 7.7 Role manager

A user is only allowed to perform a task if his user account was assigned to a role.

With the **Authorization Store** application, you can assign users or groups to roles, or remove the assignment.

To confirm the changes you made, you have to close the **Authorization Store**.

Do not delete or modify the `syngoFactoryDefault` repository in the **Authorization Store**. The system needs this folder to avoid the reintroduction of previously deleted authorizations during updates or upgrades.

(→ Page 80 *Assigning users or groups to a role*)

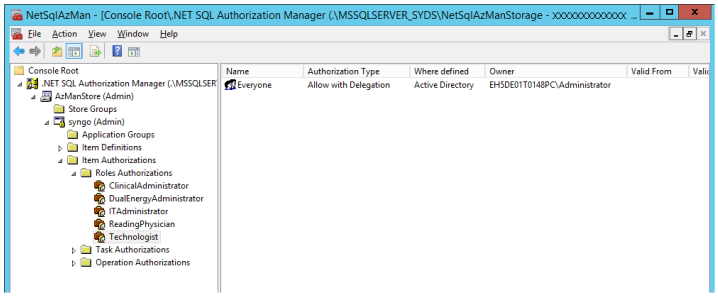
(→ Page 84 *Removing users or groups from role assignment*)

### 7.7.1 Assigning users or groups to a role

Use the **Authorization Store** to assign users or groups to a *syngo*.via role.

1. Log on to the *syngo*.via server operating system.
2. On the Windows **Start** page, search for **Authorization Store**.

   The Authorization Store opens.



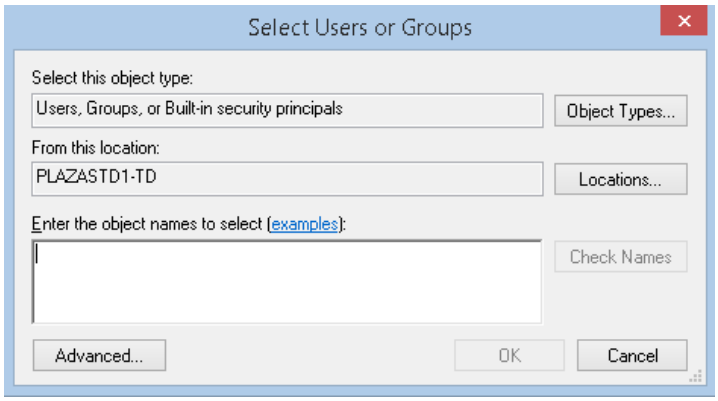

3 Expand the tree down through **.NET SQL Authorization Manager** > **AzManStore** > **syngo** > **Item Authorizations** > **Roles Authorizations** and select the role to which you would like to add the users or groups.

4 Right-click the role and choose **Manage Authorizations** from the context menu.

The **Item Authorizations** dialog box opens.

5 Click the **Add Windows Users and Groups** button.

The **Select Users or Groups** dialog box opens:



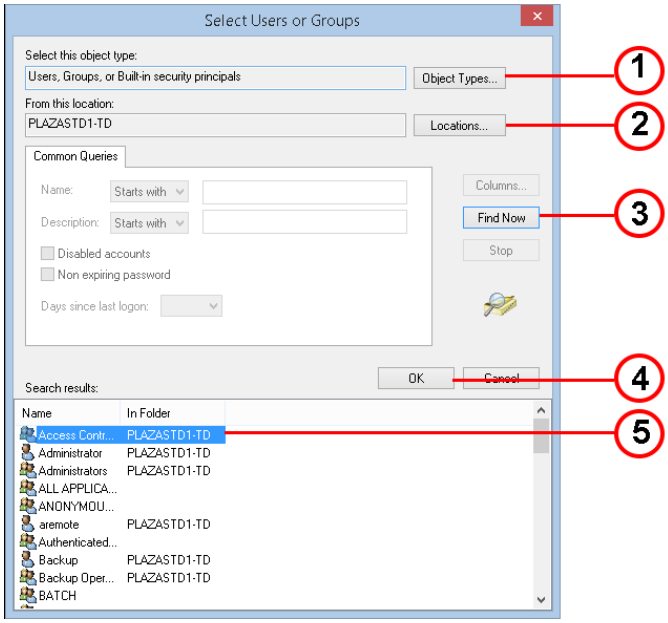
6 Enter the user or group name that you want to add by using one of the following syntax examples:

- **`DisplayName`** (example: FirstName LastName)
- **`UserName`** (example: User1)
- **`ObjectName@DomainName`** (example: User1@Domain1)
- **`DomainName\ObjectName`** (example: Domain1\User1)

You can add multiple objects by separating each name with a semicolon.

– or –

Click the **Advanced...** button and search users or groups within the **Select Users or Groups** dialog box:




(1) Open the object types definition dialog box

(2) Open the search location dialog box (computer, domain)

(3) Start the search

(4) Take over selected objects

(5) Select desired objects from the search result

Active Directory user accounts and groups can only be assigned if the domain of the *syngo*.via server trusts the domain of the Active Directory. Only global groups can be used.

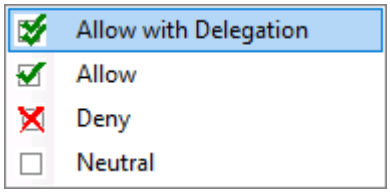
You need a domain user account to be able to select another location and to add domain user accounts from there.

7 Click the **Check Names** button and **Ok**.

The user or group is listed in the content area and is added to the role assignment.

After domain integration, the Authorization Store displays **Administrators([domain]\Administrators)** instead of **Administrators([hostname]\Administrators)**.

8 Right-click the new assignment and set the **Authorization Type** to "Allow with Delegation":




9 To confirm your changes, click **Ok**, and close the **Authorization Store**.

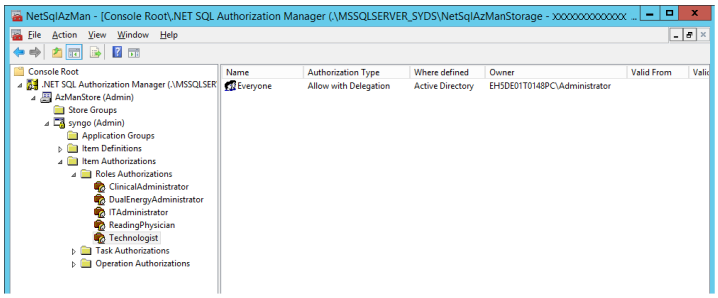
### 7.7.2 Removing users or groups from role assignment

Use the **Authorization Store** to remove users or groups from role assignment.

1 Log on to the *syngo*.via server operating system.

2 On the Windows **Start** page, search for **Authorization Store**.

The Authorization Store opens.




3 Expand the tree down through **.NET SQL Authorization Manager** > **AzManStore** > **syngo** > **Item Authorizations** > **Roles Authorizations** and select the role from which you would like to remove a user or a group.

4 Press **Del**.

The user or group is removed from the role assignment.

Always remove a user account from its role assignment before deleting it.

5 To confirm your changes, close the **Authorization Store**.

## 7.8 Access control to *syngo*.via Administration Portal based on service levels

Access to the ***syngo*.via Administration Portal** is restricted and controlled by a dedicated concept for managing service users.

There is a predefined set of service roles.

Each service role is associated with a service access level and a set of service functions that can be accessed.

Service users log on to the ***syngo*.via Administration Portal** with a service key. This key is translated by the system into the service access level and role.

Administration, authentication, and authorization for service users are done by a Siemens Healthineers proprietary solution.

A clinical user may also act as service user. However, he does not need to enter a service key for login. Instead, each *syngo*.via role is assigned to a certain service access level that contains the set of service functions offered to the user.

In *syngo*.via, there are the following service access levels:

- **FREE** (service level 1)

  General service functions based on Windows user roles, for example:

  - Everybody: site information, system status, access mode
  - Clinical administrator: clinical configuration, data preprocessing, layout management
  - IT administrator: technical configuration (for example, site information, or system status)

- **BASIC** (service level 3)

  Device Guard configuration (*syngo*.via only)

- **EXPERT** (service level 5)

  Extended service functions for clinical administrators and IT administrators (for example, file transfer, auto-report, or patient identification)

- **SIEMENS** (service level 7)

  Service functions for service users only

# 8 *syngo*.via server administration

Administrative tasks on operating system level are usually performed using a Remote Desktop Connection.

See (➔ Page 87 *Logging on to syngo.via server operating system*).

Most Windows administration tasks are performed using the Microsoft Management Console, for example, the **Server Manager** (ServerManager.msc) provides access to the **Diagnostics** and **Server Roles** management console.

Do not modify the regional and language settings of the server. The UI language of the server must remain English as set by default during installation.

(➔ Page 87 *Logging on to syngo.via server operating system*)

(➔ Page 141 *Setup of syngo.via server after installation*)

(➔ Page 142 *Update of syngo.via server*)

(➔ Page 146 *Backup and restore of the syngo.via server*)

(➔ Page 158 *Joining the syngo.via server to an Active Directory*)

(➔ Page 172 *Security settings for clients*)

(➔ Page 163 *Audit trail*)

(➔ Page 168 *Uninstallation of the syngo.via server*)

## 8.1 Logging on to *syngo*.via server operating system

For certain administrative purposes, it is necessary to log on to the *syngo*.via server on operating system level.

Certain special considerations are necessary for logging on as a service user.

Every time you log on, a logbook text file opens in a text editor allowing you to document your tasks.

Once logged on, never log off!

The *syngo*.via application server must have access to the graphics card to allow hardware 3D rendering. Once a local user has been logged on, hardware access to the graphics card is only possible for this user session (console level).

If the first user session of the *syngo*.via server operating system is taken over by another user or if the administrative account is logged off, 3D rendering switches to slow GDI rendering. See *Server-side 3D rendering performance decreased* in the *syngo*.via Administration Online Help.

(→ Page 88 *Using a Remote Desktop Connection*)

(→ Page 89 *Logging off from a Remote Desktop session*)

(→ Page 89 *Logging on locally to the server*)

### 8.1.1 Using a Remote Desktop Connection

Administrative tasks are usually performed using a Remote Desktop Connection (RDC).

1. Log on to a Windows workstation.
2. Open the RDC client by searching **Remote Desktop Connection** on the Windows **Start** page or by running the `mstsc` command.

   The **Remote Desktop Connection** dialog box opens.
3. Type the IP address or the computer name of the *syngo*.via server.
4. Click **Connect**.

   The **Windows Security** login screen opens.

5 Log on with the user credentials of a user who is member of the "Remote Desktop Users" user group (default user: "RemoteAdmin").

6 To transfer files between your local PC and the remote server, use a common share on both computers.

If all sessions are already occupied, you will be asked which user you want to log off. Never choose a user who is logged in on console level. Logging off a console user leads to a restart of the *syngo*.via application server. In this case, see (→ *Starting a Remote Desktop Connection fails in the syngo.via Administration Online Help*).

### 8.1.2 Logging off from a Remote Desktop session

After completing your administrative tasks, you log off from the **Remote Desktop Connection**.

◆ Log off with the Windows **Start** page.

The **Remote Desktop Connection** is closed.

Do not use the **X** icon from the terminal session bar to close the session, as the session will stay active. The number of user sessions which can be open at the same time is restricted.

If a Remote Desktop Connection was not closed correctly, Windows could refuse to open a new connection for another user account. In this case, see (→ *Starting a Remote Desktop Connection fails in the syngo.via Administration Online Help*).

### 8.1.3 Logging on locally to the server

When working at the server console, use a local administrative account (e.g. "AdminUser", "Administrator", or "Admin", depending on the operating system).

1 Enter the user credentials for the administrative account and click **OK**.

2 Perform the necessary administrative tasks.

3 Press **Windows Logo Key** + **L** to lock the computer.

Do not log off! Otherwise, the *syngo*.via application server will be restarted and lose access to hardware 3D rendering. See *Server-side 3D rendering performance decreased* in the *syngo*.via Administration Online Help.

## 8.2 Stopping / restarting the *syngo*.via host

**CAUTION**

Unexpected shutdown of the server for users.

**Data loss or data inconsistency possible.**

- Before any scheduled shutdown, inform all users (for example by e-mail or phone) about the scheduled downtime and give them enough time to finish their work and close workplace applications before the server is shut down.

### 8.2.1 Starting up the *syngo*.via server

- Power on the *syngo*.via server.

  The host boots and *syngo*.via applications and services automatically start.

### 8.2.2 Shutting down the *syngo*.via server

1 Inform all *syngo*.via client users about the upcoming shutdown.

2 Stop the *syngo*.via application server.

  See (→ Page 91 *Stopping / restarting the syngo.via application server*).

3 Shut down the operating system of the *syngo*.via host.

  The server powers off automatically.

### 8.2.3 Rebooting

In some cases it is necessary to reboot the system, for example, to clean memory from "dead" processes.

1 Inform all *syngo*.via client users about the upcoming reboot.

2 Stop the *syngo*.via application server.

See (→ Page 91 *Stopping / restarting the syngo.via application server*).

3 Restart the operating system of the *syngo*.via server.

The *syngo*.via server is down for several minutes.

After performing the reboot, certain server processes can have the state **Unknown** in Status Monitoring. The status of the server processes will be available after several minutes.

## 8.3 Stopping / restarting the *syngo*.via application server

The *syngo*.via application server starts automatically when you boot the system. The *syngo*.via application server provides 2D, 3D, and hybrid viewing, processing, and reading. It is one of the central functions.

**CAUTION**

Unexpected shutdown of the server for users.

**Data loss or data inconsistency possible.**

- Before any scheduled shutdown, inform all users (for example by e-mail or phone) about the scheduled downtime and give them enough time to finish their work and close workplace applications before the server is shut down.

Check the status bar of the *syngo*.via Administration Portal for users still logged on to the system. See *Icons of the syngo.via Administration Portal* in the Online Help.

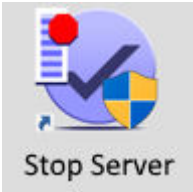
When the application server needs to be restarted, currently processed workflows are ended and rescheduled (State: *Ready*). Modifications for these workflows can be lost.

In case of a forced shutdown, check the **Message Viewer** after restarting the application server. The affected workflows, users, and patient data is listed there. Please check for messages with the name WORKFLOW_RESTARTED and Severity "warning". Inform the affected users about the loss of their modifications.

### 8.3.1 Stopping the *syngo*.via application server

In case of maintenance you must stop the *syngo*.via application server.

Killing any *syngo* process with the Windows **Task Manager** risks data loss!




1 Log on to the *syngo*.via server operating system.

2 On the server desktop, double-click the **syngo.via - Stop Server** icon:

   In case of active workflows, you are prompted to either cancel or to perform a forced shutdown.

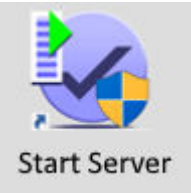
   A status window reports the shutdown sequence of the *syngo*.via application server.

3 Close the status window when the shutdown sequence is completed.

The *syngo*.via application server is stopped.

At the server stop, an open *syngo*.via Administration Portal session is closed and all *syngo* processes are stopped.

### 8.3.2 Starting the *syngo*.via application server




1 Log on to the *syngo*.via server operating system.

2 On the server desktop, double-click the **syngo.via - Start Server** icon to start the *syngo*.via application server.

3 Check started processes in **Status Monitoring**.

### 8.3.3 Restarting the *syngo*.via application server

1 Log on to the *syngo*.via server operating system.

2 On the server desktop, double-click the **syngo.via - Restart Server** icon to restart the *syngo*.via application server:

The restart functionality is deactivated while a server update is being performed.

Every day at 05:00 a.m., a Windows scheduled task restarts the *syngo*.via application server.

Every Monday at 04:59 a.m., a scheduled task restarts the operating system.

If there are active workflows, jobs or connected clients, the system will wait for 1 minute and try again. After 60 failed attempts, the restart is skipped and an error log is written. If you want to change the start time, the waiting interval, or the number of attempts, contact the Customer Care Center.

## 8.4 About *syngo*.via configuration

Most administrative tasks are performed using the *syngo*.via Administration Portal, which can be accessed through Microsoft Internet Explorer from a Windows workstation.

The following list provides an overview of the configuration windows of the *syngo*.via Administration Portal. For detailed descriptions, see (→ *syngo.via Configuration Online Help*).

- Licensing

  The license configuration comprises the following tasks:

  – Importing new license files for single server or for multi-server (cluster license)

  – Reservation of licenses

  – Inspecting the status of available licenses

  – Inspecting the usage of floating licenses

- Site Information

  The **Site Information** window provides service-related information that may be required for support cases.

- Automatic Data Deletion

  *syngo*.via provides an automatic data deletion of archived and temporary data by the system.

- Job Settings

  In the **Automatic Deletion of Jobs** section, you can configure when exactly a successfully completed job or all jobs are deleted from the **Job View**.

  In the **Automatic Retry for Network Jobs** section, you can configure the number and delay values of retries for network jobs.

- Software Update

  Software packages are retrieved from SRS with the **Software Update** of the *syngo*.via Administration Portal. Downloaded packages can be installed separately. Updates which have an impact on the *syngo*.via client, are automatically distributed to the clients.

- DICOM configuration

  With the DICOM configuration, you will specify the parameters for the DICOM communication interfaces for the DICOM nodes in the vicinity of your system. The configuration includes some general settings and the list of supported DICOM services for each node.

  – In the first step, you will configure the *syngo*.via server itself as a **Local DICOM Node**. The local DICOM properties preset the system behavior towards the configured DICOM partners.

  – Afterwards, each DICOM node is independently configured with the **Remote DICOM Nodes** configuration.

- Archive configuration

  For archiving, you can define the following settings:

  – Archives available in the *syngo*.via environment

  – Default archive

  – Specific archiving rules

- Workflow configuration

  The workflow management is a central service of *syngo*.via. It comprises the following topics:

  – Handling of DICOM modality worklists: DICOM modality worklists can be retrieved from RIS.

  – Workflow assignment: Incoming orders or images are assigned to workflow templates according to specified rules.

  – Automatic retrieval of prior data: Data of previous examinations can be retrieved from archives according to specified rules.

- Autorouting

  On the **Autorouting Rules** window, you can set up rules for automatic transfer of data to specific DICOM nodes and for archiving.

- File transfer (SL7 only)

  File transfer allows exchanging files between the *syngo*.via server and the Customer Care Center:

  – Transfer files from Smart Remote Services back-end to the local server, for example, specific software updates for troubleshooting.

  – Transfer files from the local server to Smart Remote Services, for example, auto reports or SaveLogs for troubleshooting.

  – Investigate transfer jobs.

## 8.5 Configuration of DICOM nodes

With the DICOM configuration in the ***syngo*.via Administration Portal**, you can connect several DICOM nodes in a network for data exchange, and you can specify the parameters for the DICOM communication interfaces for the DICOM nodes in the vicinity of your system.

DICOM (Digital Imaging and Communications in Medicine) is a standard for the communication between medical imaging applications. It allows you to exchange data between different systems such as PACS, scanners, and workstations.

The configuration includes some general settings and the list of supported DICOM services for each node.

DICOM configuration in the ***syngo*.via Administration Portal** can only be performed remotely if remote access rights were granted under **Technical Configuration** > **Remote Service** > **Access Control**.

DICOM configuration is done at the server and on each other system (for example, a PACS) the server communicates with. Local DICOM node configuration and remote DICOM node configuration at communicating systems have to be aligned.

The DICOM configuration is done in following main steps:

- **Local DICOM Node** configuration: the server itself is configured as a DICOM node. (→ Page 98 *Configuration of the local DICOM node*)

  The local DICOM properties preset the system behavior toward the configured DICOM partners.

- **Remote DICOM Nodes** configuration: remote DICOM participants are integrated into the DICOM network. This must be done for each DICOM node independently. (→ Page 106 *Configuration of remote DICOM nodes*)

- **DICOM Printer Configuration**: DICOM printers are configured in both the **Remote DICOM Nodes** and the **Local DICOM Node** settings. (→ *Configuring DICOM printers*)

- DICOM modality worklist (DMWL) configuration: the DMWL is configured in both the **Remote DICOM Nodes** and the **Local DICOM Node** settings. (→ *Configuring DICOM modality worklist query*)

### 8.5.1 Configuration of the local DICOM node

The local DICOM node represents the DICOM configuration of your system. It configures the source and the destination addresses of the service classes provided by your system.

*syngo*.via acts as Service Class Provider (SCP) and as Service Class User (SCU) for several service types, for example, Storage and Storage Commitment.

The **Local DICOM Node** window can be accessed from the ***syngo*.via Administration Portal** by first selecting the **Technical Configuration** workspace and then choosing **DICOM Nodes** > **Local DICOM Node** from the navigation tree.

For the local DICOM node configuration, you will use the following configuration items:

- **Local DICOM Node** window

  Interface settings of the server are configured with this window. (➔ Page 99 *Configuration of interface settings for the local DICOM node*)

- **General Settings** dialog box

  This dialog box allows the specification of transfer, connection, and data settings. These settings are valid for DICOM communication from and to the server. (➔ Page 101 *Configuration of general settings for the local DICOM node*)

- **DICOM Modality Worklist Query** dialog box

  This dialog box configures the queries which are performed to retrieve procedure information from a RIS (DMWL). (➔ *Configuring DICOM Modality Worklist Query*)

- **Printing Composing Parameters** dialog box

  This dialog box configures the LUT depending on the image type. Assigning LUTs to image types for printing in the *syngo*.via Configuration Online Help

You may also be interested in the following:

(➔ Page 96 *Configuration of DICOM nodes*)

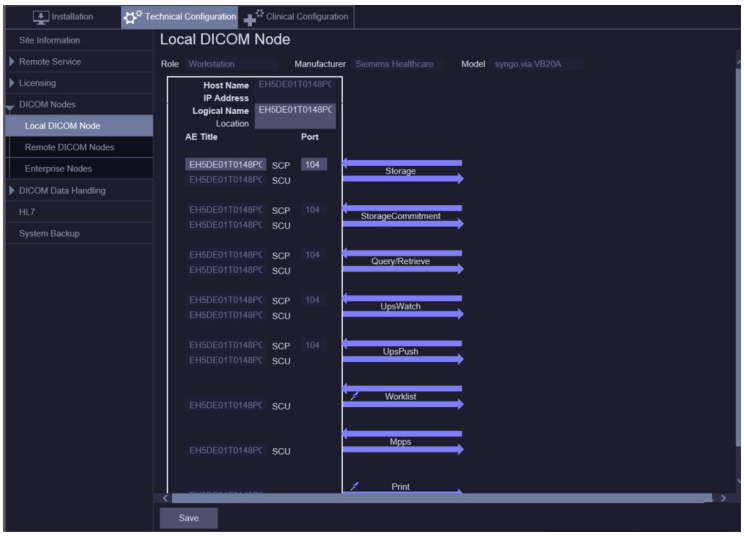
### 8.5.2 Configuration of interface settings for the local DICOM node

The local DICOM node configuration defines how *syngo*.via communicates with other nodes of your DICOM network.

Most DICOM services have to be configured at both nodes concerned. If you change the local configuration, you have to adapt the DICOM settings at remote nodes as well.

The following image displays the main window of the Local DICOM Node configuration:

You can open the **Local DICOM Node** interface settings window by first selecting the **Technical Configuration** workspace and then choosing **DICOM Nodes** > **Local DICOM Node** from the navigation tree of the ***syngo*.via Administration Portal**:




- **Role**, **Manufacturer** and **Model**

  These read-only parameters are part of the system delivery.

- **Host Name** and **IP Address**

  These read-only parameters are set in the network settings of the server.

- **Logical Name** (mandatory field)

  This name is used to display the system in any user interface. It is used, for example, in the list of possible targets for the **Export Data** or **Send to Archive** functionality of the client.

Obey the following restrictions for the logical name:

- Only letters, numbers, dashes, underscores, and periods are allowed. [a..z, A..Z, 0..9, -, _, .]
- Other characters and spaces are not allowed.
- Dash, underscore, and period are neither allowed as first nor as last character of the logical name.

- **Location**

  This parameter is the location where the system resides. You can enter free text. The location is displayed only in this configuration window.

- **Service List**

  *syngo*.via provides (SCP) and uses (SCU) several services. The arrows shown in the configuration window display the direction of the service messages configured by the corresponding line.

  - **AE-Title**

    The Application Entity Title (AE-Title) is preset to the host name in capital letters. The same AE-Title is used for all services.

    Obey the following restrictions for entering the AE-Title:

    - A maximum of 16 alphanumeric characters, hyphens, and underscores is allowed.
    - Do not use white spaces or double quotes.
    - Using capital letters is recommended.

  - **Port**

    The port number is preset to 104. The same port number is used for all services.

After changing the port number or the AE-Title, the *syngo*.via application server has to be restarted to activate the changes.

Leaving the display without saving *discards* all changes (without notification).

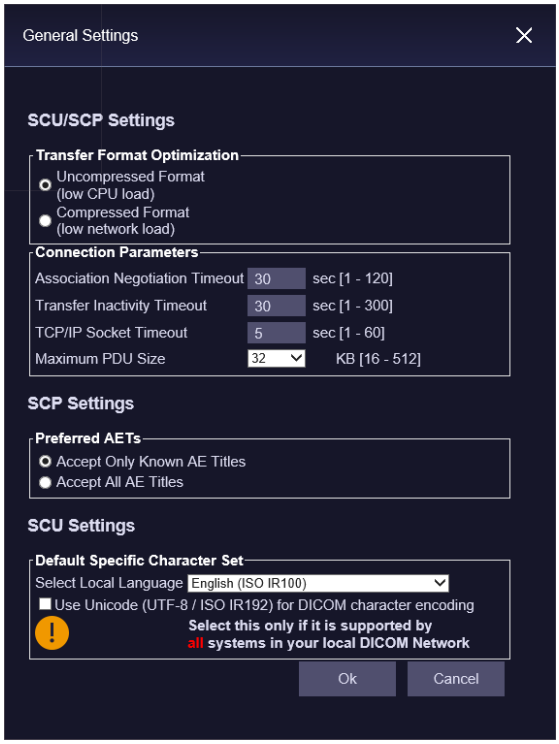
You may also be interested in the following:

(➔ Page 98 *Configuration of the local DICOM node*)

(➔ Page 96 *Configuration of DICOM nodes*)

### 8.5.3 Configuration of general settings for the local DICOM node

The **General Settings** dialog box can be accessed by clicking the **General Settings** button at the bottom of the **Local DICOM Node** window.




There are three configuration sets which are valid for all service types:

- **SCU/SCP Settings**
- **SCP Settings**
- **SCU Settings**

## SCU/SCP Settings

These settings are valid when *syngo*.via serves as Service Class User (SCU) or as Service Class Provider (SCP).

- **Transfer Format Optimizations**

  This setting influences the Association Negotiation at the beginning of each DICOM session. If the communication partner supports the preferred setting, it will be used.

  The **Compressed Format** option should be used for networks with low bandwidth only.

- **Connection Parameter**

  These parameters define general DICOM negotiation and connection settings.

  - **Association Negotiation Timeout**

    A DICOM communication starts with the Association Negotiation. The initiator (SCU) sends a list of the supported objects and transfer syntaxes to the remote system (SCP). The SCP responds with the status (accept or reject) of the objects and selects one of the proposed transfer syntaxes. If this process is not completed within the time configured with the **Association Negotiation Timeout** field, it will restart.

    The default value is “30” seconds.

  - **Transfer Inactivity Timeout**

    The timeout value configured with the **Inactivity Timeout** field is valid for message transfer and needs to be increased when performance problems in the network occur. If a timeout occurs, the whole communication is aborted.

    The default value is “30” seconds.

  - **TCP/IP Socket Timeout**

    This value defines the maximum waiting time for network connections.

    The default value is “5” seconds.

  - **Maximum PDU Size**

    If large objects are transferred between the DICOM nodes, the data will be split up into packages. The PDU (Process Data Units) size defines the size of those packages. If it is set to a small number, the traffic will increase. But a large PDU size is also used for small objects, which can decrease performance.

    For small DICOM objects like CT / MR images with 1 MB to 2 MB, the default PDU size is sufficient. But for huge DICOM objects like CR or MG images, and AX or US multi-frames, the maximal PDU size increases the transfer performance.

It is only possible to set one PDU size for the system. You must select the appropriate size depending on the amount of small or large DICOM images.

The **Maximum PDU Size** setting influences the Association Negotiation at the beginning of each DICOM session. The lowest common factor supported by both communication partners will be used.

The default value is “32” kByte.

### SCP Settings

These settings are only valid when *syngo*.via serves as Service Class Provider (SCP).

- **Preferred AETs**

  DICOM nodes identify each other using the Application Entity Title (AE-Title or AET). If the **Accept Only Known AE Titles** option is selected, *syngo*.via will only communicate with DICOM nodes which are configured in the **Remote DICOM Nodes** window. The **Accept All AE Titles** option eliminates this restriction.

  The default value is “Accept Only Known AE Titles”.

### SCU Settings

These settings are only valid when *syngo*.via acts as Service Class User (SCU).

- **Default Specific Character Set**

  In this section, you can define the language and encoding settings used for messages sent from the server to other SCPs. The setting should comply with the standard character set of your medical IT environment.

  When the Unicode character encoding check box is selected, the list becomes inactive.

Unicode encoding should only be activated if all systems in your local DICOM network support Unicode encoding. Otherwise, data corruption can occur.

You may also be interested in the following:

(→ Page 98 *Configuration of the local DICOM node*)

(→ Page 96 *Configuration of DICOM nodes*)

### 8.5.4 Configuration of remote DICOM nodes

Remote DICOM nodes are devices which communicate over the DICOM standard with *syngo*.via.

The **Remote DICOM Nodes** configuration in the ***syngo*.via Administration Portal** defines how the system communicates with other DICOM nodes. For each node, the available services and the interfaces are defined. This procedure does not modify the remote nodes themselves.

If applicable, a corresponding configuration must be performed at the remote node.

Configurations at modalities are performed by service engineers.

**CAUTION**

The connection between a remote node and *syngo*.via is temporarily unavailable, due to a system shutdown/crash or network problems.

**Messages from a remote node are not applied to data (no patient update) or data availability is not notified to remote node.**

- Only connect to remote nodes that can buffer and return messages.
- If patient (personal) data has been corrected but not propagated properly to remote nodes receiving images from *syngo*.via, although the respective study is available, resend HL7 messages from *syngo*.via to the remote node, so both systems are in sync again.
- Configure an appropriate amount of time between retries for HL7 messages on both communicating systems to ensure high probability of HL7 message application.
- Contact your Customer Service Engineer for adapting the configuration of the remote node interfaces.

In some cases, it is necessary to assign more than one AE-Title to the same DICOM service. This assignment can be done by configuring two remote DICOM nodes with different logical names, but with the same DICOM services on the same host.

The **Remote DICOM Nodes** configuration consists of the following windows:

- **Remote DICOM nodes list** window

  This window gives an overview of all configured DICOM nodes.

  (➔ Page 109 *Overview of configuration settings for remote DICOM nodes*)

- **Add New Remote DICOM Node** window

  When adding a new DICOM node, this window leads you through the template selection.

  (→ Page 112 *Adding a new remote DICOM node for configuration*)

  You can also add new remote DICOM nodes with the automatic SmartConnect configuration.

- **SmartConnect** window

  To enable other Siemens Healthineers systems to automatically connect to your system and retrieve the necessary information for remote DICOM node configuration.

  (→ *Activating SmartConnect for DICOM configuration in the Configuration Online Help*)

- **DICOM Remote Node Editor** window

  This window allows you to configure and edit the remote DICOM node based on the selected template.

  (→ Page 111 *Configuring remote DICOM nodes*)

- **Storage Settings** dialog box

  For DICOM nodes with Storage service, you can add specific settings for storage receive and storage send services.

  (→ *Configuring Storage Settings in the Configuration Online Help* )

- **Storage Commitment Settings** dialog box

  For DICOM nodes with Storage Commitment service, you can add specific settings for Storage Commitment send services.

  (→ *Configuring Storage Commitment settings in the Configuration Online Help*)

You may also be interested in the following:

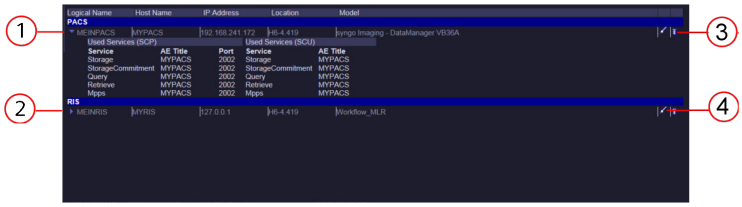
(→ Page 96 *Configuration of DICOM nodes*)

## Overview of configuration settings for remote DICOM nodes

DICOM nodes need to be mutually configured. For example, if a modality is configured in *syngo*.via as a Remote DICOM node, the modality needs to configure *syngo*.via as a Remote DICOM node, too.

The first window of the **Remote DICOM Nodes** configuration is an overview of the configured remote DICOM nodes. Each entry in each list shows the data of one node. The nodes are grouped by the assigned template (Modality, RIS, PACS, Workstation, Printer, and Customized).

The overview window of the configured remote DICOM nodes consists of the following areas:




(1) Remote DICOM node, expanded view

(2) Remote DICOM node, collapsed view

(3) **Delete DICOM node** icon

(4) **Edit DICOM node** icon

In the collapsed view, the following data is shown for each node:

- **Logical Name**
- **Host Name**
- **IP Address**
- **Location**
- **Model**

Detailed information about the used and provided services can be received from the expanded view.

Configured services can be edited by clicking the **Edit DICOM node** icon:

Configured nodes can be deleted by clicking the **Delete DICOM node** icon:

(→ Page 113 *Deleting a configured remote DICOM node*)

You may also be interested in the following:

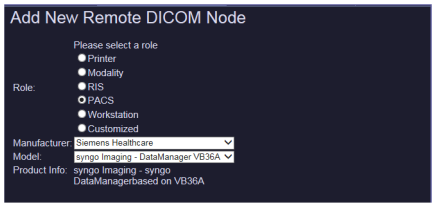
(→ Page 106 *Configuration of remote DICOM nodes*)

(→ Page 96 *Configuration of DICOM nodes*)

(→ Page 99 *Configuration of interface settings for the local DICOM node*)

## DICOM configuration templates

Each DICOM device has a specific configuration. They vary among the device roles (for example, a printer or a PACS) as well as among manufacturers and products. To support you with the configuration, the **Remote DICOM Nodes** configuration provides a set of templates. You may also create “Customized” configurations, which are not based on a template.




Templates minimize the configuration effort by restricting the customizable settings. Settings which are either not used by the DICOM node, or should not use varying port numbers or AE-Titles, are inactive. Default values are preset.

The template selection window appears only when a new remote DICOM node is added to the configuration.

- **Role**

  This list provides names of roles which are available for the different templates.

- **Manufacturer**

  This list provides names of manufacturers which produce products for the selected **Role**. If you have a product of an unlisted vendor, select “Other”.

- **Model**

  This list provides names of models which are produced by the selected **Manufacturer** and are of the selected **Role**. If you have an unlisted model, select “Other”.

- **Product Info**

  Additional information about the selected **Model** is given here.

Clicking the **Next** button leads to the main configuration window. The **Cancel** button stops the configuration without saving.

### Configuring remote DICOM nodes

The **Remote DICOM Node List** window gives an overview of all configured remote DICOM nodes within your network.

Configuring remote DICOM nodes includes several options:

- Reviewing the settings of remote DICOM nodes (➔ Page 109 *Overview of configuration settings for remote DICOM nodes*)
- Adding new remote DICOM nodes (➔ Page 112 *Adding a new remote DICOM node for configuration*)
- Modifying the configuration of remote DICOM nodes
- Deleting configured remote DICOM nodes (➔ Page 113 *Deleting a configured remote DICOM node*)

You may also be interested in the following:

(➔ Page 106 *Configuration of remote DICOM nodes*)

(➔ Page 96 *Configuration of DICOM nodes*)

### Accessing the overview of configured remote DICOM nodes

1 Log on to the ***syngo*.via Administration Portal**.

2 First select the **Technical Configuration** workspace and then choose **DICOM Nodes** > **Remote DICOM Nodes** from the navigation tree.

The **DICOM Remote Node List** window opens.

You may also be interested in the following:

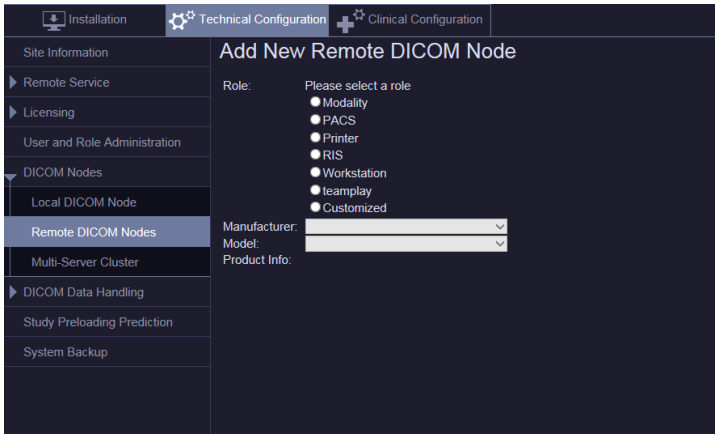
(→ Page 109 *Overview of configuration settings for remote DICOM nodes*)

### Adding a new remote DICOM node for configuration

In the **Add New Remote DICOM Node** window of the ***syngo*.via Administration Portal**, you can add new remote DICOM nodes to your DICOM network.

1 In the **DICOM Remote Node List** window, click the **Add New** button at the bottom to configure a new node.

The template selection window opens:




2 From the **Role** list, select the role (or profile) of the DICOM node.

3 From the **Manufacturer** list, select the manufacturer of the DICOM node. If the manufacturer is not listed, select “Other”.

4 From the **Model** list, select the model name of the DICOM node. If the model is not listed, select “Other”.

5 Click **Next** at the bottom of the **Add New Remote DICOM Node** window.

Your selection is used to load the appropriate configuration template.

If you could not find a template which fits to the characteristics of the DICOM node, choose “Customized” from the **Role** list.

6 Configure the node.

You may also be interested in the following:

(➔ Page 113 *Configuration of interface settings for remote DICOM nodes*)

(➔ Page 106 *Configuration of remote DICOM nodes*)

(➔ Page 96 *Configuration of DICOM nodes*)

## Deleting a configured remote DICOM node

You can delete remote DICOM nodes from your DICOM network configuration.

1 In the **DICOM Remote Node List** overview window, click the **Delete DICOM node** icon of the desired system/device.

A warning message about the consequences of deletion appears.

2 Make sure that the node is not used as a default archive, auto routing target, default printer or active RIS.

3 Confirm the warning message by clicking **Yes**.

The node is deleted.

You may also be interested in the following:

(➔ Page 106 *Configuration of remote DICOM nodes*)

(➔ Page 96 *Configuration of DICOM nodes*)

## Configuration of interface settings for remote DICOM nodes

In the **DICOM Remote Node Editor**, you define the remote interface of your system which either receives or calls services.

The interface settings window of the **DICOM Remote Node Editor** is similar to the **Local DICOM Node** window.

If you click the **Edit DICOM node** button or the **Add New** button in the **DICOM Remote Node List** overview (**Technical Configuration** > **DICOM Nodes** > **Remote DICOM Nodes**), the interface settings window opens:

(1) *syngo*.via own interface settings (read-only)

(2) DICOM service list

The arrows show the direction of the service messages, configured by the corresponding row.

(3) **Further Settings** icon (only available for certain services)

(4) Remote DICOM node interface settings

(5) **Unlock** button

In the top row, each remote DICOM node is identified by the following:

- **Role**

  This parameter is preset according to the selected template.

- **Manufacturer** and **Model**

  These parameters are preset according to the selected template. You can modify them, for example, by adding a version name.

- **Host Name** (mandatory if IP address is not provided)

  If you use a name resolution service (DNS or WINS), you can enter the host name of the DICOM node.

Remember host names must comply with the RFC 952 pattern. Underscores are not allowed.

Clicking the **nslookup** button tests whether the host name is known at the WINS or DNS. If the name is found, the corresponding IP address is prompted in a pop-up window. You can copy and paste the IP address into the **IP Address** field.

Either host name or IP address must be provided. If both are provided, the IP address has priority.

If only the **Host Name** is given, a DNS name server lookup is performed for each connection which requires an IP address. If this command returns more than one IP address, the first one reported by the Operating System is used.

- **IP Address**

  This field represents the IP address of the DICOM node. This parameter is mandatory, if you do not use a name resolution service (DNS or WINS).

  Clicking the **Test (ping)** button sends a PING command to the corresponding IP address. A TCP/IP ping can fail for the following reasons:

  – The remote host is turned off, not in the same network (or subnet), or the gateway is not configured.

  – The TCP/IP address is configured incorrectly.

  – Any networking device (router, switch, bridge,...) denies access to the other network.

- **Logical Name** (mandatory field)

  This name is used to display the system in any user interface. It is used, for example, in the list of possible targets for the **Export Data** or **Send to Archive** functionality of the client.

Obey the following restrictions for the logical name:

- Only letters, numbers, dashes, underscores, and periods are allowed. [a..z, A..Z, 0..9, -, _, .]
- Other characters and spaces are not allowed.
- Dash, underscore, and period are neither allowed as first nor as last character of the logical name.
- The logical name can have up to 64 characters.

- **Location**

  This field represents the system's location. You can enter free text. The location is displayed only in this configuration window.

- **Service List**

  Each DICOM node provides (SCP) and uses (SCU) in their specific set of services. The arrows shown in the dialog box display the direction of the service messages configured by the corresponding row. Only those messages which can be exchanged between *syngo*.via and the currently configured node are shown.

  – **AE-Title** (mandatory field)

    The Application Entity Title (AE-Title) for the DICOM node.

    For some templates, a **Default AET** button is available, which sets predefined AE-Titles for all services.

    Obey the following restrictions for entering the AE-Title:

    – A maximum of 16 alphanumeric characters and hyphens is allowed.

    – Do not use white spaces, underscores, or double quotes.

    – Using capital letters is recommended.

  – **Port** (mandatory field)

    The port number of the DICOM node.

    The port number is the TCP/IP port where the DICOM process (defined by the AE-Title) at the DICOM node is listening.

Your local settings may vary from the default values. Check the configuration of the corresponding DICOM node or ask the customer service of the device vendor.

- Clicking the **Test (C-Echo)** button sends a so-called C-Echo command to configured DICOM services of the remote node.

  The C-Echo command is a DICOM service used for test purposes. It is transferred to the selected DICOM service which is defined by the IP address, the port number, and the AE-Title. If the Test (C-Echo) is performed successfully, the configured DICOM services are verified.

  A C-Echo may fail for the following reasons:

  – The server is unknown to the remote DICOM node.

  – TCP/IP address, AE-Title, or port number is not configured correctly.

  – The DICOM process at the remote host is not running.

  – Certain products check the IP address or AE-Title of the sending system. If the AE-Title or IP address of the local system is not entered correctly there, DICOM verification fails.

  – The remote host does not support DICOM verification as a Service Class Provider (SCP).

Initially, the template disables entry fields if several services use the same port or **AE-Title**. Presets for both entry fields may be available. Click the **Unlock** button in order to edit the disabled values of the remote node. When changing the values of the predefined template, make sure that the configuration is valid.

Leaving the window without saving *discards* all changes (without notification).

Some services provide further configuration options. They can be accessed by clicking the **Further Settings** icon:

Before the detailed options can be accessed, the interface settings must be saved once.

- If the device provides a **Storage** service, there is a **Further Settings** icon next to the **Storage Settings** service. (➔ *Configuring Storage Settings in the syngo.via Configuration Online Help*)
- If the device provides a **Storage Commitment** service, there is a **Further Settings** icon next to the **Storage Commitment Settings** service. (➔ *Configuring Storage Commitment settings in the syngo.via Configuration Online Help*)

You may also be interested in the following:

(➔ Page 106 *Configuration of remote DICOM nodes*)

(➔ Page 96 *Configuration of DICOM nodes*)

### 8.5.5 About unique patient identification

Your system uses a configured set of DICOM attributes to decide which datasets belong to which patient.

The **Patient Identification** window of the ***syngo*.via Administration Portal** is used to configure this set of DICOM attributes.

(➔ Page 120 *Selecting DICOM attributes for unique patient identification*)

**CAUTION**

Patients may be incorrectly merged if only Patient ID is used to identify patients.

**Incorrect diagnosis basis.**

- The types of identification used by the system when automatically merging patients can be configured. It is recommended that at least two of the following forms of identification be used: Patient Name, Patient ID (required), and Date of Birth. If the default configuration is changed, test the new configuration to avoid unintended consequences.

**CAUTION**

Patient merge conflicts can occur without user's knowledge.

**Wrong diagnosis.**

- Check **Status Monitoring** for potential patient merge conflicts and resolve them. Use the e-mail notification system to be notified of potential patient merge conflicts.

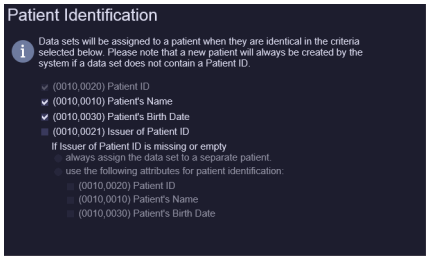
## Selecting DICOM attributes for unique patient identification

To configure the handling of patient identification you have to open the **Patient Identification** window and select the appropriate attributes.

1 Log on to the ***syngo*.via Administration Portal** as administrator.

2 First select the **Installation** workspace and then choose **First Installation** > **Patient Identification** from the navigation tree.

The **Patient Identification** window opens.




The default configuration is Patient ID + Patient's Name + Patient's Birth Date.

3 Select from the following DICOM attributes:

- (0010,0020) Patient ID

  The Patient ID uniquely identifies a patient within a hospital department. The Patient ID is required and cannot be changed.

- (0010,0010) Patient's Name.

  The Patient Name can be used as an additional patient identification key if the Patient ID is not reliable enough (e.g. because it is sometimes typed in manually at the modality).

- (0010,0030) Patient's Birth Date

  The Patient Birth Date can be used as an additional patient identification key if the Patient ID is not reliable enough.

- (0010,0021) Issuer of Patient ID

  If your system receives datasets from different hospitals or different departments within a hospital, two patients could have the same "Patient ID", assigned by different hospitals or departments. In that case, the "Patient ID" is only unique in combination with the "Issuer of Patient ID" (in HL7: "Assigning Authority").

4 Click **Save**.

If the "Data Consistency License" is available at your site, and "HL7 Patient Update" as well as "HL7 Patient ID Change" notifications are received, the attributes **Patient's Name** and **Patient's Birth Date** cannot be used for patient identification as they are not provided by HL7.

Matching of HL7 notifications is therefore always performed by Patient ID + Issuer of Patient ID/Assigning Authority, if available.

## 8.6 Data management

*syngo*.via data management contains data archiving, data deletion, and data transfer.

- Data archiving comprise the definition of target archives (for example, PACS) and the rules for data to be archiving.
- Deletion of data:
  - Automatic data deletion is used to clean up the Short Term Storage of *syngo*.via.
  - The *syngo*.via client provides a manual deletion of data (for Clinical Administrators).
- Data transfer settings comprise configurations for import and export of DICOM data, including the local media type settings.

Some PACS do not support DICOM objects with certain SOP classes. To check if archiving for these SOP classes works and to encapsulate corresponding objects, see *Checking a PACS for unsupported SOP classes* in the *syngo*.via Configuration Online Help.

### 8.6.1 About the configuration of data archiving

If your system is connected to a PACS or a corresponding DICOM node, received and generated data should be sent there for archiving.

**CAUTION**

Hardware failure such as disk crash.

**Data loss.**

- The IT Administrator is responsible for developing a concept for patient data recovery in case of defective hardware, and for the improvement of fail-safe operation of short-term and archive configuration (i.e. use redundant RAID concept).

You can make the following configuration settings for archiving in the ***syngo*.via Administration Portal**:

- Archiving rules and intervals for automatic archiving (➔ Page 123 *Configuration of auto-archiving*)

  From the Short Term Storage (STS), data marked with the "archivable" flag is sent for archiving to a PACS or other DICOM nodes.

- Autorouting rules (➔ Page 130 *Configuration of autorouting rules*)

  Rules are based on data attributes checked for when data arrives or is generated on your system. Accordingly, data will be sent to other DICOM nodes.

- Exclude from archiving rules (➔ Page 128 *Autoexcluding data from archiving*)

  Rules are based on data attributes checked for when data arrives on your system. Accordingly, data will be excluded from archiving.

- Configure DICOM encapsulation

  Some result objects can be encapsulated in DICOM objects to enable PACS systems to store them.

## 8.6.2 Configuration of auto-archiving

In the **Technical Configuration** workspace of the ***syngo*.via Administration Portal** you configure archives.

For configuring archives, you have the following general options:

(➔ Page 123 *Opening the Archive Configuration*)

(➔ Page 124 *Setting up auto-archiving*)

(➔ Page 127 *Selecting DICOM nodes for archiving*)

(➔ Page 127 *Saving changes in the Archive Configuration window*)

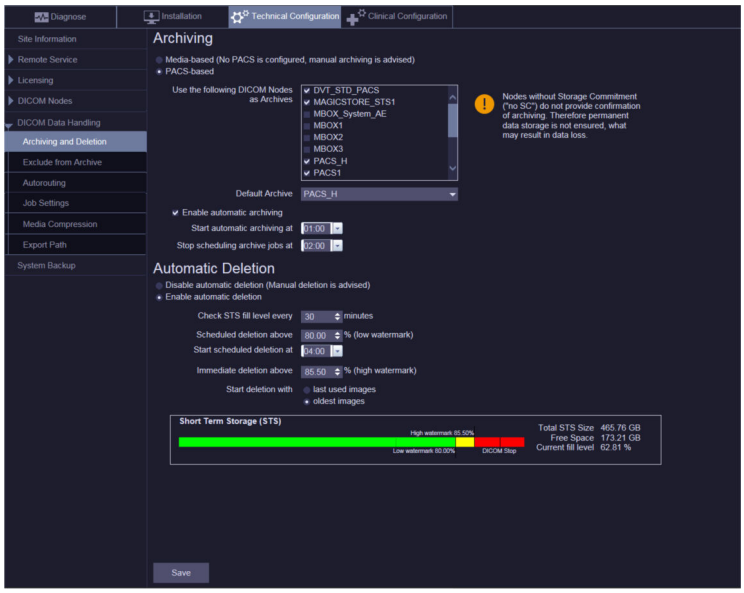
### Opening the Archive Configuration

To configure archives, you have to open the **Archive Configuration** window.

1 Log on to the ***syngo*.via Administration Portal**.

2 First select the **Technical Configuration** workspace and then choose **DICOM Data Handling** > **Archiving and Deletion** from the navigation tree.

The **Archiving** window opens:




You may also be interested in the following:

(→ Page 123 *Configuration of auto-archiving*)

## Setting up auto-archiving

You can enable or disable automatic archiving, define autorouting rules, select the target node and set the archiving time.

✓ Archive nodes are configured as remote DICOM nodes.
(➔ Page 111 *Configuring remote DICOM nodes*)

**CAUTION**

Data transfers between systems are not guaranteed.

**Loss of data if data are deleted locally before they have been successfully transferred to another system.**

- Only systems and communication protocols supporting Storage Commitment guarantee successful transfer to another system.
- In other transfers, it is necessary to verify the correct arrival of the data at the remote system itself.
- Do not delete local data until the transfer to the remote system is verified.

**CAUTION**

Archiving has been configured without using Storage Commitment. Thus, the archive flag is set based only on a transfer response.

**Data is marked with the archive flag even if it has not been archived successfully. If this data is subsequently deleted from the local system, it is irretrievably lost.**

- Only use DICOM nodes configured with Storage Commitment as an archive. If this is not possible, verify the storage of the data at the remote system. Do not delete local data until its storage at the remote system has been verified.

1. Open the **Archiving** configuration window. (➔ Page 123 *Opening the Archive Configuration*)
2. To set up an archiving strategy, select an option:

- **Media-based**: the status of incoming and locally created data is set to "not to be archived". Data will be lost if it is not manually archived!
- **PACS-based**:
  - if no default archive is set, the status of incoming and locally created data is set to "not to be archived" (except for data affected by autoexclusion rules).
  - if a default archive is set, the status of incoming data is set to "not to be archived" and the status of locally created data is set to "archivable" (except for data affected by autoexclusion rules).

**Media-based** option can be used in clinical environments without a PACS. In this case, automatic archiving will not be available and media-based archiving must be performed by the administrator.

3 To enable automatic archiving, select **Enable automatic archiving**.

4 Define a time period for scheduling automatic archiving by setting the start time (**Start automatic archiving at**) and end time (**Stop scheduling archive jobs at**).

Use time settings based on the 24-hour time notation. The default setting is from “01:00” to “02:00”.

All archiving jobs which are scheduled during the archiving time interval will be processed, even if an archiving job exceeds the end of the time interval.

Schedule archiving jobs for outside main working times to avoid interference with your daily work.

Avoid overlapping times for archiving and backup. (→ *About backup and restore in the Administration Online Help*)

You may also be interested in the following:

(→ Page 123 *Configuration of auto-archiving*)

(→ Page 127 *Selecting DICOM nodes for archiving*)

(➔ Page 130 *Configuration of autorouting rules*)

### Selecting DICOM nodes for archiving

You can select archiving destinations and define a standard archive.

1 First select the **Technical Configuration** workspace and then choose **DICOM Data Handling** > **Archiving and Deletion** from the navigation tree.

   The **Archiving** window opens.

2 In the **Use the following DICOM Nodes as Archives** list, use the check boxes to select DICOM nodes *syngo*.via will use for archiving.

3 From the **Default Archive** list, select an archive to which *syngo*.via will send data by default.

   Only DICOM nodes marked for archiving are available in the **Default Archive** list.

The status of data sent to an archive is set to "Archived".

DICOM nodes which do not support Storage Commitment (“no SC”) will not confirm successful archiving.

You may also be interested in the following:

(➔ Page 123 *Configuration of auto-archiving*)

(➔ Page 112 *Adding a new remote DICOM node for configuration*)

### Saving changes in the Archive Configuration window

To complete the configuration of archives, you have to save the changes you made in the **Archiving** window of the ***syngo*.via Administration Portal**.

1 Click **Save**.

2 Check if your changes affect autorouting or DICOM configuration.

   (➔ Page 130 *Configuration of autorouting rules*)

   (➔ Page 96 *Configuration of DICOM nodes*)

You may also be interested in the following:

(→ Page 123 *Configuration of auto-archiving*)

### 8.6.3 Autoexcluding data from archiving

Establish rules to mark incoming data objects as “not to be archived” and thus exclude them from default archiving. Create the rules by selecting appropriate conditions.

*syngo*.via applies these rules on objects received from external DICOM nodes.

**CAUTION**

Data are automatically deleted due to configurable settings.

**Loss of clinically relevant data if rules are not correctly specified.**

- Be very careful when creating "not to be archived" rules for data. This data can be automatically deleted and cannot be recovered.
- Make sure that all data (images and reports) necessary for medical purposes are completely and successfully sent to an archive.
- Do not use automatic deletion if the archive node does not support Storage Commitment.

**CAUTION**

Configurable automatic rules can become complex.

**Unexpected system behavior or loss of data due to definition of complex automatic rules.**

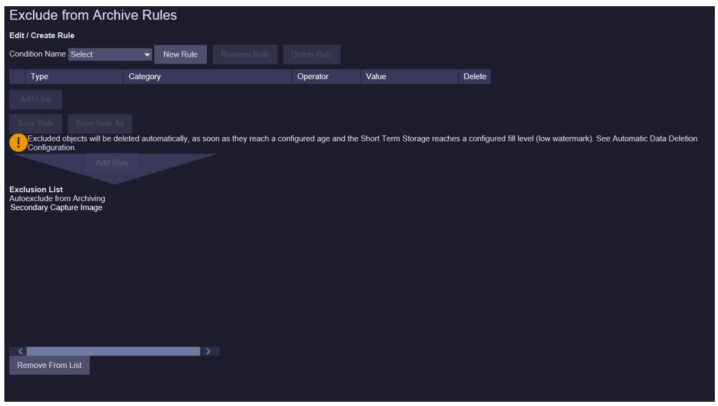
- Test all new rules to ensure that the results conform to your expectations.

✓ A corresponding condition was created. *Configuring conditions for autorouting / autoexclude* in the *syngo*.via Configuration Online Help

1 Log on to the ***syngo*.via Administration Portal**.

2 First select the **Technical Configuration** workspace and then choose **DICOM Data Handling** > **Exclude from Archive** from the navigation tree.

The **Exclude from Archive Rules** window opens:




3 Select a rule from the list and modify it, for example, by adding a new condition with **Add Line**.

– or –

To create a new rule, click the **New Rule** button, select the conditions and save the rule.

4 To add a rule to the exclusion list, select it and click **Add Rule**.

Data that match these conditions is excluded from auto-archiving.

5 To delete a rule, select a rule from the **Exclusion List** and click the **Remove From List** button.

New data that match the respective condition is no longer excluded from archiving.

Removing a rule from the exclusion list does not delete the rule itself.

### 8.6.4 Configuration of autorouting rules

In the **Autorouting Rules** window of the ***syngo*.via Administration Portal**, you can set up rules for automatic transfer of data to a specific DICOM node.

You can create, edit, or delete rules for DICOM data imported, received and retrieved and for objects created with *syngo*.via (for example, findings or reports).

- The default autorouting rules are created on the basis of certain SOP classes. When a new software version is installed, a white list with a default set of identifiers is installed on the system. (→ Page 131 *List of SOP classes installed on the system per default*)
- When upgrading your system to the latest software version, the system automatically creates all rules needed to provide the same archiving behavior that existed before the upgrade. You can later change these archiving rules. (→ Page 134 *List of SOP classes created by the system*)

To configure autorouting rules, you can do the following:

- Access the **Autorouting Rules** window in the ***syngo*.via Administration Portal**. (→ Page 131 *Opening the Autorouting Rules window for configuration*)
- Create or modify an autorouting rule. (→ Page 132 *Creating an autorouting rule*)
- Delete an autorouting rule. (→ Page 135 *Deleting an autorouting rule*)

You may also be interested in the following:

(→ Page 123 *Configuration of auto-archiving*)

### List of SOP classes installed on the system per default

The following table shows a white list with a default set of SOP classes installed on the system by default:

| Identifier | Description | Remarks |
|---|---|---|
| **SOP Class UID:**<br>**1.2.840.10008.5.1.4.1.1.104.1** | Encapsulated PDF | Will only be created if user explicitly opens the reporting template. Thus considered a clinical result. |
| **SOP Class UID:**<br>**1.2.840.10008.5.1.4.1.1.88.33** | Comprehensive Structured Report | Considered as a clinical result. |
| **SOP Class UID:**<br>**1.2.840.10008.5.1.4.1.1.88.22** | Enhanced Structured Report | Considered as a clinical result. |
| **Clinical Result** | Result Images | All items created by the user and displayed under the **Results** area of the **Series** panel, are considered a clinical result and thus relevant for reporting and follow-up. |
| **SOP Class UID:**<br>**1.2.840.10008.5.1.4.1.1.7** | Secondary Capture Image | Without using *syngo*.Reporting, only two objects are created: Structured Report (SR) and secondary capture images |

You may also be interested in the following:

(➔ Page 134 *List of SOP classes created by the system*)

(➔ Page 130 *Configuration of autorouting rules*)

(➔ Page 123 *Configuration of auto-archiving*)

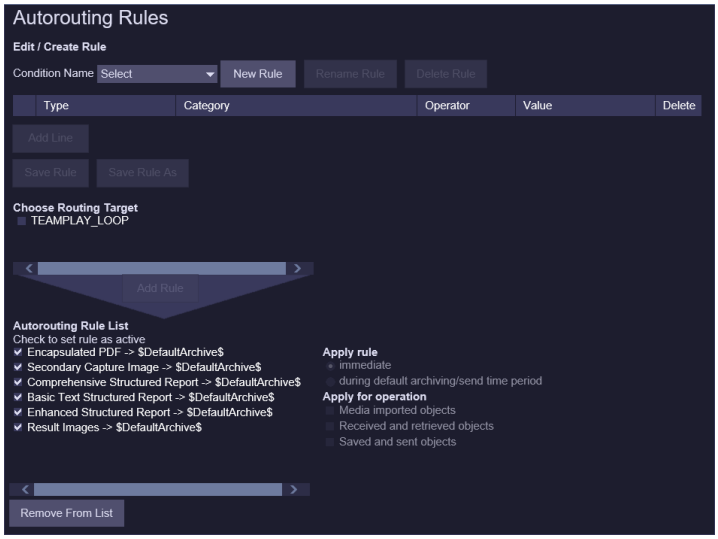
### Opening the Autorouting Rules window for configuration

For configuring rules for data transfer, you have to open the **Autorouting Rules** window.

1 Log on to the ***syngo*.via Administration Portal**.

2 First select the **Technical Configuration** workspace and then choose **DICOM Data Handling** > **Autorouting** from the navigation tree.

The **Autorouting Rules** window opens:




You may also be interested in the following:

(→ Page 130 *Configuration of autorouting rules*)

(→ Page 123 *Configuration of auto-archiving*)

## Creating an autorouting rule

You can create autorouting rules for data transfer in the **Autorouting Rules** window of the ***syngo*.via Administration Portal**.

✓ Nodes that serve as routing target support DICOM storage. (➔ Page 111 *Configuring remote DICOM nodes*)

✓ A corresponding condition was created. (➔ *Configuring conditions for autorouting / autoexclude in the Configuration Online Help*).

**CAUTION**

Configurable automatic rules can become complex.

**Unexpected system behavior or loss of data due to definition of complex automatic rules.**

- Test all new rules to ensure that the results conform to your expectations.

1 From the **Condition Name** list, select a condition.

The **Edit / Create Rule** editor is filled with the data of the selected condition.

If you select the route type **Clinical Results**, all objects created in the workflow (for example, snapshots, evidence documents) that are displayed under **Results** in the **Series** panel are transferred to the archive when the workflow is completed.

2 Select one or more routing targets by selecting the corresponding check boxes.

The **Choose Routing Target** list provides all configured DICOM nodes, which support the storage service. (➔ Page 123 *Configuration of auto-archiving*)

3 Click the **Add Rule** button.

A new rule is added to the **Autorouting Rule List**.

4 Select the check box to activate the rule.

By default, the **immediate** option is selected and autorouting is performed immediately when data arrives at *syngo*.via.

5 If you want the autorouting to be performed according to this rule only during the default archiving/send time period, select the rule from **Autorouting Rule List** and select the **during default archiving/send time period** option.

6 In the **Apply for operation** section, select the corresponding option to which the created rule is applied.

Example **Apply for operation** for routing received thin slices to a dedicated DICOM node:

The DICOM header attribute **Slice Thickness (0018,0050)** is less than "1" (unit in mm). This rule may need to be applied to **Received and retrieved objects** only.

7 Click **Save**.

If you want to see the details of a rule in the **Autorouting Rule List**, select the used condition from the **Edit/Create Condition** list.

You can check whether the created archiving rules are applied by checking the **Archived Status** in the Patient Browser.

You may also be interested in the following:

(→ Page 130 *Configuration of autorouting rules*)

(→ Page 123 *Configuration of auto-archiving*)

### List of SOP classes created by the system

The following table shows a list of DICOM objects (SOP classes) that are created by your system:

| SOP Class UID | Description |
|---|---|
| 1.2.840.10008.5.1.4.1.1.104.1 | Encapsulated PDF Storage |
| 1.2.840.10008.5.1.4.1.1.20 | Nuclear Medicine Image Storage |
| 1.2.840.10008.5.1.4.1.1.4.1 | Enhanced Magnetic Resonance (MR) Image Storage |

| SOP Class UID | Description |
|---|---|
| 1.2.840.10008.5.1.4.1.1.4.2 | Magnetic Resonance (MR) Spectroscopy Image Storage |
| 1.2.840.10008.5.1.4.1.1.481.3 | RTSTRUCT Storage |
| 1.2.840.10008.5.1.4.1.1.66.1 | Spatial Registration Storage |
| 1.2.840.10008.5.1.4.1.1.66.4 | Segmentation Storage |
| 1.2.840.10008.5.1.4.1.1.66.5 | Surface (Mesh) Segmentation Storage |
| 1.2.840.10008.5.1.4.1.1.7.4 | Multiframe True Color Secondary Capture Image Storage |
| 1.2.840.10008.5.1.4.1.1.88.11 | Basic Structured Report Storage |
| 1.2.840.10008.5.1.4.1.1.88.22 | Enhanced Structured Report Storage |
| 1.2.840.10008.5.1.4.1.1.88.33 | Comprehensive Structured Report Storage |
| 1.2.840.10008.5.4.1.1.2 | Computed Tomography (CT) Image Storage |
| 1.2.840.10008.5.4.1.1.2.1 | Enhanced Computed Tomography (CT) Image Storage |

You may also be interested in the following:

(→ Page 131 *List of SOP classes installed on the system per default*)

(→ Page 130 *Configuration of autorouting rules*)

(→ Page 123 *Configuration of auto-archiving*)

## Deleting an autorouting rule

You can delete autorouting rules for data transfer in the **Autorouting Rules** window of the ***syngo*.via Administration Portal**.

1 To delete an existing autorouting rule, select the rule from **Autorouting Rule List**.

2 Click the **Remove From List** button.

Removing a rule from the list does not delete the rule itself.

To delete a rule, you select the condition name and click **Delete Rule**.

3 Click **Save**.

You may also be interested in the following:

(→ Page 130 *Configuration of autorouting rules*)

(→ Page 123 *Configuration of auto-archiving*)

### 8.6.5 Manual data deletion from Short Term Storage (STS)

In the **Patient Browser**, you can manually delete data from the Short Term Storage (STS). For manual data deletion, you need appropriate user rights.

The Short Term Storage (STS) is not an archive. It stores recently acquired data and data needed for current studies until the corresponding workflow is closed and data are archived.

In the **Patient Browser**, you can do the following:

- Delete data from the STS:
  - Delete objects to manually clean the system if auto deletion is disabled.
  - Delete specific objects, for example, series that need to be resent or images with insufficient image quality.
- Protect patient data from deletion, for example, reference examinations

In the ***syngo*.via Administration Portal**, you can configure the rules for automatic data deletion.

(→ Page 137 *Configuring automatic data deletion from STS*)

### 8.6.6 Configuring automatic data deletion from STS

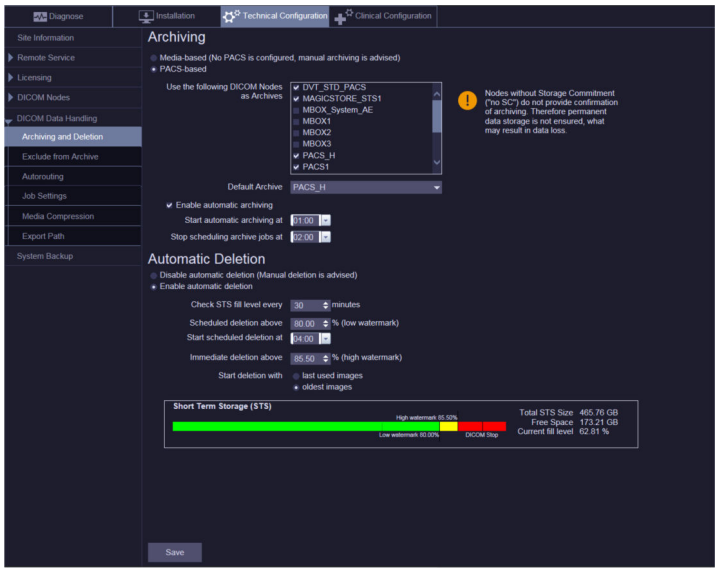
Image data transferred to or acquired or created at your system is saved in the STS. To ensure that there is sufficient space for image data in the STS, images have to be deleted regularly according to configured rules.

The rules for automatic data deletion from the STS can be configured in the ***syngo*.via Administration Portal**

1 Log on to the ***syngo*.via Administration Portal** as administrator.

2 First select the **Technical Configuration** workspace and then choose **DICOM Data Handling** > **Archiving and Deletion** from the navigation tree.

The **Automatic Deletion** window opens:




3 Specify the deletion strategy, the fill level, and the data deletion time interval.

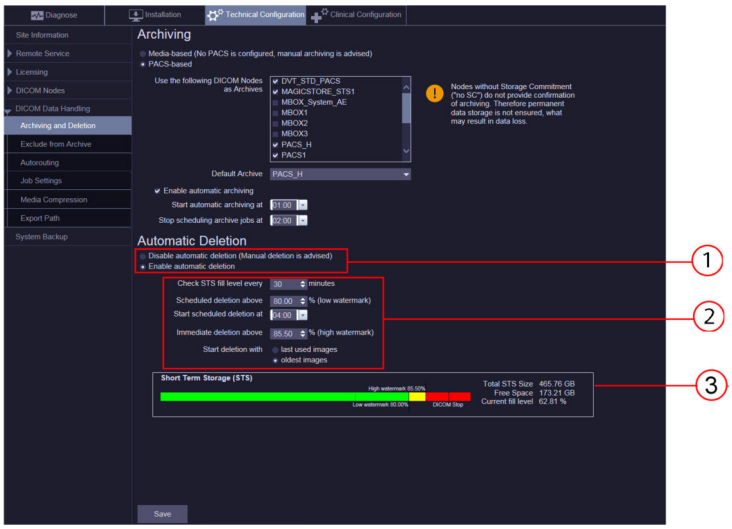
(→ Page 138 *Configuration items on the Automatic Data Deletion window*)

4 Click **Save**.

DICOM objects that were received but could not be processed because of an error are stored in the `C:\Windows\Temp\syngoTfFailedInstances` folder. Files older than 5 days are automatically deleted from this folder twice a day.

## Configuration items on the Automatic Data Deletion window

In the **Automatic Deletion** window of the ***syngo*.via Administration Portal** you set the rules for data deletion from the STS.




(1) **Deletion options** list

(2) Tools for setting the fill level and the time interval

(3) **Short term storage (STS) fill level** color bar

### (1)Deletion options

| | |
|---|---|
| **Disable automatic deletion** | Your system will not automatically delete any data.<br>The STS will run full if data is not manually deleted! |
| **Enable automatic deletion** | Your system will delete data according to the rules for automatic data deletion. |

**(2) Tools for setting the fill level and the time interval**

| | |
|---|---|
| **Check STS fill level every** | Defines how often the system should check if conditions for automatic deletion are met. The default value is “30” minutes. |
| **Scheduled deletion above** | Defines the fill level size which will initiate scheduled data deletion (low watermark). The default value is “80.00”%. |
| **Start scheduled deletion at** | Defines the start time for scheduled nightly data deletion. The default value is “04:00”. |
| **Immediate deletion above** | Defines the fill level size which will initiate immediate data deletion (high watermark). The default value is “85.00”%.<br>The remaining storage in the red range must be sufficient for at least three days of system operation. The maximum value is 94% to ensure some remaining storage space. |
| **Start deletion with** | • **least used images**<br>The images with the oldest access time are deleted first.<br>• **oldest stored images**<br>The oldest stored images are deleted first. |

**(3) Short term storage (STS) fill level color bar**

The current fill level is depicted by a dotted line on the color bar. You can specify certain fill level limits which are represented as low and high watermarks.

The low watermark is the limit the fill level reaches during scheduled deletion.

The high watermark is the fill level limit after immediate deletion.

### 8.6.7 About configuration for data import and export

*syngo*.via allows users to export and import DICOM data. You can adjust settings for data transfer in the *syngo*.via Administration Portal and in the **Configuration Panel** on clients.

*syngo*.via Administration Portal: The general settings in the *syngo*.via Administration Portal take effect on the whole system:

- You can set the default media compression for export.

  See (→ Page 140 *Configuring default media compression for export*).

- You can configure paths where exported DICOM images can be stored.

  See *Configuring the DICOM export path* in the *syngo*.via Configuration Online Help.

Configuration Panel: The settings in the **Configuration Panel** take effect on the **Export Data** dialog box:

- You can set the displayed number of recently used nodes to export DICOM data to network.

  See *Setting the displayed number of recently used nodes* in the *syngo*.via Configuration Online Help.

- You can manage media burning profiles, for example to write DICOM data on a CD.

  See *Defining media profiles* in the *syngo*.via Configuration Online Help.

- You can define media types and corresponding storage capacities, for example if you use special CDs in your institution.

### Configuring default media compression for export

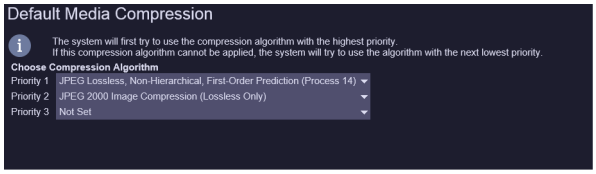
You can set the default media compression for export. These settings are used for exporting data to CD, DVD, or the file system at a *syngo*.via client. This configuration is valid for all *syngo*.via clients.

When exporting data, *syngo*.via compresses each file separately. *syngo*.via uses the compression algorithm defined with **Priority 1** first. If this algorithm cannot be used for the type of media (according to the standard defined in DICOM Part 3), the system uses the next algorithm. If none of the defined algorithms is applicable, the data is exported uncompressed.

1 Log on to the ***syngo*.via Administration Portal**.

2 First select the **Technical Configuration** workspace and then choose **DICOM Data Handling** > **Media Compression** from the navigation tree.

The **Default Media Compression** window opens:




3 In the **Default Media Compression** window, choose up to three compression algorithms.

Some of the proposed media compression algorithms use lossy compression. If these algorithms are used, the image quality may no longer be adequate for diagnosis. A warning box will prompt the user to confirm the selection of such a compression algorithm.

4 Click **Save**.

## 8.7 Setup of *syngo*.via server after installation

During initial setup of *syngo*.via, the main task for the administrator is to prepare the *syngo*.via server for the local IT environment.

Adaptation concerns the following topics:

- Network Configuration (➔ Page 157 *Network configuration*)
- Security Settings (➔ Page 193 *Data and system security*)
- Backup Settings (➔ Page 150 *Configuring backup settings*)
- User Management Configuration (➔ Page 67 *User management*)
- DICOM Configuration (➔ *Configuration of DICOM nodes in the Configuration Online Help* )
- Short Term Storage Configuration (➔ *Configuring automatic data deletion from STS in the Configuration Online Help*)

- Preparation of your system for Smart Remote Services (SRS) (→ Page 223 *Smart Remote Services*)
- Active Directory Integration (→ Page 160 *Active Directory policies for syngo.via*).
- Configuration of the Remote Service Board
- Modification of IP address or server name
- Time synchronization

If there is a problem with the *syngo*.via server which cannot be solved, contact Customer Care Center.

## 8.8 Update of *syngo*.via server

The *syngo*.via server comprises hardware, driver, firmware, operating system, database and application software manufactured by Siemens Healthineers, and software components of other vendors.

The *syngo*.via server has devices connected, for example printer, keyboard, mouse, or microphone. These devices are typically supplied with a vendor-specific driver and management software, which must be installed on the *syngo*.via server and/or client.

All kinds of software, that is, server and client software, firmware, driver, operating system, database, application, require updates for improvement, security or stability reasons.

When users log on to their clients, they are informed (from time to time) about pending updates on the server.

**CAUTION**

Failed system updates can be time-consuming.

**System availability can be impacted**

- Always calculate a sufficient time buffer for updates or upgrades.

The following software update mechanisms exist:

- *syngo*.via application server update

  This software update mechanism provides Siemens Healthineers software updates for the *syngo*.via server and integrated client. The software updates are offered on the **Software Update** page of the *syngo*.via Administration Portal.

- Windows update

  This software update mechanism provides updates for the Windows operating system, for MS Office, and more. The software updates are offered by **Windows Update**.

- Vendor-specific updates

  Software for connected devices such as printers is updated by vendor-specific procedures.

The IT Administrator needs to validate the system after the installation of updates. The System Monitoring Status should be the same as before.

The IT Administrator needs to check the general functionality of *syngo*.via as learned in the *syngo*.via training course.

If errors occur, updates need to be removed from the system and the IT Administrator needs to contact the Customer Care Center.

### 8.8.1 Restriction to installation of other software (Software Blacklist *syngo*.via)

Once the server system has been handed over to the customer, no software must be installed on the *syngo*.via server that does not comply with the rules and restrictions described in the "Software Blacklist *syngo*.via".

The blacklist specifies which software is allowed to be installed on the server, and which not.

The latest available revision of the "*syngo*.via Software Blacklist" is provided in the LifeNet syngo Information Center (https://bps-healthcare.siemens.com/sic), in the section **System Security**.

Each blacklist entry refers to a Services Knowledge Base (SKB) entry, which gives details on the restrictions.

Failure to observe the restrictions described in the Software Blacklist *syngo*.via may cause software malfunctions or system outages when performing software updates.

### 8.8.2 Updating the *syngo*.via application server

Update packages are downloaded and installed using the **Software Update** window of the *syngo*.via Administration Portal.

Always refer to the **Instructions** provided as link in this window.

The following is a generic description of the tasks related to the update/upgrade process.

Updates of the *syngo*.via application server can **only** be started from a local session or from a remote desktop session at the server.

Be aware that *syngo*.via is not able to accept image transfers during the software/update installation. If connected modalities do not resend data automatically after the downtime, data needs to be resent manually.

In case of any unexpected incidents or problems during the update, please contact the Customer Care Center.

### Starting *syngo*.via update

1 Log on to the *syngo*.via server using a local or remote desktop session.

2 Verify that the *syngo*.via server is running on the required version as specified when following the **Instructions** link.

The *syngo*.via version can be identified on the *syngo*.via Administration Portal login page.

3 Double-click the *syngo*.via Administration Portal icon on the desktop and log on. Do not open the *syngo*.via Administration Portal from a remote node.

4 First select the **Installation** workspace and then choose **Software Update** from the navigation tree.

5 Verify that the update package is available in the Software Update (Status: Ready for Install).

6 Select the required update packages and click the **Install** button. See *Installing software packages* in the *syngo*.via Administration Online Help

The *syngo*.via Administration Portal is closed and you are redirected to an Installation Page.

### Performing pre-update and post-update steps

When installation has been invoked, the *syngo*.via FieldUpdater tool starts automatically to perform the pre-update steps. The tool guides you through the update process. The installation is controlled by the *syngo* installer.

As soon as the *syngo*.via package installation is completed, the *syngo*.via FieldUpdater automatically performs the post-update steps. It pauses only in case of hold-up.

- Reboot the *syngo*.via server.

The *syngo*.via client application detects the new server version automatically and performs the necessary updates.

For details, refer to package specific update information in the Software Update window.

## 8.9 Backup and restore of the *syngo*.via server

The Backup & Restore toolset provides automatic and manual backup and restore functionality for the server system and the MSSQL database. Backups include:

- *Server system (drive image of system partition)*
  - The operating system (OS)
  - OS configuration settings (for hardware components, network settings, localization settings)
  - OS patches and hotfixes
  - Clinical applications
  - Application configuration (for example, DICOM configuration)
- *Database*

  Patient and workflow data (but no image data)

Siemens Healthineers does not provide a backup mechanism for *syngo*.via clients. It is in the responsibility of the Administrator to care for client backups.

The Backup & Restore toolset consists of the following parts:

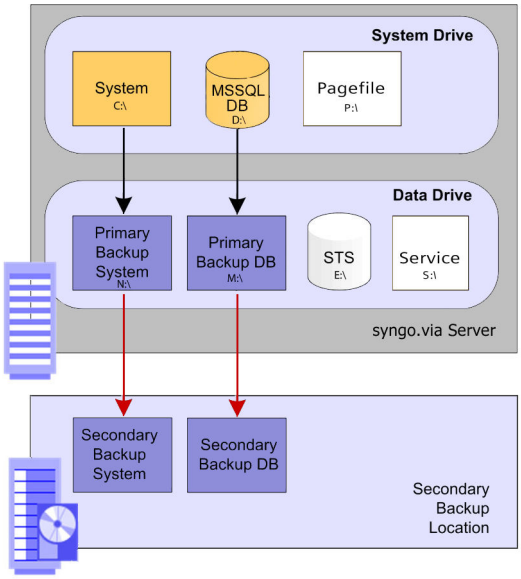
- Configuration in the Administration Portal
- Command scripts
- Windows Backup
- ManagementStudio (for MSSQL server backup)
- STS consistency tool

By default and if enabled, the *syngo*.via server automatically performs a backup of the system and the database every day at 3 a.m. Manual backups can be performed anytime.

There is no backup of the image data stored in the Short Term Storage (STS)! In case of a major problem, unarchived data from the STS may be lost!

The STS Consistency tool must be used to check for inconsistencies with the STS after restoring the database. See *STS Consistency Tool* in the *syngo*.via Configuration Online Help.

The following diagram illustrates a sample setup of server drives and backup locations (the setup varies based on the hardware used):




The **system drive** holds the operating system, the MS SQL database, the configuration settings, and the applications.

The hard disk configuration depends on the hardware setup of your system, but all systems are based on redundant hard drives (RAID).

The **primary backups** of both data areas are stored on complementary disks.

**Primary backups** are only kept for a limited amount of time depending on size and configuration (typically 2 generations for the database backup; for the operating system, it depends on the target size). Therefore, restoring older backups requires a secondary backup.

(→ Page 150 *Configuring backup settings*)

(→ Page 151 *Recovery procedures*)

### 8.9.1 About secondary backups

A **secondary backup** is a copy of the primary backup. Siemens Healthineers does not provide any special mechanism to create secondary backups. It is in the responsibility of the Administrator to care for secondary backups.

A secondary backup should be used to integrate the local backups into your own backup and storage management. To create secondary backups, copy the following folders to the secondary backup location:

- N:\WindowsImageBackup for the Windows backup
- M:\BackupRestore for the database backup

External mass storage devices as well as network shares can be used as secondary backup locations.

**Secondary backups** are not part of the *syngo*.via backup. The IT administrator is responsible for a secondary backup. If no secondary backup is made, the *syngo*.via system and the database may be lost in the event of a hardware failure.

**CAUTION**

Backups onto non-redundant hardware are not sufficient for data security.

**If a hardware failure or other severe failures happen, a massive loss of data can occur if backups have not been performed or if non-redundant hardware was used for backups.**

- Set up a routine for secondary backup of database and configuration items on external (removable) media at regular intervals and based on a backup concept.
- Set up a backup concept for patient data routed from modalities over *syngo*.via to long-term archive (PACS).
- Regularly check that backups are performed properly.
- Ensure that critical data is additionally stored on redundant hardware (RAID).

**CAUTION**

Hardware failure such as disk crash.

**Data loss.**

- The IT Administrator is responsible for developing a concept for patient data recovery in case of defective hardware, and for the improvement of fail-safe operation of short-term and archive configuration (i.e. use redundant RAID concept).

The time needed to run a secondary backup depends on the media type of the secondary backup location (for example, DVD, USB-Disk, NFS mount point).

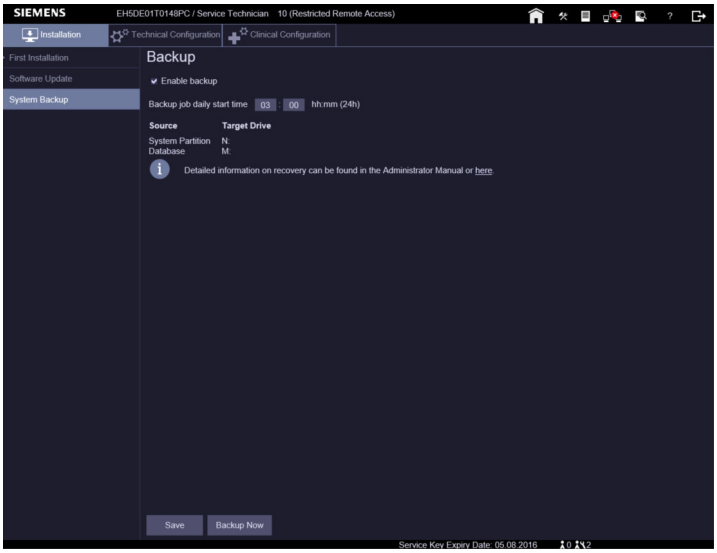
### 8.9.2 Configuring backup settings

You can use the Administration Portal to configure the Backup & Restore toolset according to your needs.

✓ The *syngo*.via application server is running.

1 Log on to the *syngo*.via Administration Portal.

2 First select the **Installation** workspace and then choose **System Backup** from the navigation tree.




3 If the **Backup Configuration** area is dimmed, select the **Enable backup** check box.

4 In the **hh:mm (24h)** text fields, enter the exact time of the day (hour and minute) when the backup should be performed. Use the 24 hour clock format.

5 Click **Save** to apply the changes.

6 If you want to take a backup immediately, click the **Backup Now** button.

The backup task is configured with the selected parameters in the Windows Task Scheduler. The task is located under **Task Scheduler Library** > **Siemens** > **Backup_syngo.via**. The backup process is started at the configured time every day. It consists of two successive steps:

- Backup of the *syngo*.via system partition
- Backup of the MSSQL database

### 8.9.3 Recovery procedures

There are several scenarios which require a recovery of the system, the database, or even both.

When software errors occur, the following recovery strategies are available for you:

- Recovery of corrupted files

  See *Recovering corrupted files* in the *syngo*.via Administration Online Help.

- Recovery of C: partition in case of corrupt OS or application

  See (➔ Page 152 *Recovering the C: partition*).

If it is required to perform a rollback to an older version of *syngo*.via, always recover in the following order: C: partition and MSSQL database. For database recovery, please contact the Customer Care Center.

In case of hardware errors which require a recovery, you have to call Customer Care Center. The following cases should only be handled by them:

- Recovery if RAID is damaged
- Recovery if system drive is damaged
- Recovery if complete hardware is damaged
- Recovery if data drive is damaged

During disk, system or database recovery, the system cannot be used.

You do not have to reinstall the Windows operating system before running the recovery.

If you had to replace the hard disk, make sure that the disk is at least as large as the disk that contained the backed up data. It is not possible to use a smaller disk, even if the required amount of disk capacity is small.

After a system recovery, workflows in status **To be Assigned** are reset to the status **Ready** and the job queue is empty. Also workflows in status **Scheduled**, **Ready**, **In Work**, **In Progress** and **Saved** are reset to the status **Ready**. Workflows that were not mapped or were in status **Completed** remain unmapped without any state.

### 8.9.4 Recovering the C: partition

The recovery of the operating system, applications, and configurations should be done with a complete server recovery.

For this purpose, a Backup & Restore tool set is provided on the system disk and can be accessed during start-up.

The **Re-image your computer** wizard utilizes backup packages stored on the server, network drives, or removable media.

If the recovery tool is not available (for example, due to disk failure), contact the Customer Care Center.

If a rollback to an older major version of *syngo*.via is required, contact the Customer Care Center. Rollback may include reverting database changes, which can only be performed by Siemens Healthineers.

### Starting the Recovery environment from the system disk

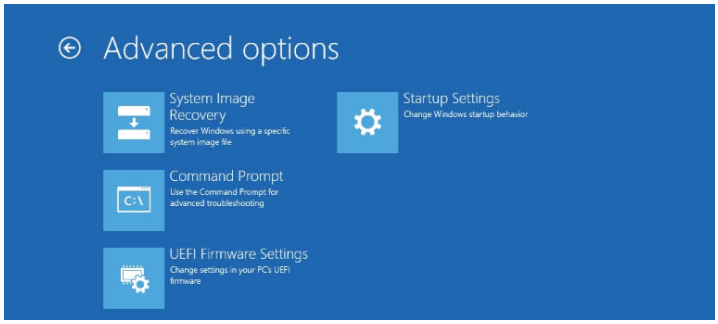
The screenshots given in this section are examples for Windows Server 2016.

1 Shut down and restart the host computer.

2 From the **Windows Boot Manager** menu, choose **syngo.via OS-Recovery**.

You can also boot from a Recovery DVD or an operating system DVD to access the system recovery environment.

3 Choose **Troubleshoot** and then **System Image Recovery** at the next screens.

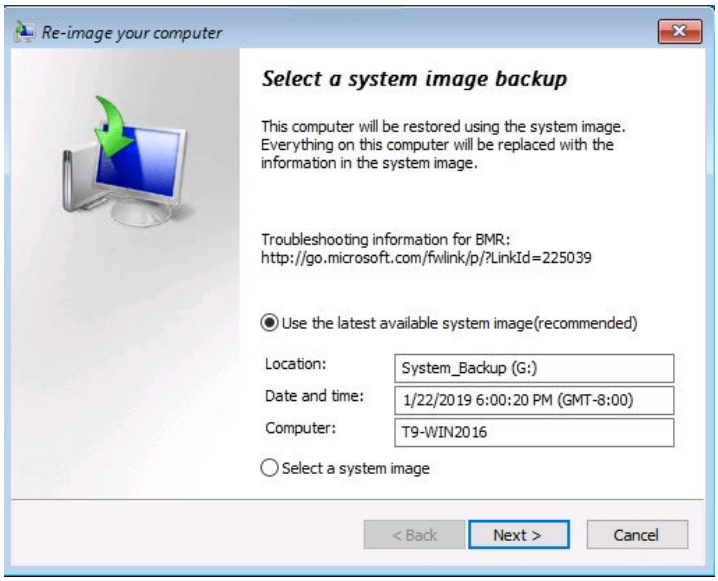



4 Choose an administrator account to continue, and enter the password for this account.

### Selecting a system image backup

The **Re-image your computer** wizard offers 2 options:

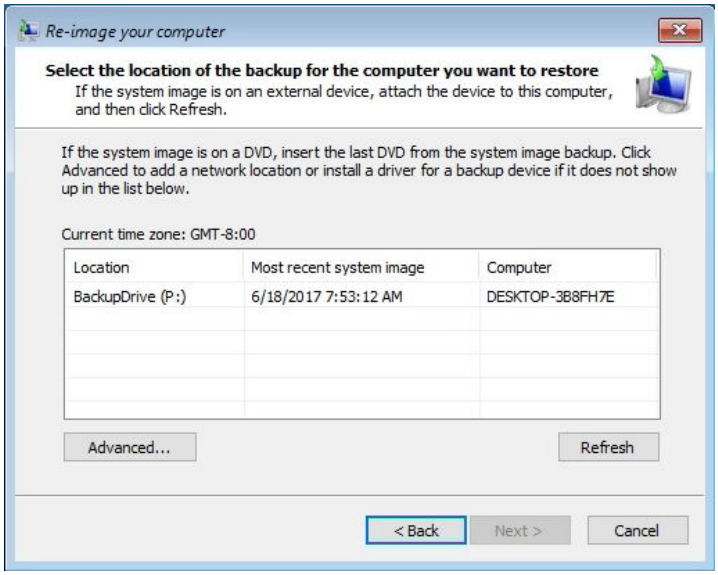
1 Click **Use the latest available system image (recommended)** to install the latest backup from the primary backup location and click **Next**.




=> Continue with *Starting the recovery*.

– or –

Click **Select a system image** to install an older backup, or a backup from a network share or removable media and click **Next**.




**Subsequent page when selecting the 2nd option**

The system recovery environment may display a different time zone for the creation time of available backups.

You cannot reuse old backups from a different OS version!

2 To connect a network drive, click the **Advanced** and the **Search for a backup on the network** buttons.

3 Confirm with **Yes**, and specify the location of the backup (`\\Servername\Foldername`). Click **OK**.

4 Enter the username and the password of a user account with access rights to the specified network location. Click **OK**.

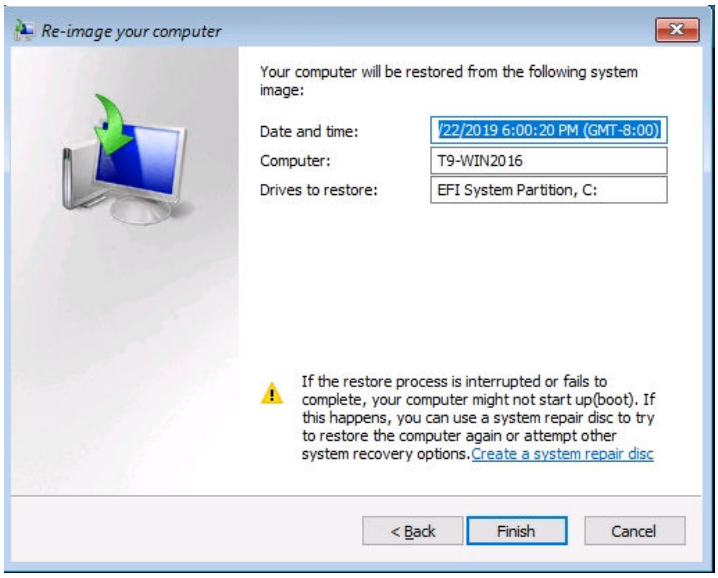
5 Select the desired backup location from the list and click **Next**.

The **Choose additional restore options** page appears.

Nothing to be selected here. Do not use the **Format and repartition disks** option.

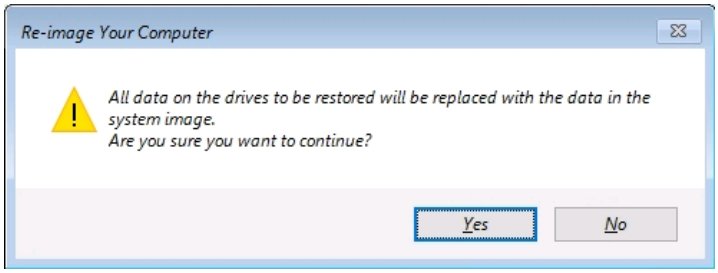
## Starting the recovery

1 Click **Next**.




2 Confirm the backup selection by clicking the **Finish** button.

A message appears:




3 Click **Yes** to continue.

All data from the system drive will be deleted and overwritten with the data from the backup.

The re-image process might take some time.

### Completing the recovery

To complete the recovery process, perform the following steps:

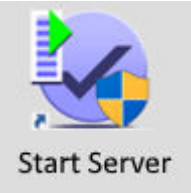
1 Restart the system and wait until the application server (APS) has started; messages can be ignored.

You will be automatically logged on to the server.

2 On the server desktop, double-click the **syngo.via - Stop Server** icon to stop the *syngo*.via application server.

3 Execute the following commands from a ***syngo*.via Server Shell**:

- `syngo.RemoteServices.Workflow.WfAdmin.exe storeWorkflowUids file="%MED_LOG%\Workflow\WorkflowRestore.xml" dataserver=SQL`
- `syngo.Services.Workflow.DeploymentHelper.exe -i`




4 Double-click the **syngo.via - Start Server** icon to start the *syngo*.via application server.

The system is reset to the status at backup time. Consider checking user accounts and passwords, configurations, etc.

## 8.10 Network configuration

*syngo*.via is integrated in your local, clinical network environment in such a way that it is accessible both by the medical workplace and the Smart Remote Services (SRS) back-end.

Depending on your network environment, *syngo*.via is equipped with one or two network connections for the medical network and one for the remote service board.

The *syngo*.via server supports two network cards with link aggregation. Therefore, *syngo*.via has one IP address within the medical network. However, the remote service (iLO) board needs an additional IP.

After the initial installation, the IP address and the network settings are configured. Later changes are possible, but a special procedure must be followed.

See *Changing IP address or server name* in the *Administration Online Help*.

*syngo*.via is operable in a DHCP environment, but must be equipped with a fixed IP address.

For information about the firewall configuration, see:
(→ Page 207 *Communication ports*).

## 8.11 Joining the *syngo*.via server to an Active Directory

Adding a server to an active directory provides several enhancements:

- It enables single sign-on for domain users.
- It facilitates user management.
- Security policies are centrally managed.
- Software distribution by AD policy is possible.

(→ Page 159 *Adding a server to a domain*)

(→ Page 160 *Adapting the Active Directory settings*)

(→ Page 160 *Active Directory policies for syngo.via*)

### 8.11.1 Adding a server to a domain

✓ You know the credentials of the domain administrator.

✓ *syngo*.via must have reading access to your clinical domain Active Directory (AD).

1 Ensure that the Active Directory policies required by *syngo*.via are not overwritten.

See (➔ Page 160 *Active Directory policies for syngo.via*).

2 Log on to the *syngo*.via server operating system.

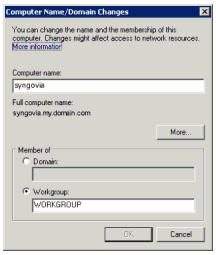
3 Stop the *syngo*.via application server.

4 Open the **Control Panel** in an icon view, and choose **System**.

5 Click the **Advanced system setting** link to open the **System Properties** dialog box.

6 Click the **Computer Name** tab card and click the **Change...** button.

7 In the **Computer Name/Domain Changes** dialog box, select the **Domain** option and enter the domain name you would like to join.




8 In the **Windows Security** dialog box, enter the user credentials of the domain administrator and click **OK**.

9 Confirm the dialog boxes which appear with **OK** to reboot the server.

10 Restart the server.

### 8.11.2 Adapting the Active Directory settings

1 Log on to the Active Directory domain controller.

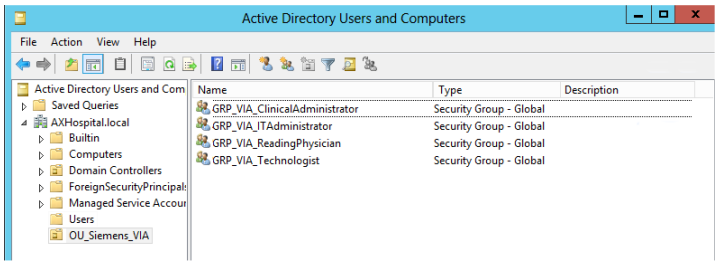
2 Open the **Control Panel** and select **Active Directory Users and Computers**.

3 Select the node of your domain.

4 Create a *syngo*.via organization unit within your domain.

5 Create global security groups for each *syngo*.via role and configure the membership of your domain users accordingly.

See (→ Page 73 *Access rights and roles*).




6 Adapt the role mapping of *syngo*.via and add the domain groups to the corresponding *syngo*.via role.

See (→ Page 80 *Role manager*).

### 8.11.3 Active Directory policies for *syngo*.via

When joining an Active Directory, the Local Security Policies of *syngo*.via can be overwritten. Check your policies before integrating the *syngo*.via server to your Active Directory domain. If necessary, change the Group Policy for the *syngo*.via server.

For maximum performance, *syngo*.via needs several Windows user groups with the “Create Global Objects” privilege.

To check the currently applied policies, proceed as follows:

1 On the Windows **Start** page, open a command prompt and enter **gpresult /H c:\temp\GPReport.html** and press **Enter**.

The GPReport.html file is saved.

2 Open the GPReport.html file.

The file opens the **Group Policy Results** page.

All overwritten policies are listed.

3 Search and check the following settings for overrides:

- User Rights Assignment settings
- UAC Security options
- Remote Desktop Services

If necessary, go to the specified location and configure the options.

### User Rights Assignment settings

| Policy name | Contains | Location |
|---|---|---|
| Perform volume maintenance tasks | • Local System<br>• Network Service | Computer Configuration\Windows Settings\Security Settings\Local Policies\User Rights Assignment |
| Deny log on locally | • Remote Desktop Users | Computer Configuration\Windows Settings\Security Settings\Local Policies\User Rights Assignment |
| Create global objects | • Administrators<br>• Local Service<br>• Network Service<br>• Service | Computer Configuration\Windows Settings\Security Settings\Local Policies\User Rights Assignment |

| Policy name | Does not contain | Location |
|---|---|---|
| Allow log on through Remote Desktop Services | • Administrator | Computer Configuration\Windows Settings\Security Settings\Local Policies\User Rights Assignment |

Local administrators can only log on at console level and not with Remote Desktop Connections. Any local administrator has to be added to the “Administrators” user group and cannot be a member of the “Remote Desktop Users” user group.

Remote administrators can only log on with Remote Desktop Connections and not on console level. Any user who should act as a remote administrator must be a member of the “Administrators” and the “Remote Desktop Users” user groups.

For better distinction of local administrators from remote administrators, Remote Desktop Users should be labeled with “Remote_...”

### UAC Security options

A system not protected by UAC is vulnerable to being exploited by malware and inexperienced or careless users could render the *syngo*.via server useless.

| Policy name | Option selected | Location |
|---|---|---|
| User Account Control: Behavior of the elevation prompt for administrators in Admin Approval Mode | • all others except: Elevate without prompting | Computer Configuration\Windows Settings\Security Settings\Local Policies\Security Options |
| User Account Control: Run all administrators in Admin Approval Mode | • Enabled | Computer Configuration\Windows Settings\Security Settings\Local Policies\Security Options |

### Remote Desktop Services

| Policy name | Option selected | Location |
|---|---|---|
| Allow users to connect remotely using Terminal Services | • Enabled | Computer Configuration\Administrative Templates\Windows Components\Remote Desktop Services\Remote Desktop Session Host\Connections |

## 8.12 Audit trail

Whenever patient data is accessed or processed by a user, the action is logged and stored in the so-called audit trail, if audit trail is enabled.

By evaluation of audit trail records, it is possible to trace which actions are done to the data of a specific patient.

(➔ Page 163 *Actions logged in the audit trail*)

(➔ Page 164 *Audit trail content*)

(➔ Page 165 *Audit trail storage*)

(➔ Page 166 *Audit trail archive*)

(➔ Page 167 *Audit trail evaluation*)

The audit trail record is subject to security and data protection.

In *syngo*.via it is possible to disable and enable the audit trail using scripts. The audit trail is enabled by default. (➔ Page 167 *Enabling and disabling auditing*)

### 8.12.1 Actions logged in the audit trail

The following actions are logged according to the audit record trigger events as specified in Integrating the Healthcare Enterprise (IHE):

- Access to patient data: create, modify, delete, or read
- Query/retrieve of image data, or reports
- Access to protected procedures: create, modify, delete, or read
- Patient data sent or received over network transfer
- Patient data imported or exported
- Captured screenshot

- Actors start/stop: system, Workflow, log on, log off
- Security configuration activities
- Security-sensitive application events: for example, unsuccessful login, access attempt by unauthorized user

Audit trail records from the ***syngo*.via Administration Portal** provide a supervision instrument to the administrator. The following actions are logged:

- Start of service session
- Stop of service session
- Change of the remote access mode
- Change of the DICOM node configuration

The audit trail only logs actions that are performed in the *syngo* software. Actions that are performed in Windows or a third-party software are not logged.

### 8.12.2 Audit trail content

The audit trail consists of numerous audit trail records.

In order to reduce the number of generated audit trail records, all accesses to a single patient are summarized in a single audit record at the level of studies.

The system stores the following information within each audit trail record:

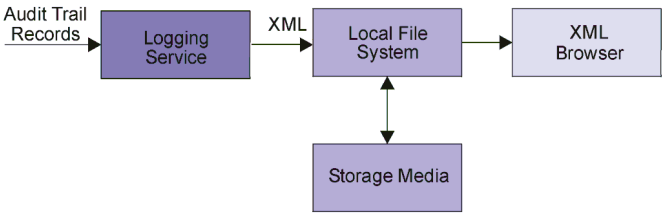
- Host name or IP address of the server node from where the audit was triggered
- AET of local node when applicable
- Event creation date and time
- User ID (for example, account name)
- Type of audit event
- In case the actor has had access to patient data: patient identifier

- User name or service key, IP address and service level at the start of a ***syngo*.via Administration Portal** service session
- User name or service key, IP address and source of termination (service user, administrator, or timeout) at the end of a ***syngo*.via Administration Portal** service session

### 8.12.3 Audit trail storage

All actions to be logged are collected by the Logging Service.

Audit trail records are stored in the local file system in syslog format. (➔ Page 166 *Audit trail records on the local file system*)




The local file system is used as logging repository.

As soon as audit trail records land in the audit trail repository, they should not be modified.

In order to protect the audit trail repository against modification and deletion, the administrator can define Access Control Lists using the Windows operating system. These lists can restrict access to the audit trail repository for certain Windows user or user groups (by default only the “Administrators” group).

Before deleting audit trails to create free disk space, export the files to an external archive for on demand access.

In the ***syngo*.via Administration Portal**, you can configure automatically forwarding the audit trails to an external Syslog Server.

### Audit trail records on the local file system

On the local file system, audit trails are stored in **%MED_LOG%\Audit\AuditMessages.log** (default configuration is **C:\Store\log\Audit\AuditMessages.log**).

- When the log file reaches the maximum file size of 10 MB, a copy of the file is created and a new **AuditMessages.log** file is started. The copy is compressed to save disk space and is stored in the same folder. The file name contains a time stamp: AuditMessages_yyyy-MM-dd HH-mm-ss.zip
- When the log file folder exceeds 500 MB, a warning message is sent to the administrator by e-mail, if configured. The warning is a reminder to move the existing log files to an external archive. If the folder exceeds 700 MB, the **Audit trail** component in **Status Monitoring** changes to "Faulted" status.

A local file system is used as an audit trail repository. For this reason, no audit trail records can be generated for the following actions:

- Exporting an audit trail file to a storage medium (DVD, CD-RW or Network-Share)
- Deleting an audit trail file from the local file system

## 8.12.4 Audit trail archive

It is advisable to archive audit trails in regular intervals on removable media, such as DVDs. You can use the writer for optical disks of the Windows operating system.

Alternatively, you can store audit trails on a network share.

Store backup media containing audit trail records in a fireproof location.

To comply with HIPAA (Health Insurance Portability and Accountability Act, USA only) requirements, retain your audit trail records for at least six years.

### 8.12.5 Audit trail evaluation

To trace which actions are done to a specific patient, you can evaluate the audit trail records.

Only authorized users are allowed to inspect the audit trail records.

If the audit trails are stored on the local file system, you can evaluate the audit trail logs in the **Audit Messages** tab of the **Save Log Viewer**.

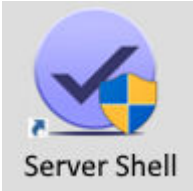
### 8.12.6 Enabling and disabling auditing

If you have administrator rights, you can access and execute scripts for enabling and disabling audit trails.

Consider reading the read-me.txt file in C:\Program Files\Siemens \syngo\bin\AuditScripts.

✓ Users interrupted their work.

1 To stop the application server, double-click the **syngo.via - Stop Server** icon on the Windows desktop of the server.




2 Double-click the ***syngo*.via Server Shell** icon.

3 To enable auditing, type:

```
syngo.common.starter -IKM.IS_BE -AuditEnable
```

4 To disable auditing, type:

```
syngo.common.starter -IKM.IS_BE -AuditDisable
```

5 To start the application server, double-click the **syngo.via - Start Server** icon on the Windows desktop of the server.

Enabling or disabling of audit trails only takes effect after the next start of the client, because the Config.net items are cached.

## 8.13 Uninstallation of the *syngo*.via server

If your institution wants to remove *syngo*.via from operation, it has to be uninstalled. Please make sure that relevant patient data is being transferred to another system before uninstallation.

Make sure to follow your institution's specifications for removing an IT system from operation.

Before uninstalling the server, the following steps must be performed:

- Uninstalling *syngo*.via clients from workstations

  Open the *syngo*.via Administration Portal: In the **Asset view** of the **Status monitoring** window under **Site Report** > **Customer Site** > **syngo.via** > **Client**, the host names of all workstations with a *syngo*.via client are listed.

- Uninstalling or removing front-end integrations (for example, image callup from RIS or PACS client)

- Stopping data inflow from DICOM connections (for example, the scanner), and removing automatic send rules, if established

  At all systems that connected to the *syngo*.via server, the *syngo*.via DICOM node entry should be removed.
- Removing Siemens Healthineers special transfer protocols for scanners (CT Fast Transfer, MR Fastlink), if established
- Removing HL7 connections, if established
- Removing PACS integration (if a PACS or LTS connection is established)
- Deleting patient data from the D:\ partition (DB_Data) and the E:\ partition (MED_Images) permanently: for example, by formatting the partitions while not using the quick format option (The formatting, especially for the E:\ partition, may take some hours.)
- Asking the local administrator to remove the server from the domain (if the server is integrated in the domain)

If you wish to not only format the partitions, but rather to overwrite them, you should use an appropriate tool.

# 8 *syngo*.via server administration

# 9 *syngo*.via client installation

The *syngo*.via client is the medical user interface of *syngo*.via. It is used for reading images as well as for the preparation of reports.

The client software can be downloaded from the *syngo*.via server and must be installed on each client computer.

- (➔ Page 175 *About the installation of syngo.via clients*)
- (➔ Page 177 *Installing the prerequisites for syngo.via clients*)
- (➔ Page 179 *Installing syngo.via clients using the Admin Client Installation Page*)
- (➔ Page 182 *Installing syngo.via clients using a software deployment infrastructure*)

For installation and updates, standard Microsoft Windows deployment tools are used. These tools automatically check for updated software versions on the *syngo*.via server, and download and install updates.

(➔ Page 187 *Updates of clients or secondary consoles*)

(➔ Page 189 *Uninstalling syngo.via clients*)

In case the Device Guard blocks a client installation, you need to disable the Device Guard, install the client, enable the Device Guard back again and create a reference scan of the computer. During this process, some reboots are required.

(➔ Page 198 *Windows Device Guard for the server*)

The *syngo*.via client supports different screen resolutions, orientations, and multi-monitor usage.

(➔ *syngo.via Basic Application Online Help*)

# 9 *syngo*.via client installation

Security management of the clients is in the responsibility of the IT administrator:

- (→ Page 172 *Security settings for clients*)
- (→ Page 191 *Communication Ports at clients*)
- (→ Page 192 *Hints and Troubleshooting*)

## 9.1 Security settings for clients

The IT administrator is in charge of his own security management for the clients. He can install compatible virus protection software, download patches and hotfixes to fix program bugs, and configure firewall settings.

### 9.1.1 Virus protection for clients

The IT administrator is responsible for installing compatible virus protection software on *syngo*.via clients and keeping them up to date.

The IT administrator needs to validate the system after the installation of virus protection software, scan engines or virus patterns.

Virus protection products of major vendors (Trend Micro, McAfee, Symantec, and Sophos) have been tested for usage with *syngo*.via. A current list of tested virus protection software is published in the Release Information.

Virus protection products known to affect the *syngo*.via stability, performance, or functionality will be announced by Siemens Healthineers. Do not install blacklisted virus protection programs! Please refer to the corresponding blacklist in the LifeNet *syngo* Information Center.

To avoid false positives, we recommend to exclude the folder `C:\Windows\Installer\*.*` from scanning when configuring the virus scanner at clients.

### 9.1.2 Updates for Windows operating system

Patches and hotfixes can improve data security by fixing program bugs, errors, security gaps, and other vulnerabilities. Therefore, they can protect your system from attacks caused by malicious software.

The installation of all kinds of Windows updates for client operating systems is performed by the IT administrator according to the customer's enterprise-wide IT security strategy.

Windows updates may include updates for prerequisites of the *syngo*.via clients. Known problems regarding incompatibility between Windows updates and *syngo*.via will be published by Siemens Healthineers on regular basis.

Refer to the corresponding blacklist in the LifeNet syngo Information Center.

### 9.1.3 Updates of third-party software on clients

The IT administrator is responsible for updates of existing third-party software on clients (for example, office applications).

Prerequisites of the *syngo*.via client software should only be updated by the automatic update mechanism of *syngo*.via.
(→ Page 187 *Updates of clients or secondary consoles*).

### 9.1.4 Updates for *syngo*.via clients

Updates regarding the *syngo*.via client are initiated as soon as an outdated client connects to the *syngo*.via server.

(→ Page 187 *Updates of clients or secondary consoles*)

The IT administrator is responsible for the installation of recommended graphics card drivers for *syngo*.via clients.

Siemens Healthineers publishes information about incompatible client graphics card drivers which turn out to affect the stability, performance, or functionality of *syngo*.via in the Blacklist published in LifeNet *syngo* Information Center.

### 9.1.5 Firewall settings

The Windows Firewall is used to protect the system against intrusions. It can be configured separately for each network adapter.

The Windows Firewall is able to block both incoming and outgoing traffic.

To enable communication with the *syngo*.via application server, the IT administrator has to open the following ports at the network firewall (X → Y means that X will connect to the port at system Y, either permanently or temporarily):

| Service/Function | Port number |
|---|---|
| *syngo*.via client → *syngo*.via server | 32912, 32914, 3389, 8090, 80 |
| *syngo*.via server → *syngo*.via client (VNC) | 5800, 5900, 5901, 5902, 5903 |

The Windows Firewall blocks network share access across subnet boundaries by default. You can change the corresponding settings in the **Advanced Settings** of the Windows Firewall.

One of the **Inbound Rules**, "File and Printer Sharing (NB-Session-In)", needs to be adapted. On the **Scope** tab card, the **Remote IP addresses** option has to be changed to "Any IP addresses" or the required subnets.

### 9.1.6 Settings for VNC connections

The Collaboration tool (also known as Expert-i) relies on VNC technology. Therefore, security settings must allow VNC connections.

For VNC ports, see (→ Page 207 *Communication ports*).

## 9.2 About the installation of *syngo*.via clients

The *syngo*.via client software can run wherever necessary, for example, on a modality console or other workstations. The client is installable on standard hardware.

Refer to the *syngo*.via Data Sheet for a list of Windows operating systems on which the *syngo*.via client can run, including system and service pack requirements.

The *syngo*.via client hosts a lightweight, .NET-based application UI. Business logic and 3D-image processing services are hosted by the *syngo*.via server. The software prerequisites and the client software must be installed at the workplace.

If the client installation is aborted without further notice, please consider to add an exception to the proxy configuration of the Internet Explorer to make sure that no proxy is used for the communication to the *syngo*.via server.

### 9.2.1 Prerequisites for client installation

The *syngo*.via client relies on a set of prerequisites, which vary according to the operating system. Local administrative rights are needed for the initial installation of the prerequisites.

- Microsoft Visual C++ runtime libraries 8.0, 10.0, 12.0, and 14.0
- Microsoft .NET Framework 4.7.2

### 9.2.2 Installation scenarios for clients

Due to the prerequisites, the following installation scenarios are possible:

- Full *syngo*.via client installation by administrator

  The administrator installs the prerequisites and the client. The application is then available for all users without additional installation.

  Updates are installed immediately as soon as the *syngo*.via client detects the version of the application server, and the server is running a newer version than the client is.

- Distributed *syngo*.via client installation

  The client installation is distributed by the clinical software distribution system using, for example, Active Directory mechanisms. This procedure is recommended when a considerable high number of client machines and users are going to use the *syngo*.via client.

### 9.2.3 Language settings for clients

The language of the *syngo*.via client is set at startup time.

To set up support of certain languages on client PCs, see (→ *Troubleshooting language settings in the Administration Online Help*).

In order to assure proper visualization of Chinese characters within *syngo*.via, it is required to set the regional settings of the operating system at the client (**not** at the server) to the Chinese language before installing the *syngo*.via client.

### 9.2.4 Monitor setup

Approved calibrated monitors are required for diagnostic workstations.

*syngo*.via clients support different screen resolutions, orientations, and multi-monitor usage.

(→ *Workplaces and monitors in the Application Online Help*)

(→ *Monitor configuration in the Configuration Online Help*)

Monitors that are used for medical reporting must be calibrated before use!

## 9.3 Installing the prerequisites for *syngo*.via clients

You can install the prerequisites from the Deployment Server. Administrative rights are required for installation.

If the client installation cannot be provided using Active Directory, the administrator will have to install the prerequisites for users without administrative rights.

When installing from the **Admin Client Installation Page**, it is recommended to install the client setup that includes prerequisites. If appropriate, you can nevertheless install the prerequisites separately as described.

### 9.3.1 Installing client prerequisites from the Deployment Server

The Deployment Server provides a prerequisites-only installation package for administrators.

✓ The installation of clients has been prepared. See *Preparing client installations* in the *Administration Online Help*

1 Log on to the client PC with administrator user credentials.

On Windows with activated User Account Control (UAC), every prerequisite installation requires a separate confirmation. If you start the Internet Explorer with the “Run as administrator” option, the prerequisite installation needs only one UAC confirmation.

For security reasons, close the Internet Explorer when the installation is finished.

2 Start the Internet Explorer and enter the following address: **`http://<syngo.via-Server>`**. Specify the location of the **`<syngo.via-Server>`** by a fully qualified domain name or IP address.

If you want to activate communication encryption, you must enter the fully qualified domain name. See (➔ Page 216 *Encryption of client/server communication*)

3 Click the **Admin Client Installation Page** link.

The **Admin Client Installation Page** opens:

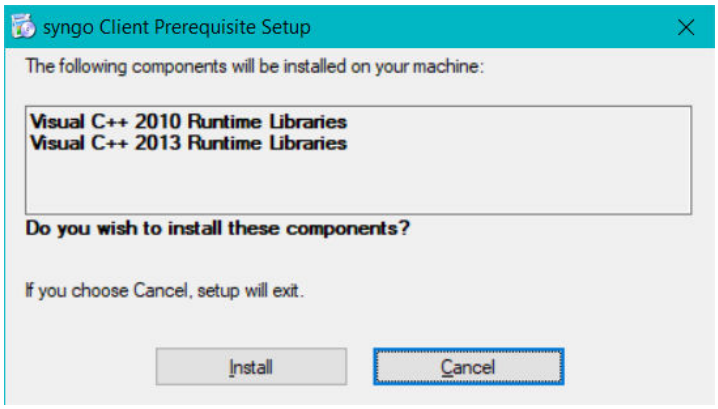
This page provides installation links for the prerequisites, the *syngo*.via client, Adobe Reader and DICOM Viewer. In case of an existing RIS/PACS integration, you might want to install the *syngo*.via client to the specific path of the existing client.

4 Click the **Install prerequisites only** button.

5 When a **File Download** dialog box appears, click **Run**.

6 When a **Security Warning** dialog box appears, click **Run**.

The **syngo Client Prerequisites Setup** window opens and displays the prerequisites which will be installed.




7 Click the **Install** button to start the installation of the listed prerequisites.

A dialog box confirms the completion of the installation. A reboot might be necessary.

8 Click **OK** to end the prerequisite setup.

## 9.4 Installing *syngo*.via clients using the Admin Client Installation Page

The installation of *syngo*.via clients can be performed by using your Internet Explorer.

✓ The installation of clients has been prepared. See *Preparing client installations* in *syngo*.via Administration Online Help.

1 Log on to the client PC.

2 Start the Internet Explorer and enter the following address: **`http://<syngo.via-server>`**. Specify the location of the **`<syngo.via-server>`** by a fully qualified domain name or an IP address.

If you want to activate communication encryption, you must enter the fully qualified domain name. See (➔ Page 216 *Encryption of client/server communication*)

3 Click the **Admin Client Installation Page** link.

The **Admin Client Installation Page** opens:

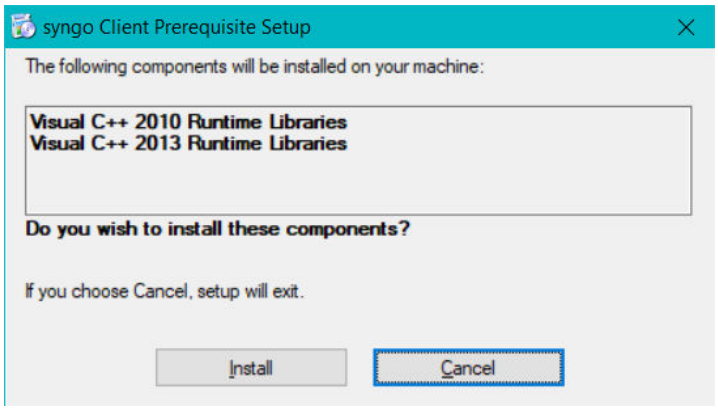
This page provides installation links for the prerequisites, the *syngo*.via client, Adobe Reader, and DICOM Viewer.

4 Click the **Install syngo.via client** button for standard installation.

5 When a **File Download** dialog box appears, click **Run**.

6 When a **Security Warning** dialog box appears, click **Run**.

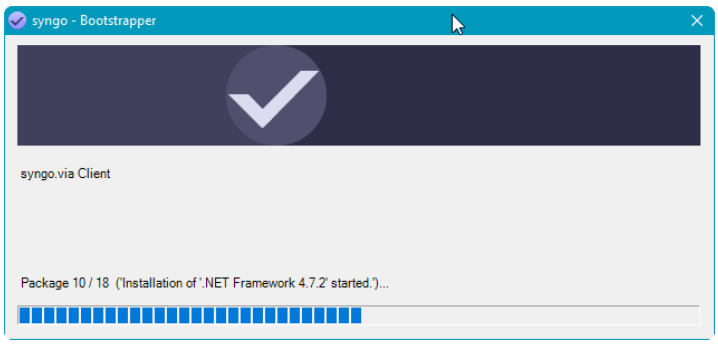
The **syngo Client Prerequisite Setup** window opens and displays the prerequisites which will be installed.




7 If a prerequisite is missing, click the **Install** button to continue.

The installation packages are downloaded from the server to the local computer. Depending on the missing prerequisites, the installation can take up to 30 minutes.

Windows with activated User Account Control (UAC) will prompt to confirm the installation of every prerequisite. Do not leave the installation unattended, because a timeout of the user request will be interpreted as abort.




8 Wait until the *syngo*.via client is installed and configured. This can take up to 5 minutes.

9 If the Windows Firewall displays a **Windows Security Alert**, click the **Unlock** button.

The *syngo*.via client is installed to the program files folder and can be used by all users of the current PC.

## 9.5 Installing *syngo*.via clients using a software deployment infrastructure

You need to include all of the files listed below for a complete client installation. Otherwise, the client startup check detects missing files, and attempts to reinstall them.

The installation packages are located on the *syngo*.via server under the folder:

```
C:\Program Files\Siemens\syngo\DeploymentServer
\RTC_Prereq
```

**The installation of clients consists of the following packages:**

- **Prerequisites** (should be installed in advance):
  - vc8redist, vc10redist, vc12redist, vc14redist (Microsoft Visual C++ Runtimes 8.0, 10.0, 12.0, and 14.0)
  - DotNetFx472 (Microsoft .NET Framework 4.7.2)

- **Bootstrapper** (choose one .msi according to your environment):

  Four different .msi files are provided, acc. to the following syntax: Bootstrapper_<target>@<syngo.via-server>.msi

  <target> means:

  – syngo: installs to the destination folder: C:\Program Files (x86)\Siemens\syngo

  – syngo.via: installs to the destination folder: C:\Program Files (x86)\Siemens\syngo.via

  <syngo.via-server> means:

  – IP address: use this if your IT infrastructure has no DNS

  – Real hostname: use this if your IT infrastructure uses the DNS service for hostname to IP resolution

- **Client application software** (choose one .msi to specify the destination):

  – syngo.via_Client_syngo.msi: installs the client in the folder: C:\Program Files (x86)\Siemens\syngo

  – syngo.via_Client_syngo.via.msi: installs the client in the folder: C:\Program Files (x86)\Siemens\syngo.via

- **Further client packages**:

  – From the Expert-i folder: VNC (Siemens - Repack)_x86.msi for in-house collaboration

  – From the TeamViewer folder: Both .MSIs are required for remote assistance

  – From ..\Store\syngo32_Package: syngo.FlightRecorder.msi for recording of incidents

  – From ..\Store\syngo32_Package: syngo.Client.DeviceGuard.CatalogFiles.msi (only if client is installed on Windows 10, or Windows Server 2016, or newer)

**You can perform the installation in one of the following ways:**

- Use the *syngo*.via server as a deployment server, or copy the installation files to your own deployment server and trigger installation using an `msiexec` command.
- Use a single silent installer file that downloads all necessary installation files (including the prerequisites) from the *syngo*.via server during execution.
- Use the **Active Directory Administrative Center** if your client belongs to a Domain controller.

### 9.5.1 Using msiexec to install *syngo*.via clients

You can trigger the download and the installation of the single installation packages using the *syngo*.via server as deployment server or after copying the installation files to your deployment server.

Prior to installation it is recommended to uninstall the previous version of the client.

1. Make sure the prerequisites are installed.
2. To install the bootstrapper, specify the file by the following syntax:

```
msiexec /i http://<syngo.via-server>
/DeploymentServer/RTC_PreReq
/Bootstrapper_<target>@<syngo.via-server>.msi
ALLUSERS=1 /qn
```

   Replace <syngo.via-server> by the IP address or the host name of the *syngo*.via server, and replace <target> with *syngo* or *syngo.via*.

There are two ways to install the client application files. We recommend to use the first one, because this will include all needed packages:

3 By using the client bootstrapping service:

```
"C:\Program Files (x86)\siemens\<target>\bin
\CUS\syngoClientBootstrapping.exe" -update
<syngo.via-server>
```

– or –

By using an msiexec command with destination:

```
msiexec /i http://<syngo.via-server>
/DeploymentServer/RTC_PreReq/syngo.
via_Client_<target>.msi ALLUSERS=1 /qn
```

In both cases, replace <syngo.via-server> with the IP address or the host name of the *syngo*.via server, and replace <target> with *syngo* or *syngo.via*.

The provided msiexec commands are just examples. To receive an overview of possible parameters (for example, how to enable logging for troubleshooting), enter: `msiexec -h`

### 9.5.2 Using a silent installer to install *syngo*.via clients

With a silent installer you can install prerequisites, bootstrapper and the client application software at once.

1 On the server, navigate to the folder: `C:\Program Files \Siemens\syngo\DeploymentServer\RTC_Prereq`

2 Depending on your infrastructure, choose `syngoSilent_SFX_en_<IP>.exe` or `syngoSilent_SFX_en_<hostname>.exe` and distribute the corresponding file to the client.

3 Execute the file.

Silent installation starts. All required packages are downloaded from the server and installed.

### 9.5.3 Using Active Directory/Group Policy to install *syngo*.via clients

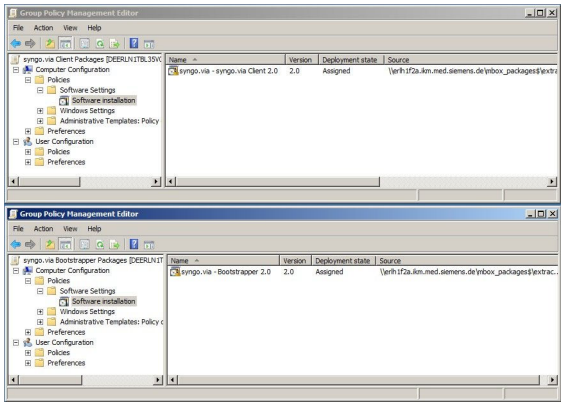
✓ The client computers have access to the network share where the .msi file is stored.

✓ The client computers belong to a Domain controller.

✓ The client computers have access to the *syngo*.via server that is used as installation source.

1. Start the **Active Directory Administrative Center** and create a new **Organizational Unit**, for example "syngo.via Clients".
2. Add the desired client computers to the "syngo.via Clients" Organizational Unit.
3. Start the **Group Policy Management** (gpedit.msc).
4. For the domain where the *syngo*.via client computers belong to, create a Group Policy Object, for example "InstallClient".
5. From the context menu of the "InstallClient" Group Policy Object, choose **Edit**.
6. In the **Group Policy Management Editor**, open the tree down to **Computer configuration** > **Policies** > **Software Settings** > **Software installation**.
7. Right-click **Software installation** and choose **New** > **Package** from the context menu.
8. Navigate to a network share, select the `Bootstrapper_<Inst_Path>@<IP or hostname>.msi` file and set the **Deployment state** to "Assigned".
9. In the **Group Policy Management** window, right-click the "*syngo*.via Clients" Organizational Unit and choose **Link an Existing GPO...**from the context menu.
10. Select the "InstallClient" Group Policy Object.

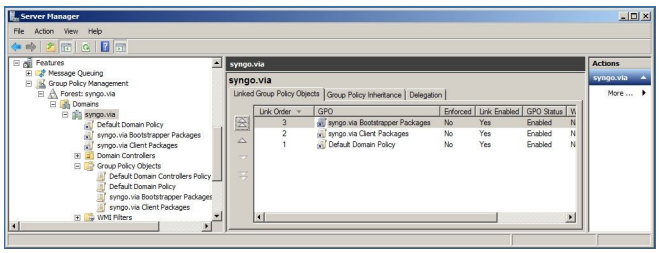
    The *syngo*.via client application is installed when computers in the "syngo.via Clients" Organizational Unit start. It is available to all users who log on to the computer.




If you want to immediately update Group Policies, call **`gpupdate.exe`** on the *syngo*.via server and **`gpupdate.exe /force`** on the client computers.

11 Repeat the steps for the *syngo*.via client.

12 When the installation is completed, sort the objects accordingly.




The higher the link order the earlier the installation, thus resulting in the bootstrapper having a higher number than the client.

## 9.6 Updates of clients or secondary consoles

After an update has been applied to the server, the relevant software packages also have to be distributed to clients by an update mechanism.

The following software update mechanisms can be used:

- Automated *syngo*.via update
- Customer infrastructure for software distribution and installation
- Manual client update/upgrade by command line

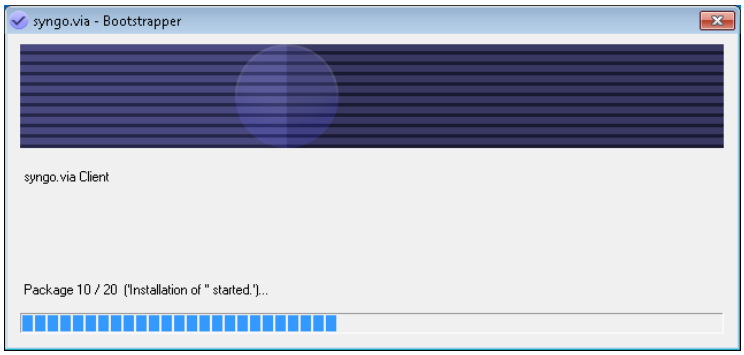
For *syngo*.via only:

- No software may be installed that does not comply with the rules and restrictions described in the *syngo*.via blacklist.
- Siemens Healthineers offers a validation service for customers who want to install third-party software together with client software but are unsure whether it will work or not.
- Please check the Siemens Healthineers price book to request such a service.

### 9.6.1 Automated *syngo*.via update

When the client connects to the *syngo*.via server, it detects the current software version of the server. If the server is running a newer software version than the client, the client will prompt the user to confirm the software update.

After confirmation, the new software will be installed.



### 9.6.2 Customer infrastructure for software distribution

The standard Active Directory mechanisms “Redeploy” and “Remove” can be used to update or to uninstall the *syngo*.via client application.

### 9.6.3 Manual client update/upgrade

If a *syngo*.via client needs to be updated or upgraded, for example after a *syngo*.via hotfix installation on the server, this can be started on the client by running the command:

```
C:\Program Files (x86)\Siemens\syngo.via\bin
\CUS\syngoClientBootstrapping.exe -UPDATE
<via.server>
```

Replace **`<via.server>`** with the *syngo*.via server name or IP address.

The **`msiexec -update`** option is not supported.

## 9.7 Uninstalling *syngo*.via clients

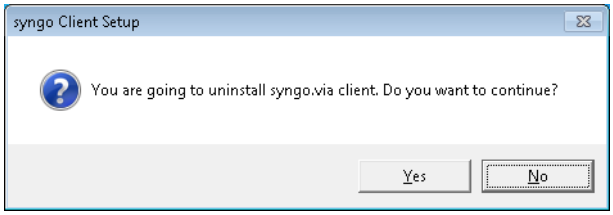
The most convenient way to uninstall a *syngo*.via client is to use the **Uninstall** link provided by the Windows **Start** page.

Another possibility to uninstall clients is to use Active Directory.

### 9.7.1 Uninstalling *syngo*.via clients using the Uninstall *syngo*.via link

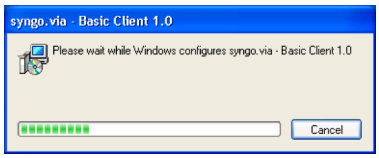
1. Log on at the client PC as user with administrative rights.
2. On the Windows **Start** page, search for **Uninstall syngo.via**.
3. Right-click the **Uninstall syngo.via** link and choose **Run as administrator** from the context menu, if the client PC runs Windows with activated User Account Control (UAC).

   The **syngo Client Setup** dialog box is displayed, asking you to confirm the uninstallation of the *syngo*.via client.




4 Click **Yes** to confirm the uninstallation.

5 Wait until the *syngo*.via client configures the uninstallation.




6 Wait until the *syngo*.via client files are removed.

The *syngo*.via client is uninstalled and physically removed from the system.

The prerequisites installed by the administrator remain on the system. To uninstall prerequisites, use **Add/Remove Programs** of Windows.

### 9.7.2 Uninstalling *syngo*.via clients using Active Directory

✓ The `Remove_RTC.exe` has been copied from C:\Program Files \Siemens\syngo\DeploymentServer\RTC_Prereq to a shared folder.

To provide an Active Directory-based deinstallation of *syngo*.via clients, you need to add a script to the "Shutdown Group Policy Object" (GPO) at the Active Directory domain controller:

1 On the Windows **Start** page, search for **`gpedit.msc`** and open the **Group Policy Management** window.

2 In the **Group Policy Management** window, select the "Default Domain Policy" entry, right-click and choose **Edit** form the context menu.

3 Choose **Computer Configuration** > **Windows Settings** > **Scripts (Startup/Shutdown)** and double-click the **Shutdown GPO**.

The **Shutdown Properties** dialog box opens.

4 Add a path to the shared folder containing **`\Remove_RTC.exe`**.

**`<syngo.via-Server>`** is the FQDN of the *syngo*.via Server, **`<Share-Name>`** is the name of the network share containing the file **`Remove_RTC.exe`**. This is a self-extracting executable file containing the files run.bat and a corresponding Remove_RTC.vbs script.

## 9.8 Communication Ports at clients

Specific TCP/IP ports must be opened in the router or network firewall to enable the communication of *syngo*.via with clients, the SRS, and other instances of the medical environment.

The Windows Firewall of the *syngo*.via server is automatically preconfigured after installation. Ensure that the ports mentioned below are opened at network firewalls and routers between the communicating instances. See (➔ Page 174 *Firewall settings*)

For maximal security, close all ports that are not needed. Refer to the manual of the router or network firewall for how to proceed. Ensure that the ports are open for *syngo*.via as described.

To avoid slow TCP/IP communication, it is recommended to create a group policy object for all users in the domain to disable WPAD access.

For a current list of communication ports, see (➔ Page 207 *Communication ports*).

## 9.9 Hints and Troubleshooting

In the *Administration Online Help* you will find hints and troubleshooting on the following topics:

- Troubleshooting language settings at clients
  - Enabling the East Asian languages support
  - Checking for East Asian fonts
  - Setting up the interface language
  - Setting up the default character set of the syngo.via server
  - Configuring OPENLink for integration with the RIS
- Setting idle session time
- Installing the client fails
- Starting the client fails
- Server-side 3D rendering performance decreased
- Sleep modus timeout longer than expected

# 10 Data and system security

*syngo*.via stores and processes personal data that is subject to the provisions of data protection:

- Electronic Protected Health Information (PHI)
- Personally Identifiable Information (PII)

The data is stored for 3-12 months, depending on the specific usage in your organization.

*syngo*.via uses various techniques to ensure a high level of security:

- To support you in complying with legal requirements such as HIPAA (Health Insurance Portability and Accountability Act, USA only)
- To protect against malicious software
- To protect against hacker attacks and unauthorized access
- To provide maximum security for data, images, and the system

The data and system security strategy is also valid for *syngo*.via options like WebViewer or WebReport. Detailed information about differences regarding the security strategy can be found in the administrator help or release information of these options.

After installation of the *syngo*.via server, you must change the passwords of the administrative user accounts.

For an improved system security, you should set the password length for user accounts to a minimum of 14 characters.

# 10 Data and system security

Connecting *syngo*.via to the Internet can potentially put at risk the data security of the system. Intrusion by virus, malware, or spyware can cause loss or inconsistency of data.

- It is recommended to install an up-to-date virus protection program on the server and clients. In addition, a firewall is recommended to protect clients.

  (→ Page 200 *Virus protection strategy*)

  (→ Page 202 *General virus protection settings*)

  (→ Page 201 *Virus protection for syngo.via server*)

- Always install the latest updates of all kinds of required software.

  (→ Page 172 *Security settings for clients*)

  (→ Page 187 *Updates of clients or secondary consoles*)

Completely fail-safe data security and protection can never be guaranteed in any technical system.

- You are responsible for installing and maintaining appropriate data security and protection measures.
- You have to comply with all applicable laws and regulations.
- Utilize all the capabilities of the system to ensure the highest possible level of data security.
- Avoid any situation that may increase the risk of a breach of data security.

(→ Page 196 *Security strategy and responsibility*)

(→ Page 67 *User management*)

(→ Page 69 *Authorization*)

(→ Page 195 *System Hardening — Secure configuration of the syngo.via server*)

(→ Page 207 *Communication ports*)

(→ Page 216 *Encryption of client/server communication*)

The security settings for *syngo*.via server and *syngo*.via clients are handled separately. The secured access to patient health information is covered by Audit Trail.

(→ Page 163 *Audit trail*)

## 10.1 System Hardening — Secure configuration of the *syngo*.via server

The medical industry is nowadays one of the most attacked industries worldwide.

The reduction of the attack surface is one of the security controls implemented in the current version.

*syngo*.via addresses the growing IT security risks, among other approaches, by hardening of the server machine. The local hardening is activated by the installation, and by upgrade from former versions. In case of domain integration, your domain policies will apply to the server machine.

The hardening is based on **S**ecure **T**echnical **I**mplementation **G**uides (STIG) which are developed and maintained by Defense Information Systems Agency of the USA. The STIGs describe recommendations on the technologies, and environment secure configurations.

- The STIGs are used as configuration standards, for example, by the US Department of Defense.
- The STIGs contain technical guidance to “lock down” information systems and software that might otherwise be vulnerable to a malicious computer attack.

For more information, see http://iase.disa.mil/stigs/Pages/a-z.aspx.

The following STIGs are considered for *syngo*.via servers:

- Microsoft Windows Server 2016 STIG
- Microsoft Windows Server 2012/2012 R2 MS STIG
- Microsoft Windows Server 2008 R2 MS STIG
- Microsoft Windows Firewall STIG and Advanced Security STIG

- Microsoft .NET Framework 4 STIG
- Microsoft SQL Server 2016 Database STIG
- Microsoft SQL Server 2016 Instance STIG
- Microsoft SQL Server 2012 STIG (DBMS and DB instance)
- Web Server Security Requirements Guide (SRG)
- Microsoft Internet Explorer 11 STIG
- Adobe Acrobat Reader DC Continuous Track STIG

The reduction of the attack surface of *syngo*.via servers might impact some specific administration workflows.

Customers who have the server integrated to their domain can adapt the configuration by Domain GPOs or by Local GPOs (GPOs = Group Policy Objects). If needed, the configuration of a locally applied hardening of a server can be adapted by the IT Administrator using local policies.

## 10.2 Security strategy and responsibility

*syngo*.via and its options have been engineered to facilitate a flexible and efficient security management. The corner stone of that concept is the seamless integration of the *syngo*.via system into the existing security strategy of the local IT environment.

The customer IT administration is responsible for the security management of the *syngo*.via system!

The Security strategy is also valid for *syngo*.via WebViewer. Detailed information about differences can be found in the Administrator Manual or Release Information of this option.

**CAUTION**

Unauthorized access to the system.

**System can become non-operational; loss of patient data.**

- This medical device is designed to be operated in a protected network environment. We strongly recommend to not directly connect the device to public networks.
- The IT Administrator is responsible for the network security at the site and for the security of optional infrastructure, such as desktop-virtualization environments. Consult the corresponding manuals for secure setup, and update as required.
- Ensure that only authenticated devices, i.e. belonging to the healthcare enterprise, are connected to the network.
- Set up firewalls and user-account password protections for both server and client.
- Do not allow users to change configuration files.
- Update virus protection software as required.

**CAUTION**

The installation of other non-system software interferes with the system software.

**Harms to the server system and loss of all patient data.**

- The installation of software not provided from Siemens Healthineers is not allowed on the server system.
- Problems arising due to interactions with non-Siemens Healthineers software are not the responsibility of Siemens Healthineers.

Once the server system has been handed over to the customer, no software must be installed on the *syngo*.via server that does not comply with the rules and restrictions described in the “syngo.via Software Blacklist”.

The latest available revision of the Software Blacklist is provided in the LifeNet syngo Information Center, “System Security” section.

## 10.3 Windows Device Guard for the server

Windows Device Guard is available when you have the *syngo*.via server installed on Microsoft Windows Server 2016 or Windows 10 Enterprise.

Windows Device Guard is a set of software security features that will lock your system down so that it can only run **trusted software** that is defined in the code integrity policy.

To ensure maximum security of your system, the Device Guard is enabled automatically during installation of *syngo*.via and an initial code integrity policy is created.

The code integrity policy file **SIPolicy.p7b** is stored in the following directory: C:\Windows\System32\CodeIntegrity

According to the code integrity policy, only software that meets one of the following requirements is allowed to run on your server:

- Software signed with known SHA256 certificates

  For example, all *syngo*.via updates

- Software that is part of a catalog file signed with a known SHA256 certificate

  For example, Adobe Acrobat Reader

- Unsigned software that is installed on your system during the creation of the code integrity policy

  The software will be added to the code integrity policy by its hashes

In the ***syngo*.via Administration Portal**, you can disable and enable again the Device Guard, and update the code integrity policy.

### 10.3.1 Status of the Device Guard

To check the current status of the Device Guard (enabled, disabled, waiting for reboot, ...), use the following tasks in the ***syngo*.via Administration Portal**:

- **Status Monitoring**
- **Device Guard Configuration** (*syngo*.via only)

### 10.3.2 Installation of additional software on the server

Depending on the situation, one of the following procedures may help when installing further software on a server with Device Guard protection:

- Enabling/disabling the Device Guard
- Installing unsigned software that is blocked by the Device Guard
- Installing signed software with unknown certificates
- Updating the code integrity policy of the Device Guard
- Troubleshooting

Details on these Device Guard procedures are provided in *Administration Online Help, Device Guard section*.

For the Microsoft deployment guide for Windows Defender Application Control, refer to: https://docs.microsoft.com/en-us/windows/device-security/device-guard/device-guard-deployment-guide

## 10.4 Virus protection strategy

Virus protection is vital to protect your system and your data from malicious viruses, worms, or trojans. It is recommended to install and maintain a virus protection program.

The IT Administrator is responsible for the virus protection of the *syngo*.via server and the *syngo*.via clients. You have to purchase the licenses, maintain the installation, configuration, and update the virus protection program used.

After the installation of updates, the IT Administrator needs to check the general functionality of *syngo*.via as learned in the *syngo*.via training course. The System Monitoring Status should be the same as before.

If errors occur, updates need to be removed from the system and the IT Administrator needs to contact the Customer Care Center.

Virus protection products that turn out to affect the *syngo*.via stability, performance, or functionality will be announced by Siemens Healthineers. Do **not install** blacklisted virus protection programs! Refer to the corresponding blacklist in the LifeNet *syngo* Information Center.

**CAUTION**

Use of an anti-virus software that is not provided by Siemens Healthineers.

**Malicious software can cause harms to the system and loss of all patient data.**

- The administrator is responsible for the configuration of the anti-virus software.
- Configure and regularly update the anti-virus software.

## 10.5 Virus protection for *syngo*.via server

You may install a virus protection product of your choice on the *syngo*.via server, provided that you have verified the compatibility with *syngo*.via. If the product is incompatible with *syngo*.via, it has to be replaced by one recommended.

Please note that virus protection suites (for example, suites including an active firewall or intrusion detection application) are not supported and must not be installed on the *syngo*.via server!

Virus protection products of some major vendors (Trend Micro, McAfee, Symantec, and Sophos) have been tested for usage with *syngo*.via. The list of tested and recommended virus scanners is published in the Release Information.

Do not install blacklisted virus protection programs! Please refer to the corresponding blacklist in the LifeNet *syngo* Information Center.

After installing a virus scanner, restart the complete server host to ensure proper function of *syngo*.via.

It is your responsibility to install and update virus protection software.

For detailed information about the configuration of virus protection software, see (→ Page 202 *General virus protection settings*).

Never delete or repair infected files automatically. Manual deletion prevents data loss if a false positive occurs. Regularly check and verify infected files and delete them manually. In case of an infection, contact the Customer Care Center.

Ensure that proper virus protection solutions are installed at all computers in your clinical environment.

## 10.6 General virus protection settings

For current information about changes in the recommended configuration of virus protection programs, refer to the **Services Knowledge Base** available using the LifeNet *syngo* Information Center.

In general, the following settings are recommended for virus protection programs installed on the *syngo*.via server.

It is recommended to scan the system regularly for viruses, worms, or trojans:

- Automatic real-time scan during open and save functions. Follow the recommended configuration settings to reduce the impact of real-time scans on the system performance.
- Schedule scans of all files at a time with less clinical routine work.
- Manual scan of all files whenever appropriate.

Configure your virus protection program to issue a warning if any infected file is found on your system.

For virus protection settings of the *syngo*.via options, for example, *syngo*.via WebViewer, see the corresponding administrator help or release information.

### 10.6.1 Settings for real-time scans

- It is recommended to perform real-time scans at all times.
- Scan files while reading from or writing to drives.
- Scan all local hard drives including the boot sectors.

  Do not scan floppy drives or network drives as this may lead to performance issues.
- Certain folders and their subfolders should not be scanned during **real-time scan** as this may lead to performance issues and false positives:
  - `C:\ISPACE\*.*` (if present)
  - `C:\Program Files\Siemens\*.*`
  - `C:\Program Files (x86)\Siemens\*.*`
  - `C:\store\*.*`
  - `C:\sysmgtmt\*`
  - `C:\Windows\Installer\*.*`
  - `D:\SQL_DATA\*.*`
  - `D:\MSSQL\MSSQL13.MSSQLSERVER_SYDS\*.*`
  - `D:\journal\*.*`
  - `E:\frontier\*` (if present)
  - `E:\journal\*.*`
  - `E:\storagefw\*.*`
  - `E:\sysmgtmt\*`
  - `M:\*.*`
  - `N:\*.*`
  - `O:\*.*` (if present)
  - `S:\*.*`

  However, most of these folders **must be scanned** during a scheduled full scan!

In the LifeNet *syngo* Information Center, regularly check the Knowledge Base for an updated list of folders to include in or exclude from virus scans.

- Only scan for *default file types*.

  Only default file types should be scanned as scanning all files may lead to performance issues. However, scan all file types during scheduled full scans!

- Do not scan *compressed files*.

  No compressed files should be scanned as this may lead to performance issues. However, scan compressed files during scheduled full scans!

- Deactivate *heuristic search*.

  Heuristic search should not be activated as the risk of false positives may arise.

- Deactivate advanced intrusion detection/prevention (IDS/IPS) and firewall features.

  Virus protection suites (for example, suites including firewall and intrusion detection applications) are not supported. Deactivate additional features.

- If you are able to define a *default warning text* in case an infected file is found, set it to “Virus Scan Alert!”.

- Only the following actions should be performed if an infected file is found:

  – Set the found file to quarantine.

  – Write an event to the event log.

  To prevent data loss in case of false positives, do not delete or repair infected files automatically. You have to check files manually and delete them if necessary.

- Only the following actions should be performed if spyware, adware, dialers, hack tools, trackware, password crackers, trojans, joke programs, or key loggers are found:
  - Set the found file to quarantine.
  - Write an event to the event log.
  - In case of remote administrator tools, ignore findings but create events.
  - In case of other unwanted programs, ignore findings but create events.

  To prevent data loss in case of false positives, do not delete or repair infected files automatically. You have to check files manually and delete them if necessary.
- It is recommended to *log all events*. Set the anti-virus event log to a maximum size of 50 MB. Store events for at least 14 days.

  You have to check the event log on a regular basis for security reasons.

### 10.6.2 Settings for scheduled or on-demand full scans

- It is recommended to perform a full system scan at least once a week during less system utilization.
- Scan all local hard drives including the boot sector.

  Do not scan floppy drives or network drives as this may lead to performance issues.
- Do not scan the following folders and subfolders as this may lead to performance issues:
  - `E:\storagefw\*.*`
  - `M:\*.*`
  - `N:\*.*`

In the LifeNet *syngo* Information Center, regularly check the Knowledge Base for an updated list of folders to include in or exclude from virus scans.

- Scan *all file types*.
- Scan *compressed files*. If there are compressed files in a compressed file, do not scan more than three levels.
- Deactivate *heuristic search*.

  Heuristic search should not be activated as the risk of false positives may arise.
- Deactivate advanced intrusion detection/prevention (IDS/IPS) and firewall features.

  Virus protection suites (for example, suites including firewall and intrusion detection applications) are not supported. Deactivate additional features.
- If you are able to define a *default warning text* in case an infected file is found, set it to "Virus Scan Alert!".
- Only the following actions should be performed if an infected file is found:
  - Set the found file to quarantine.
  - Write an event to the event log.

  To prevent data loss in case of false positives, do not delete or repair infected files automatically. You have to check files manually and delete them if necessary.
- Only the following actions should be performed if spyware, adware, dialers, hack tools, trackware, password crackers, trojans, joke programs, or key loggers are found:
  - Set the found file to quarantine.
  - Write an event to the event log.
  - In case of remote administrator tools, ignore findings but create events.
  - In case of other unwanted programs, ignore findings but create events.

  To prevent data loss in case of false positives, do not delete or repair infected files automatically. You have to check files manually and delete them if necessary.

- It is recommended to set anti-virus scanner *system utilization* to 50%, medium, or balanced.
- It is recommended to *log all events*. Set the anti-virus event log to a maximum size of 50 MB. Store events for at least 14 days.

  You have to check the event log on a regular basis for security reasons.

  Refer to (→ Page 172 *Security settings for clients*) for more security details for clients.

## 10.7 Communication ports

Ports are an essential part for the communication between systems in a network. A port is a logical construct that identifies a specific process or a type of network service.

Security aspects require to have all ports being closed which are not required/essential for the system to communicate. This is usually handled by blocking rules on firewalls.

For your *syngo*.via system, the following firewalls are of concern:

- The Windows firewall of your *syngo*.via server:

  This firewall is pre-configured after installation, but still requires site-specific adaptations.
- Router and network firewalls at your site:

  Specific TCP/IP ports must be opened there to enable the communication of *syngo*.via with clients, with the SRS, and with other instances/nodes of the medical environment.
- Firewalls of the remote/target systems you want to communicate with:

  Ensure that the corresponding ports are also opened there.

For maximal security, close all ports that are not needed. Refer to the manuals of the router or network firewalls for how to proceed.

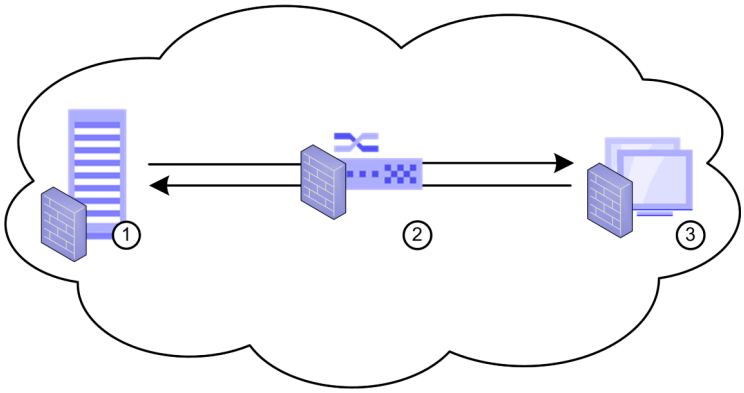
On the other hand, ensure that the ports mentioned below are opened at all firewalls between the communicating instances, i.e. Windows, network, and router firewalls.

Some of the mentioned ports are site-configurable and may vary depending on the needs of the particular installation.

In the tables below, X → Y means that X will connect to the port at system Y, either permanently or temporarily.

For an updated list of communication ports, see the *syngo*.via Release Information.

### 10.7.1 Ports used for *syngo*.via client – *syngo*.via server communication




(1) *syngo*.via server

(2) Internal network router or firewall

(3) *syngo*.via clients

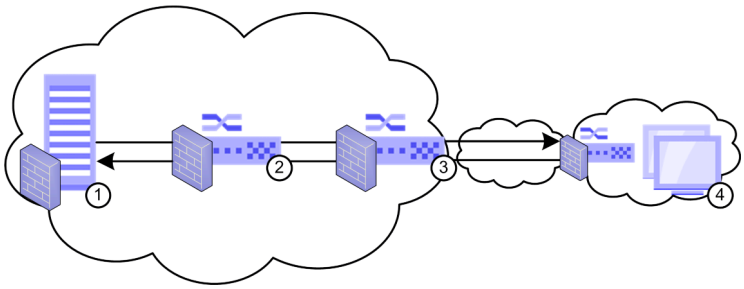
To enable communication with the *syngo*.via application server, the following ports need to be open:

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via server ← *syngo*.via client (Online Help) | Inbound | HTTP | 8090 |
| Remote Desktop connection:<br>*syngo*.via server ← SRS (MSTS)<br>*syngo*.via server ← *syngo*.via client | Inbound | TCP<br>UDP | 3389 |
| *syngo*.via server ← *syngo*.via client (Login Dialog, Online Help)<br>*syngo*.via server ← Reporting Client Application | Inbound | HTTP<br>HTTPS | 80<br>443 |
| *syngo*.via server ← *syngo*.via client (Remote Assistance) | Inbound<br>Inbound | TCP SSL | 11080<br>11081 |
| *syngo*.via server ← *syngo*.via client (VNC, Expert-i collaboration)<br>*syngo*.via server ← SRS (VNC) | Inbound | TCP | 5800<br>5900<br>5901<br>5902<br>5903 |
| *syngo*.via server ← *syngo*.via client (Basic Communication) | Inbound | TCP | 32912<br>32914 |

On Windows 7 servers, you have to adapt the Windows 7 Firewall settings to provide share access across subnet boundaries.

In the **Advanced Settings** of the Windows Firewall, open the **Inbound Rule** “File and Printer Sharing (NB-Session-In)”. On the **Scope** tab, set the **Remote IP addresses** option to “Any IP addresses”.

### 10.7.2 Ports used for *syngo*.via – SRS




(1) *syngo*.via server

(2) Internal router, open ports here

(3) Customer gateway, open ports here

(4) Smart Remote Services back-end

To enable your system to perform all SRS-based services, the following communication ports must be open:

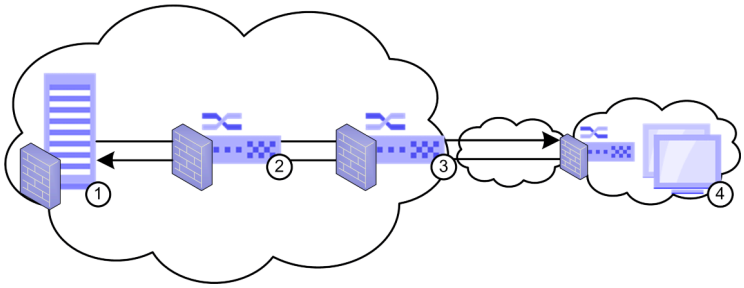
| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via server ↔ SRS (MNP) | Inbound | TCP | 8226-in |
| | Outbound | | 8227-out |
| | | | 8228-out |
| | | | EvtMgt: |
| | | | 12061-out |
| | | | 13001-in |

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via server → SRS (Administration Portal related services) | Outbound | SMTP | 25 |
| Remote Desktop connection:<br>*syngo*.via server ← SRS (MSTS)<br>*syngo*.via server ← *syngo*.via client | Inbound | TCP | 3389 |
| *syngo*.via server → SRS (FTP / Remote service FTP) | Outbound | FTP | 20<br>21 |
| *syngo*.via server → SRS (SMTP)<br>autoreport transfer and mail notification to SRS | Outbound | SMTP | 25 |
| *syngo*.via server ← *syngo*.via client (VNC)<br>*syngo*.via server ←SRS (VNC) | Inbound | TCP | 5800<br>5900<br>5901<br>5902<br>5903 |
| *syngo*.via server ↔ SRS (FTP) | Inbound<br>Outbound | TCP | 20<br>21 |
| *syngo*.via server ← SRS (HTTP)<br>*syngo*.via server ← SRS (HTTPS) | Inbound | HTTP | 80<br>443 |
| *syngo*.via server → SRS (LifeNet – *syngo* Information center) | Outbound | HTTP | 8080 |

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via server → SRS (Remote Assistance) | Inbound | TCP | 11080 |
| | Outbound | HTTP | 8080 |

If some SRS-based services are not available, the Customer Care Center can use the **Connection Check Tool** to check for closed ports.

### 10.7.3 Ports used for *syngo*.via Remote Service Board – SRS




(1) *syngo*.via server

(2) Internal router, open ports here

(3) Customer gateway, open ports here

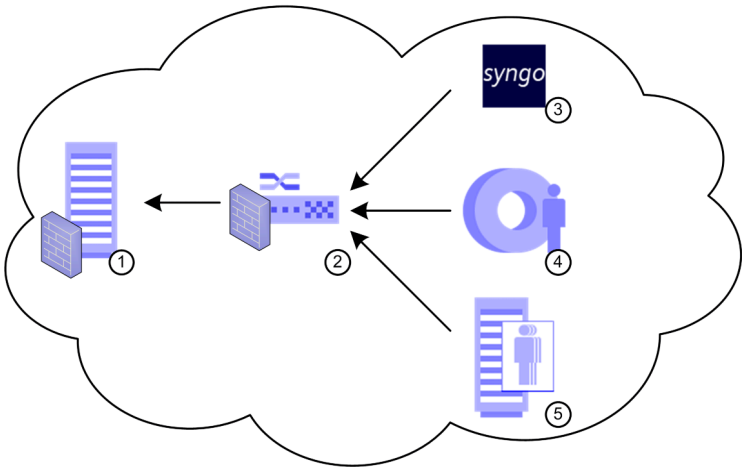
(4) Smart Remote Services back-end

When the *syngo*.via server utilizes a Remote Service Board, the following communication ports must be open:

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| SRS → *syngo*.via Remote Service Board (SSH, telnet) | Inbound | SSH | 22 |
| | | telnet | 23 |
| *syngo*.via Remote Service Board ← SRS (HTTP, HTTPS) | | HTTP | 80 |
| | | HTTPS | 443 |

See *Configuring the Remote Service Board* in the Online Help.

### 10.7.4 Ports used for *syngo*.via – Medical environment




(1) *syngo*.via server

(2) Internal router or firewall, open ports here

(3) DICOM node

(4) DICOM modality

(5) RIS/HIS

The following ports are closed by default at the *syngo*.via server firewall.

To enable *syngo*.via to receive messages and data from other instances of the medical environment, you have to open the following ports at the Windows server firewall, and at the router and network firewalls:

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via (OPENLink) ← HIS/RIS<br>Additionally for Multi-Server solutions | Inbound<br>Outbound | HTTP internal<br>HL7 | 8080<br>9973<br>9971 |
| *syngo*.via server ← DICOM nodes | Inbound | DICOM | 104 |
| *syngo*.via server ← HL7 messages<br>(internally used only) | Inbound | HL7 | 9974<br>9975 |
| *syngo*.via server → RIS<br>(default port on RIS for HL7 messages) | Outbound | HL7 | 9977 |
| *syngo*.via server ← *syngo*.via server<br>(License information, for multi-server solutions) | Inbound | TCP (Flexera-internal) | 27000<br>27010 |
| *syngo*.via server ↔ SQL server | Inbound<br>Outbound | TCP | 1433 |
| *syngo*.via server ← CT<br>(Direct Image Transfer / Fast Transfer) | Inbound | TCP | 5445 |
| *syngo*.via server ← MR (Report Access) | Inbound | SOAP | 8888 |

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via server ↔ scanning workplace<br>(open only port 5559, Status Monitoring Application) | Inbound<br>Outbound | HTTP<br><br><br><br>TCP | 5555<br>5559<br>5560<br>8889<br>9000<br>9995 |
| *syngo*.via server → Domain Controller<br>(Synchronization with Domain Controller) | Outbound | LDAP / TCP / UDP | 389 |

To enable **additional** services and functions, the following ports must be opened:

| Service/Function | Direction | Protocol | Port number |
|---|---|---|---|
| *syngo*.via server or Acquisition Workplace (CT RT Engine) → external LAP-System | Inbound<br>Outbound | TCP | 6661 - 6670 |
| Only for WebOptions: *syngo*.via server ← Web Access, CIFS for WebViewer, WebReport, Licensing, Index Manager, Authentication | Inbound | TCP | 445<br>4510 |
| Only for WebOptions: *syngo*.via server ← WebViewer (view medical images on mobile devices) | Inbound | HTTPS<br>TCP, SSL | 4443<br>4475 |

## 10.8 Encryption of client/server communication

In order to protect patient health information (PHI), you can enable encrypted communication between *syngo*.via server and the connected clients. Thus, channels that may contain PHI data are encrypted.

**CAUTION**

Unencrypted client-server transfer of patient health information.

**Patient health information will be vulnerable in case of unauthorized network access.**

- Set up encrypted client/server communication.
- Set up encrypted DICOM communication.
- Protect your network by a firewall.

Prior to enabling encryption, a certificate must be installed. The certificate is usually obtained from a certificate authority (CA).

Please note that you are responsible for acquiring, installing and maintaining the certificates.

You must stop the application server before switching encryption on or off.

After switching encryption on or off you must restart the client **twice**. On the first restart, the internal configuration of the client is updated and an error message is displayed. Confirm it and start the client a second time.

- (→ Page 217 *Configuring encrypted client/server communication*)
- (→ Page 219 *Validating certificates for encrypted communication*)

X.509 certificates contain a public key and an identity (an organization, a hostname, or an individual), and should be signed by a certificate authority (CA).

When a certificate is signed by a certificate authority, or validated by another means, you can rely on the public key it contains and establish a secure communication with another party. Furthermore, you can validate documents that are digitally signed by the corresponding private key.

If your organization wants to use a self-signed certificate for encrypted client/server communication, you have to make sure that each connected client trusts this certificate.

That is, on each connected client, the certificate must be available in the **Trusted Root Certification Authorities** certificate store of the **Local Computer** account.

### 10.8.1 Configuring encrypted client/server communication

Encryption of the client/server communication is switched off by default.

In order to configure and enable the encryption, you must perform two tasks in sequence:

- Binding certificates in IIS Manager
- Switching encrypted communication on

**CAUTION**

Security certificates may expire.

**Encrypted client/server communication will be blocked, when the server certificate expires.**

◆ Renew security certificates in time.

✓ The server name (environment variable %MED_SERVER%) used at the client must be identical to the common name given in the certificate. Clients need to use the fully qualified domain name of the server.

It might be necessary to re-install a client with the correct server name in order to ensure this. In this case, install the RTC from the deployment server using the last DNS Name entry in the certificate's “Subject Alternative Name” field as the host name in the URI.

✓ A valid x.509 server certificate (Enhanced key usage = Server Authentication) including its private key is available in the Personal certificate store of the Local Computer account. Besides, it is recommended to use a key length with 2048 bit and the used thumbprint algorithm should be stronger than SHA1.

### Binding certificates in IIS Manager

1. Log on to the server operating system.
2. Notify all clinical users and stop the application server. (➔ Page 91 *Stopping / restarting the syngo.via application server*)
3. Open the **Internet Information Services (IIS) Manager**.
4. Navigate to the **Home** page, **IIS** section (or filter the view by "server").
5. Check under **Server Certificates** if your certificate is available. If not, import your certificate from the Personal certificate store into IIS.
6. Navigate to **Sites**, right-click **Default Web Site**, and choose **Bindings...** from the context menu.
7. Select the **https** type and click **Edit**.
8. Select your SSL certificate and click **OK**.
9. Right-click **Default Web Site** and choose **View Applications**, double-click **/Reporting** and open **SSL Settings**.

The location of the **Reporting** application may vary, depending on your operating system.

10 Select **Require SSL** and click **Apply**.

### Switching encrypted communication on

1 On the server desktop, double-click the ***syngo*.via Server Shell** icon.

When entering `syngo.Common.Communication.Tools.EncryptionConfiguration.exe /?` you get help on parameters and options.

2 Enter the command `syngo.Common.Communication.Tools.EncryptionConfiguration /list` and press **Enter**.

All certificates are listed that have a private key and are part of the personal store of the APS.

3 Copy the thumbprint value of your selected certificate.

4 Enter the command `syngo.Common.Communication.Tools.EncryptionConfiguration /set EncryptCommunication=On Certificate Thumbprint=<value you copied>` and press **Enter**.

Encryption is activated.

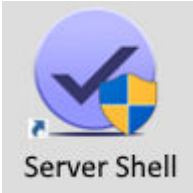
5 Restart the application server.

To deactivate encryption, enter the command `syngo.Common.Communication.Tools. EncryptionConfiguration /set EncryptCommunication=Off` and press **Enter**.

## 10.8.2 Validating certificates for encrypted communication

A valid certificate is a prerequisite for activating communication encryption. With the encryption tool, you can identify problems that occur during certificate validation.

When encryption is activated, the certificate obtained is automatically validated.




1 On the Windows Server desktop, double-click the ***syngo*.via Server Shell** icon.

2 Enter the command `syngo.Common.Communication.Tools.EncryptionConfiguration /list` and press **Enter**.

All certificates that have a private key and are part of the personal store of the APS are listed.

3 Copy the thumbprint value of your selected certificate.

4 Enter the command `syngo.Common.Communication.Tools.EncryptionConfiguration /validate <value you copied>` and press **Enter**.

The certificate is validated according to standard criteria, such as expiration of validity period, certificate revocation, completeness of certificate chain and correct DNS identity of the remote endpoint.

If no parameter is given, the currently configured certificate thumbprint is taken as the default.

The message **Certificate was successfully validated** is shown if the certificate is valid.

If the certificate is not valid, the message contains an error description and the associated remedy for the following cases:

- Valid but revoked certificate
- Out-of-date certificate
- Invalid common name certificate
- Untrusted certificate
- Entered non-existing certificate thumbprint

5 If possible, solve the problem as described or contact the certificate authority to obtain a new certificate.

### 10.8.3 Replacement of self-signed *syngo*.via certificates

*syngo*.via comes with self-signed certificates for public key encryption. Certificates ensure strong security for the product and its applications in your organization, including authentication, encryption, and data integrity.

These self-signed certificates are generated during the installation of *syngo*.via. They are used for the following components and services:

- IIS (HTTPS certificate, for server identification, *syngo*.via Administration Portal call-up, and Reporting application)
- RDP (listener port for remote desktop session on the server)
- *syngo*.via WebViewer
- encrypted DICOM communication
- HP Management Homepage
- MS SQL Server (database)

Use the **Microsoft Management Console** on the server, with the "Certificates" snap-in.

You can identify the self-signed certificates under **Local Computer** > **Personal** > **Certificates** by the friendly name "syngo_server".

It is a good and reoccurring practice for IT Administrators to replace self-signed certificates by certificates issued by the trusted Certificate Authority (CA) used in your organization.

A replacement might have the following benefits:

- IT Administrators can adapt to higher security requirements
- It prevents from warnings while calling Web sites or services from *syngo*.via server (for example, *syngo*.via WebViewer, *syngo*.via Administration Portal, and HP System Management Homepage)
- It may remove findings of network security scanners regarding untrusted certificates

In computer security, a chain of trust is established by validating each component of hardware and software from the end entity up to the root CA certificate.

Details on the replacement procedures are provided in the *Administration Online Help, Certificate section*.

### Further readings

- Encrypting client/server communication (→ Page 217 *Configuring encrypted client/server communication*)
- Encrypting DICOM communication

# 11 Smart Remote Services

The Smart Remote Services (SRS) infrastructure provides a secure data link that connects your medical systems to the service experts in the Customer Care Center.

Over SRS, the performance and condition of your equipment can be monitored in real time. It makes a broad range of proactive and interactive services available – including fast error identification, remote repair and software updates, preventive maintenance, and collaboration services.

Most of the services that formerly required on-site visits are now available by data transfer due to automatic reporting or by remote access to your system.

The connection to the SRS can be established by two different ways:

- SRS router

  Through a dedicated SRS router within the customer network.

- VPN tunnel

  A virtual network adapter on your system's server will be used.

The Customer Care Center can only access the system from a remote location if you explicitly grant remote access.

The following prerequisites must be fulfilled:

- A minimum broadband Internet connection bandwidth for uncompromised service support with 2000 kBit/s downstream and 512 kBit/s upstream. Otherwise, certain support services may not be provided, and the agreed remote response time cannot be guaranteed.
- To enable your system to perform SRS-based services, specific communication ports need to be opened and the SRS has to be configured.
- A dedicated router is only needed if you want to use the **SRS Router** option.

# 12 First-level support

The administrator is the first-level support for clinical users. If a user encounters a problem with *syngo*.via, the administrator shall first try to solve it himself using **Status Monitoring** or the **Message Viewer**.

Many issues can also be resolved quickly, for example, by restarting the *syngo*.via server.

Here is an overview of how the administrator can support clinical users:

- Using troubleshooting tools

  (➔ Page 226 *Troubleshooting tools*)

- Accessing information in the LifeNet *syngo* Information Center

  https://bpshealthcare.siemens.com/sic

- Accessing the Services Knowledge Base

  https://skb.healthcare.siemens.com

- Creating SaveLogs for analysis
- Performing a Client-Server Connection Test
- Providing remote access/support:
  - by Remote Assistance desktop sharing with the Customer Care Center

    (➔ *Remote Assistance in the Administration Online Help*)

  - by Expert-i collaboration with clinical users

    (➔ *Working with Expert-i in the Application Online Help*)

- Advising users to capture snapshots or create videos with the "syngo Flight Recorder" to facilitate investigations into incidents

  (➔ *syngo Flight Recorder in the Application Online Help*)

# 12 First-level support

The service for the hardware and the operating system is in the responsibility of the clinical IT department.

## 12.1 Troubleshooting tools

The following tools provide support in general problem determination and troubleshooting:

- Status Monitoring

  Use **Status Monitoring** to check the system status and to identify which application processes and system components do not work properly.

  See (→ Page 60 *Status Monitoring*).

  If a process has failed, you receive detailed information concerning the impact of the failed process.

- Message Viewer

  Use the **Message Viewer** to find the corresponding message to an identified error condition. In addition, you receive suggestions for further analysis and corrective actions.

  See (→ Page 62 *Message Viewer*).

- HP iLO

  Use the Hewlett Packard Integrated Lights-Out (iLO) board to establish a connection to the *syngo*.via server. You can access the BIOS setup and select an item from the Advanced Boot Options. The iLO board is also useful if a Windows Remote Desktop connection cannot be established.

  See (→ *syngo.via Administration Online Help*)

- Third-party tools

  Affiliated software and hardware vendors (for example, of the remote service board, the database, or the hardware vendor) provide additional tools for monitoring and service. For further information, refer to the user documentation of these tools.

- Client-Server Connection Test

  Use the **Client-Server Connection Test** to test the connections between the client and the servers. Different test steps for client hardware and software, network latency and bandwidth, and server hardware and software are executed.

  See (➔ *syngo.via Administration Online Help*).

- OPENLink

  Use **OPENLink** to identify network problems between the RIS and *syngo*.via interfaces. You can trace the activity on the network and/or on data mapping level and restart the connections, interfaces, and the server.

  See (➔ Page 60 *Status Monitoring*).

- STS Consistency tool

  Use the **STS Consistency** tool to detect and repair inconsistencies between the Short Term Storage (STS) and the database.

  See (➔ *About the STS Consistency Tool in the syngo.via Configuration Online Help*).

# 12 First-level support

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The original language of this document is English.

Made in Germany

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Published by Siemens Healthcare GmbH / Print No. P02-004.621.02.01.02 / © Siemens Healthcare GmbH, 2010 – 2019
Date of first issue: 2019-03